

Artemis HPC Training

Introduction to the Artemis HPC

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Sydney Informatics Hub

A Core Research Facility



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HPC Access

For login details: <http://bit.ly/2xcUrYw>

HPC Hostname: hpc.sydney.edu.au

Example: `ssh -Y <user_name>@hpc.sydney.edu.au`



Image: thecultureconcept.com

Introduction

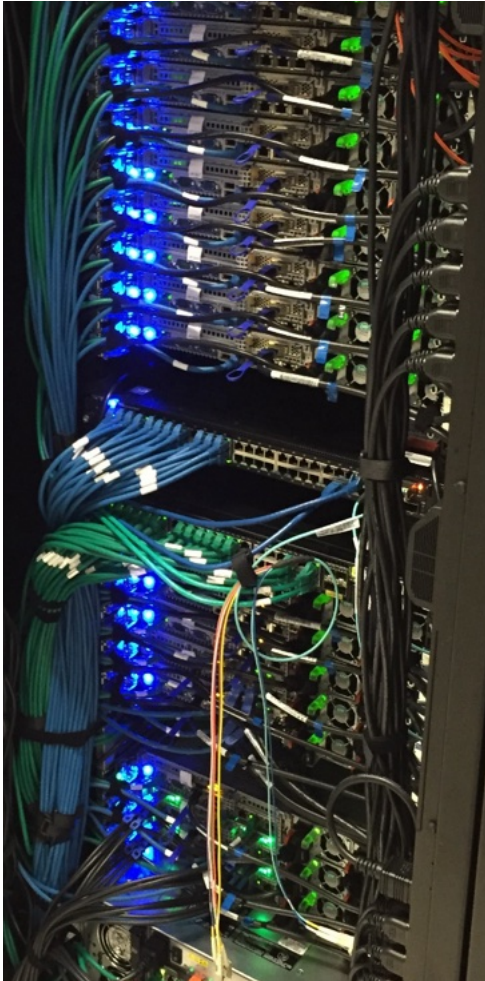
- Sydney Informatics Hub (a Core Research Facility) alongside ICT
- Under the Research Portfolio, Core Research Facilities provide access to high-end research infrastructure, personnel and services: supporting quality research outcomes for USyd researchers
<http://sydney.edu.au/research/support/facilities.shtml>
- Provides Artemis HPC and associated training and support

Overview of today's course

- Connecting to Artemis
- Artemis directory structure and 'projects'
- PBS Pro job scheduler
- PBS scripts
- Submitting and monitoring jobs
- Checking job output and resource usage
- Getting accounts on HPC and RDS via RDMP

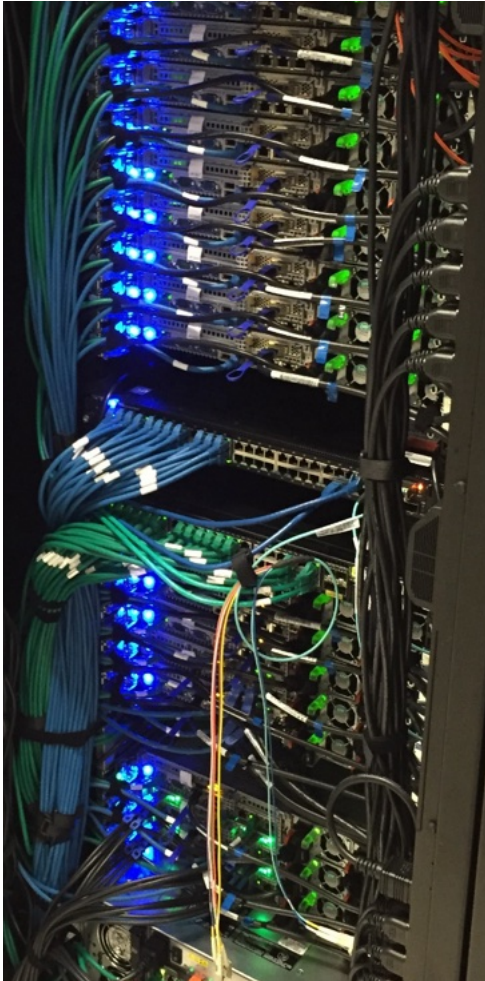


Artemis HPC



- HPC stands for 'High performance computing'
- Artemis is the name chosen for USyd's HPC cluster
- In 2016, Artemis was expanded:
 - From 1,512 cores and ~8 TB RAM
 - To 4,264 cores and > 18 TB RAM
 - Upgraded to the latest PBS Pro job scheduling software
- In 2017, Artemis will be further expanded, adding more:
 - CPU
 - GPU
 - Hard disk space

Artemis HPC



- Artemis is ideal for any computational analysis requiring:
 - Long walltime
 - High RAM usage
 - Analysis requiring multiple cores or multiple nodes
 - Large number of single-core jobs
 - Large input or output data
- Artemis is for everyone!
 - Goal to **centralise** USyd computational resources
 - This **reduces the overheads** of Faculty or School-based servers, so researchers do not have to worry about maintaining/updating the hardware and software
 - Provides a great **incentive to funding bodies** to fund projects where HPC is provided to the project at no cost by the host institution

Your use of the Artemis HPC

- No cost to USyd researchers!
- Access requires a unikey and valid RDMP (later)
- **Publications involving use of Artemis or other Sydney Informatics Hub services MUST acknowledge these free resources (this forms part of the user agreement)**
- Suggested acknowledgement:
“The authors acknowledge the Sydney Informatics Hub and the University of Sydney’s high performance computing cluster Artemis for providing the high performance computing resources that have contributed to the research results reported within this paper.”
- Benefits of acknowledgement:
 - Publications with appropriate HPC acknowledgment are eligible for the **SIH HPC Publication Incentive**
 - Acknowledgements demonstrate research outputs arising from the USyd investment into Artemis and support staff: this drives further funding meaning ongoing upgrades and maintenance for our HPC and services

SIH HPC Publication Incentive

- New scheme, run quarterly for 2017
- **Up to \$500 cash prize for each of 3 winners!**
- To be eligible, papers must:
 1. Be published in a peer-reviewed journal
 2. Be authored or co-authored by a current University of Sydney staff member or student
 3. Include data analysed on **Artemis HPC or NCI Raijin**
 4. **Acknowledge the Sydney Informatics Hub** at the University of Sydney for providing access to HPC
 5. Ensure that the publication is listed under your Research Outputs on IRMA
 6. Not have been entered in a previous round of the incentive
- Contact sih.info@sydney.edu.au for more information

Connecting to Artemis

You connect to Artemis remotely: from any computer, any location, any operating system

You just need a terminal client, and if you are on an external network, you will first need to connect to VPN

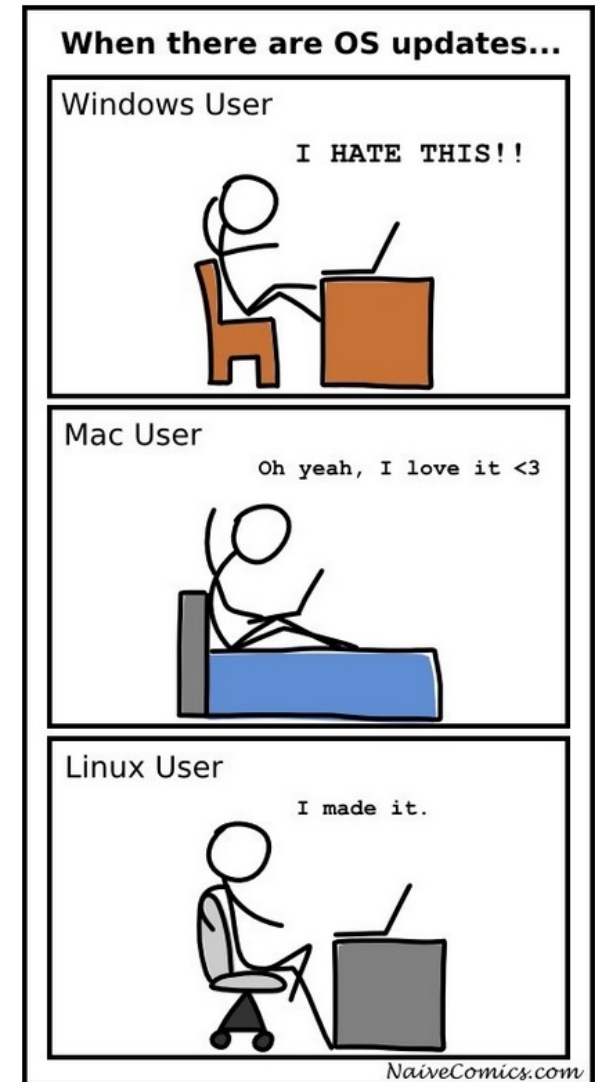


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Accessing the Artemis HPC from your computer

- **Mac/Linux:** use the existing Terminal application
- **Windows:** install a terminal client – many available with various features, eg:

Client	License	Benefit
Xwin	Provided by USyd	Edit text in a graphical window
Cygwin	Free	Approximates a native Linux environment on Windows PC
Putty	Free	Easily change size and colour of text



Connecting to Artemis from: Mac

- Optional: Install XQuartz: <https://www.xquartz.org/>
- Go to Finder → Applications → Utilities
- Double-click Terminal
- Type the following into Terminal then hit enter:

`ssh -Y ict_hpctrain<N>@hpc.sydney.edu.au`

NOTE:

If you see text enclosed in arrows, this means replace the text with content relevant to you

NOTE:

The -Y option only works if you installed XQuartz

Connecting to Artemis from: Mac

- Optional: Install XQuartz: <https://www.xquartz.org/>
- Go to Finder → Applications → Utilities
- Double-click Terminal
- Type the following into Terminal then hit enter:

`ssh -Y ict_hpctrain<N>@hpc.sydney.edu.au`

NOTE:

The `-Y` option only works if you installed XQuartz

- Type the password then press 'enter'

NOTE:

If you see text enclosed in arrows, this means replace the text with content relevant to you

```
Stephens-MacBook-Pro:~ stephenlevo$ ssh -Y ict_hpctrain10@hpc.sydney.edu.au
The authenticity of host 'hpc.sydney.edu.au (10.250.96.204)' can't be established.
RSA key fingerprint is da:c3:1e:26:24:fc:f9:1c:74:99:99:fb:85:26:d7:50.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'hpc.sydney.edu.au,10.250.96.204' (RSA) to the list of known hosts.
ict_hpctrain10@hpc.sydney.edu.au's password:
Last login: Fri Apr 24 10:39:55 2015 from vlan-2657-10-16-233-8.5 student-wireless-sydney.edu.au
-----
Welcome to the University of Sydney HPC Linux Cluster - Artemis
-----

** NOTE: All jobs must request a project the user is a member of.
Please specify -P <project> on the command line,
or #PBS -P <project> in your PBS scripts.

** Please don't run jobs on the login node!!! Use the queuing system

** Repeat! If you run jobs on the login nodes they will be subject to
random termination if they cause load problems. Please don't do it.

[ict_hpctrain10@login2 ~]$
```

If connecting for the first time you will be asked if you trust the authenticity of host – type 'yes' then hit enter key

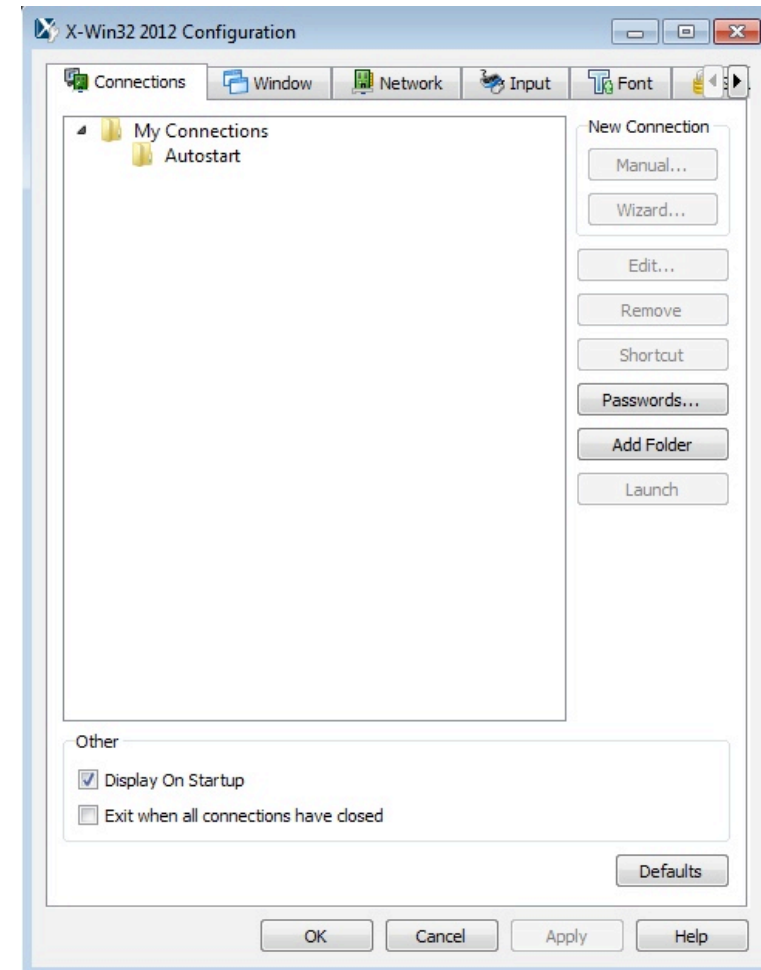
Logon success

Where to download software for your own Windows machine?

- Xwin: <http://softserv.usyd.edu.au/data/X-Win32/X-Win32-14.0/>
 - Requires unikey credentials
 - Follow the instructions to install license server details
- Putty: <http://www.putty.org/>
 - Download and run 'putty.exe'
- Cygwin: <https://www.cygwin.com/>
 - Download and run 'setup-x86_64.exe' (64 bit OS) or 'setup-x86.exe (32 bit OS)
 - During setup, select additional packages to install. Recommended: openssh, x-org server, xinit

Connecting to Artemis from: Windows with Xwin

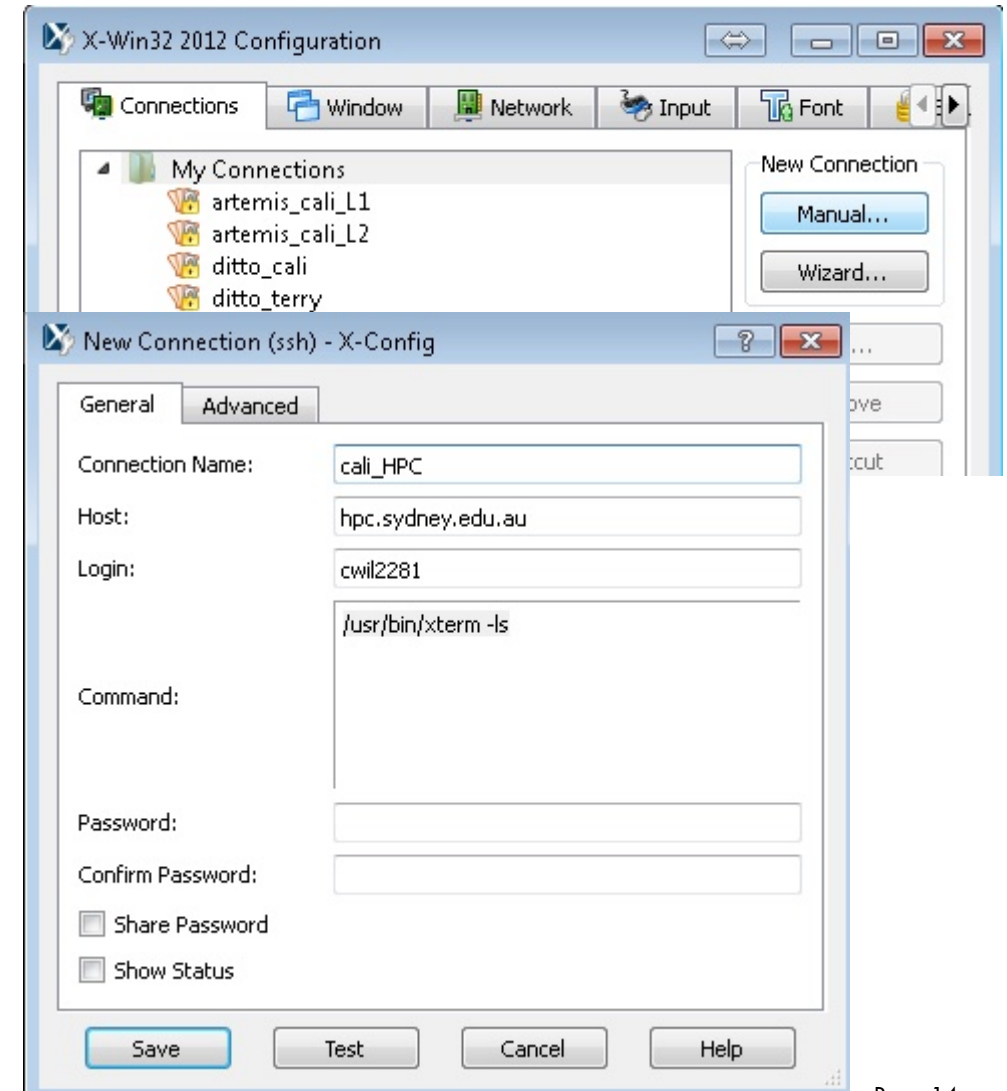
- Download and install X-Win: <http://softserv.usyd.edu.au/data/X-Win32/X-Win32-14.0/>
- Click the windows icon in the lower-right hand corner of your screen
- Type X-Win into the search bar
- Click the X-Win32 or X-Config options
- A configuration window will appear



Connecting to Artemis from: Windows with Xwin

- Along the right hand side under 'New Connection' select 'Manual'
- Select 'ssh' for connection method then click 'Next'
- In the 'Connection Name' field, enter any descriptive name eg `<name_HPC>`
- In the 'Host' field, enter:
`hpc.sydney.edu.au`
- In the 'Login' field, enter your unikey
 - Training unikey: `ict_hpctrain<N>`
- In the 'Command' field, enter:
`/usr/bin/xterm -ls -fa 'consolas' -fs 11`
- Click 'Save' then 'Launch'

Letter 'l'
not numeral 1

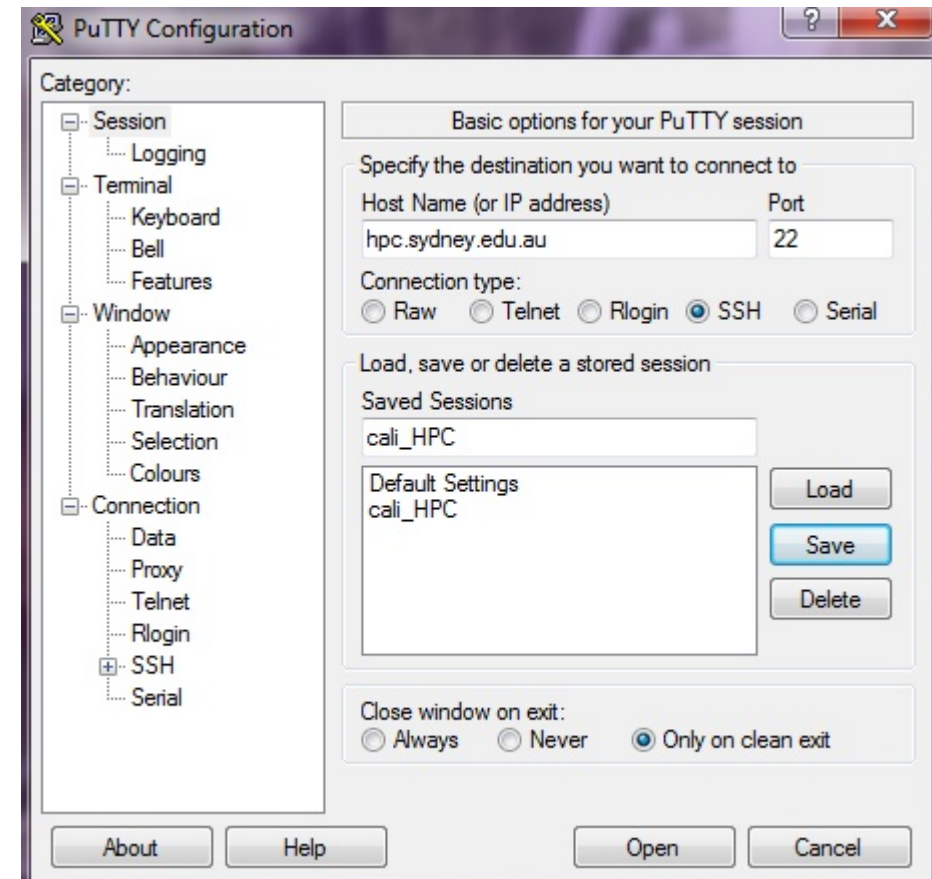


Connecting to Artemis from: Windows with Putty

- Launch Putty: A configuration window will appear
- In the 'Host Name' field, enter
hpc.sydney.edu.au
- Leave port as 22 (default)
- Type a name for your connection into 'Saved Sessions'
- Under 'Connections', expand 'SSH' and click on 'X11'
- Select the check box next to 'Enable X11 forwarding'
- Click 'Session' (top left)
- Click 'Save' (VIP step!) then 'Open'
- At 'login as' enter your unikey (or ict_hpctrain<N>)
- Enter your unikey (or training) password

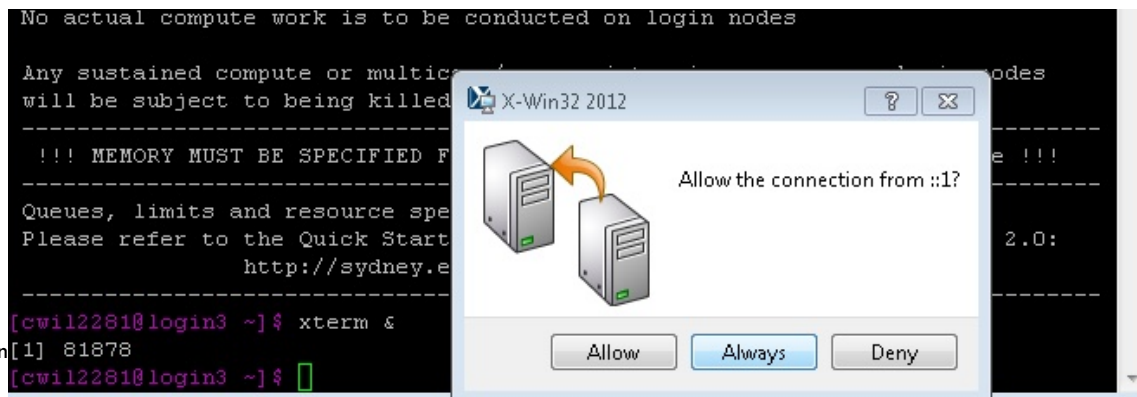
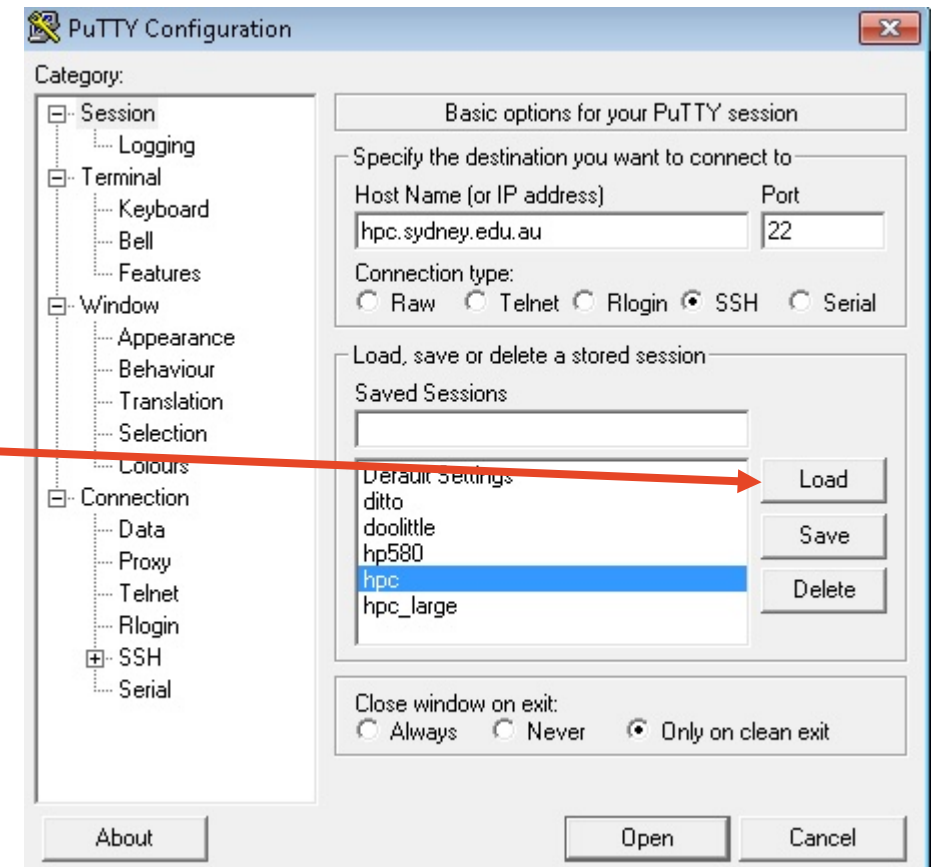


<http://www.putty.org/>



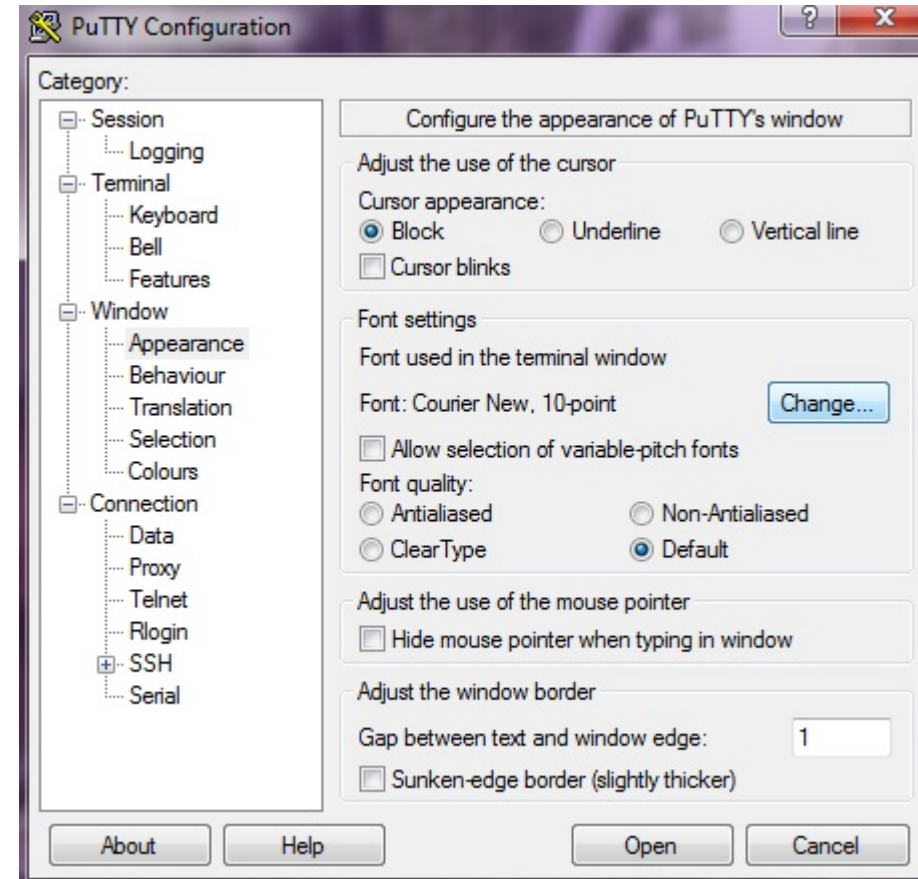
Connecting to Artemis from: Windows with Putty

- At 'login as' enter your unikey (or ict_hpctrain<N>)
- Enter your unikey (or training) password and hit enter
- Once you have a saved session, you can launch it by selecting the saved session name, clicking 'Load' then 'Open'
- The first time you use an X11 command, you will be asked if you allow the connection. Select 'Always'



Adjusting the terminal font size/colour on Putty

1. Open your Putty configuration window again
2. Click on your saved session name
3. Click 'Load'
4. Under 'Window', select 'Appearance'
5. Under 'Font Settings', click 'Change'
6. Change the font as desired
7. Click 'OK'
8. Click 'Session'
9. Click 'Save'
10. Click 'Open' to launch your new-look terminal



Artemis directory structure and ‘projects’

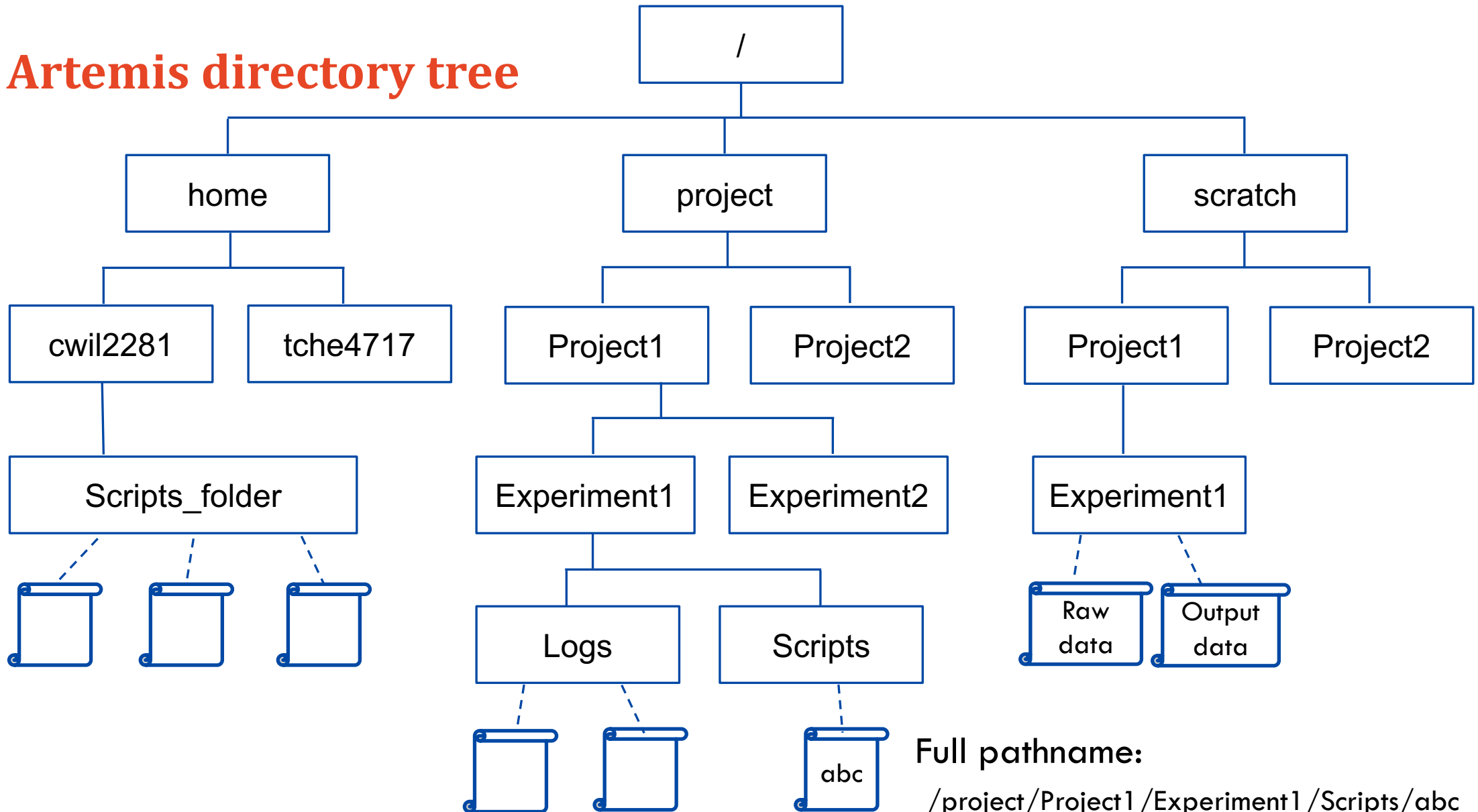
All user accounts on Artemis are linked to at least one ‘project’

Projects are registered using a Research Data Management Plan (rdmp.sydney.edu.au)



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Artemis directory tree



Full pathname:

/project/Project1/Experiment1/Scripts/abc

*File suffixes are irrelevant to Linux, but useful to you

Artemis directory tree

- When you first log in, you are situated in your home directory:

`/home/<unikey>`

`/home/ict_hpctrain<N>` ←

- Home directories have a tiny 10 Gb quota
- Do not do any project work in here: use instead your **project directory**

REMEMBER:

If you see text enclosed in arrows, this means replace the text with content relevant to you

Artemis directory tree

- All project directories are located in:
/project
 - Each project has its own directory with 1 Tb space
 - This is a **shared directory amongst all members of the project**
 - To ensure quota is not exceeded, **check usage** and **utilise scratch space**
 - Project is not permanent storage and is not backed up, so **utilise Research Data Store (RDS, later)**
- You (or your Lead CI) selected the short project name in the RDMP
 - You can find out which project(s) you're a member of by typing 'groups' on the command line.
 - Projects you are a member of begin with RDS- and end with -RW

```
[ict_hpctrain1@login3 ~]$ groups  
linuxusers HPC-ACL RDS-ICT-HPCTRaining-RW RDN-USERS RDN-CORE-Training  
RDS-CORE-Training-RW  
[ict_hpctrain1@login3 ~]$
```



Remember these
names!

Artemis directory tree

- All project directories are located in:
/project
- The `ict_hpctrain<N>` accounts are members of two projects, which means this account has access to two project directories:
 - **/project/HPCTRAINING**
 - **/project/Training**

← “short names”

```
[ict_hpctrain1@login3 ~]$ groups  
linuxusers HPC-ACL RDS-ICT-HPCTRAINING-RW RDN-USERS RDN-CORE-Training  
RDS-CORE-Training-RW  
[ict_hpctrain1@login3 ~]$
```

Remember these
names!

Artemis directory tree

- All project directories are located in:
`/project`
- Check usage with the `pquota` command:

```
[[ict_hpctrain1@login3 Training]$ pquota
Disk quotas for user ict_hpctrain1 (uid 508194):
  Filesystem    used    quota    limit    grace    files    quota    limit    grace
    /home    2.191M    10G    11G      -        34         0         0         -
Disk quotas for group RDS-ICT-HPCTRAINING-RW (gid 12524):
  Filesystem    used    quota    limit    grace    files    quota    limit    grace
    /project   125.7G     1T   1.074T      -       5342         0         0         -
Disk quotas for group RDS-CORE-Training-RW (gid 14206):
  Filesystem    used    quota    limit    grace    files    quota    limit    grace
    /project   50.26G     1T   1.074T      -       1592         0         0         -
[[ict_hpctrain1@login3 Training]$
```

Artemis directory tree

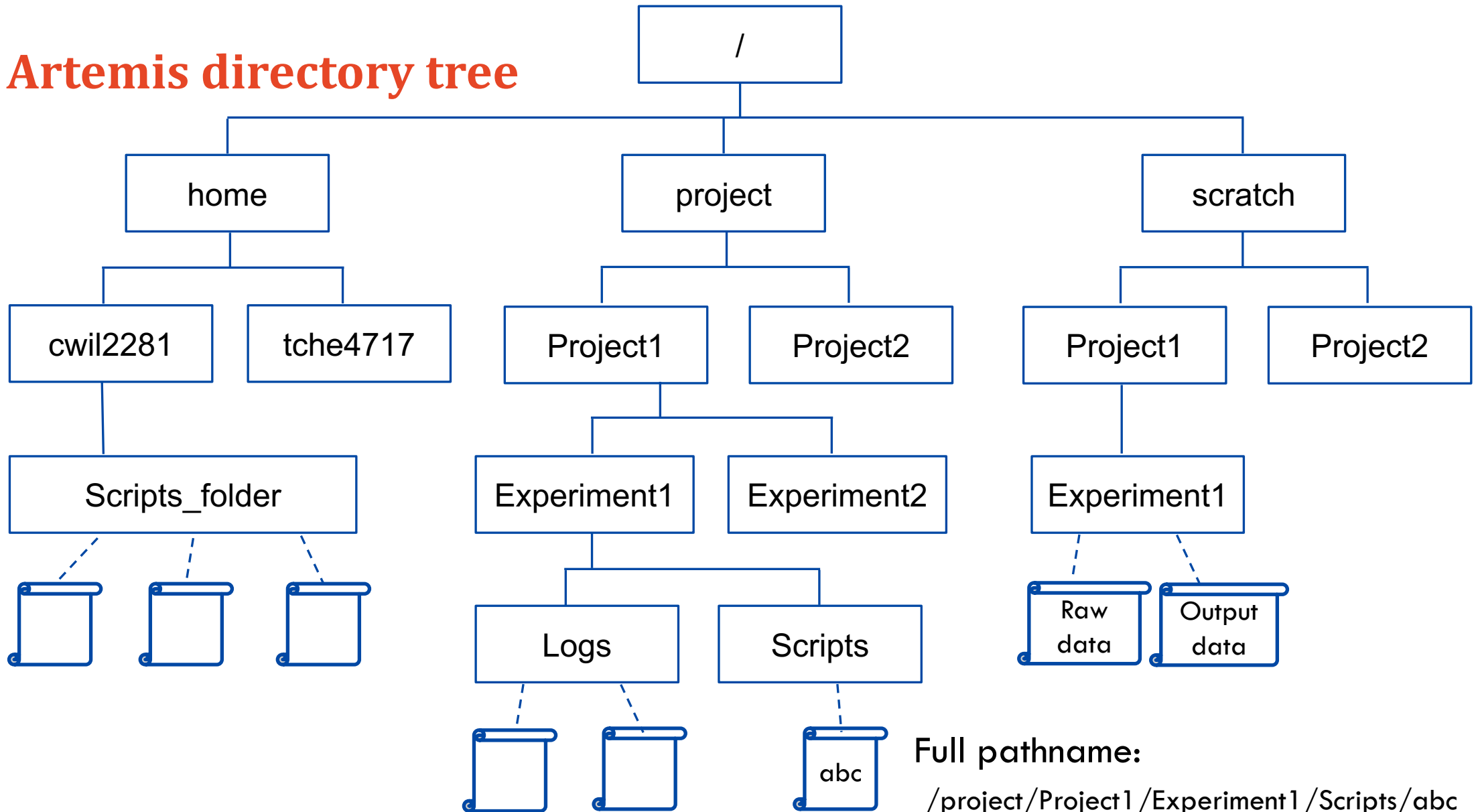
- Each project has a corresponding 'scratch space' situated in:
`/scratch`
 - Same project name is used, eg your project is called 'Training', you will have access to:
`/scratch/Training`
`/project/Training`
 - Using scratch space for input and output files is recommended because:
 - Disk quota is vast and not capped to 1 Tb like project directories
 - Ensures there is space to write your output files:
 - If your quota is exceeded during an analysis, your job will likely fail and/or the output will be non-existent or truncated
 - Remember project is shared with other group members, so there may be other group members filling up the quota at the same time as you
 - **Scratch is not permanent storage:** after your job completes, back up output to RDS (see the data transfer course!) and delete what is no longer needed. System managers clean up scratch periodically!

Artemis directory tree

- All software is installed in:
`/usr/local`
- To see a list of installed software, run:
`module avail`
- If the software you want is not installed:
You can request any open-source (free) software, as long as it has a Linux command line version
 - Go to the ICT Self-Service Portal at https://sydney.service-now.com/selfservice/ict_services.do
 - Select ICT Services → Research → High Performance Computing Request
 - Under ‘Category’, select ‘Application (new software or different version)’
 - Use the text box to name the software and version you want installed, including the download URL

You will be sent TrackIT notifications regarding the status of your request

Artemis directory tree



Full pathname:

`/project/Project1/Experiment1/Scripts/abc`

*File suffixes are irrelevant to Linux, but useful to you

Change into your project directory

If using your own HPC account, your project directory is:

`/project/<YourProject>`

The long way: `/project/RDS-<FAC>-<YourProject>-RW`

If using a training account, your project directory for today is:

`/project/Training`

The long way: `/project/RDS-CORE-Training-RW`

- ❖ Short and long project names are both equally valid
- ❖ Some programs only work with the long name, so it is worth knowing both versions

Change into your project directory

When you see this bullet point, it means 'do something'

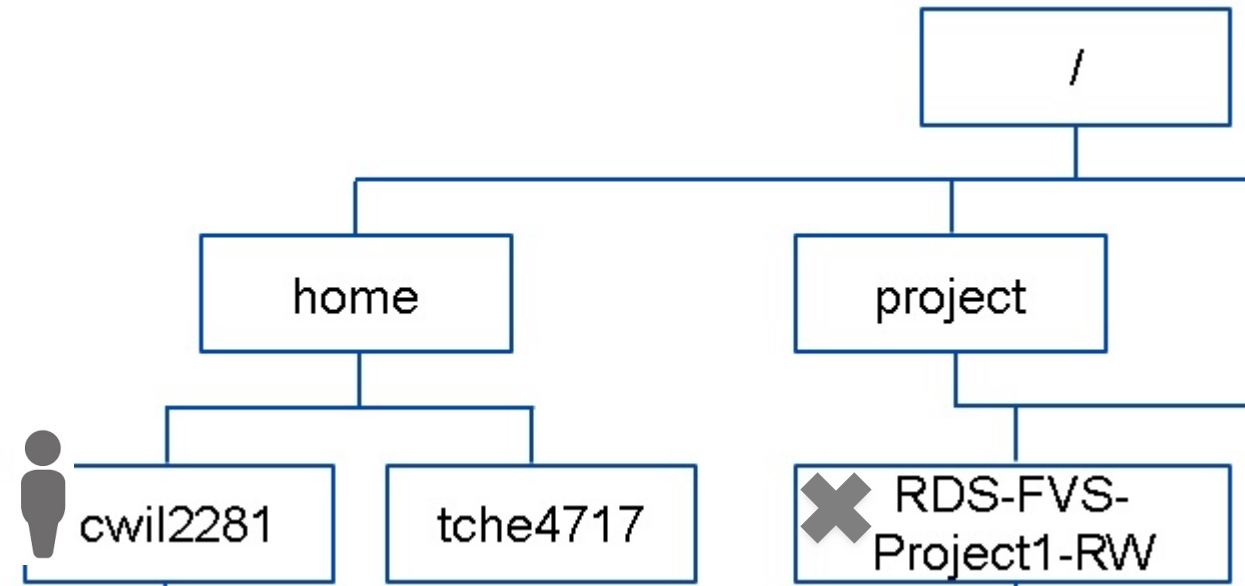
➤ Change into your project directory now

If using your own HPC account:

`cd /project/<YourProject>`

If using a training account:

`cd /project/Training`



Make a working directory for today

- Make a directory, call it your name
- Change into it
- Work in here for the rest of today

```
mkdir <YourName>
```

```
cd <YourName>
```

(remember to replace the content inside the arrows with your details)

- We will use this location to store our raw data and to write output files

Very important:

You are sharing the same project space with other users today – you all have read and write permissions to each other's directories!!

So please be sure to only work within your own named directory.

Prepare the input data

- Today, the sample data can be found here:
`/project/Training/sample_data.tar.gz`
- Everyone: please double check you are in your working directory before running the extract command, because you are all members of the same 'project' today
- eg: I would be in `/project/Training/stephen`

- Extract the archive:

```
tar -zxvf ../sample_data.tar.gz
```

- Move all files out of 'Sample_data'

```
mv sample_data/* .
```

- Delete the empty 'Sample_data' directory

```
rmdir sample_data
```

If your system has decompressed the archive (depending on your settings this may happen automatically), use instead:

```
tar -xvf Sample_data.tar
```

The “..” here means “the directory above this one. Only run this command from your own directory:
`/project/Training/<YourName>`

PBS Pro

Job scripts

Artemis 'queues'

Text editing on Artemis HPC



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PBS Pro

- We now have our directory set up with input data – we can submit jobs to Artemis!
- Why “submit jobs” and not just “run jobs”?
- Unlike other single-node Linux servers you may have used, Artemis is a **cluster** of computers
- Jobs are allocated to nodes on the cluster by **scheduling software** – in this case, PBS Pro
- PBS Pro manages the workload on Artemis by scheduling jobs to run at different times and on different nodes, to ensure the cluster is being used to optimal efficiency
- The practical difference to you:
 - Instead of running your job directly on the command line to be immediately executed, you must submit it to PBS Pro
 - You can submit your job using a PBS script (covered today), directly on the command line for single-command jobs, or by entering commands in ‘interactive mode’
 - The scheduler may run your job right away, or it may be placed in a queue and start later

PBS Pro

- When your job starts is determined by:
 - What resources you have requested (RAM, cores, walltime)
 - What resources are currently available on the HPC
 - What other jobs are in queue
 - 'Fair share' user model – factors recent usage to ensure no monopoly on resources
- Resource requests and relevant job information is provided to PBS Pro using 'directives'
 - Either saved in a PBS script or entered manually on the command line

PBS scripts

- Now we will create and examine your first PBS script
- Copy the file 'index.pbs' and call the new file 'basic.pbs'
`cp index.pbs basic.pbs`
- Open basic.pbs with your **preferred text editor...**

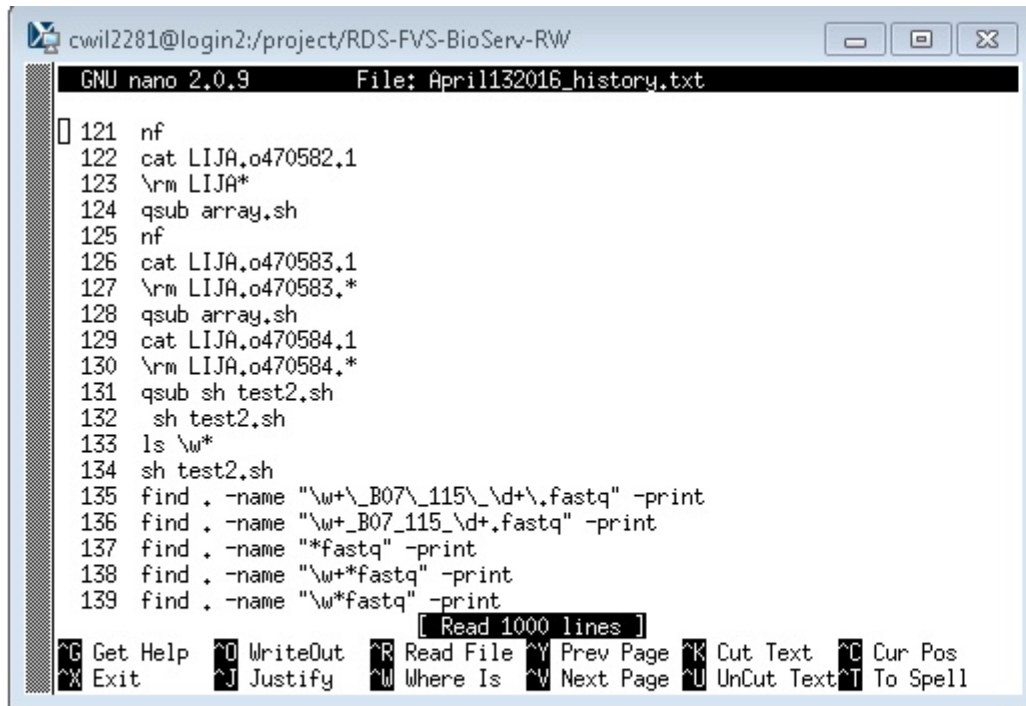
Copying scripts is a great way to avoid retyping directives for every new job! Just be sure to modify the directives to suit each analysis

Text editors on the HPC

Editor	Ease of use	Power/flexibility	GUI option*
GEdit	High	Low	Yes
Nano	Moderate to High	Low	No
Emacs	High with gui Moderate without GUI	High	Yes
Vi/vim	Low	High	No

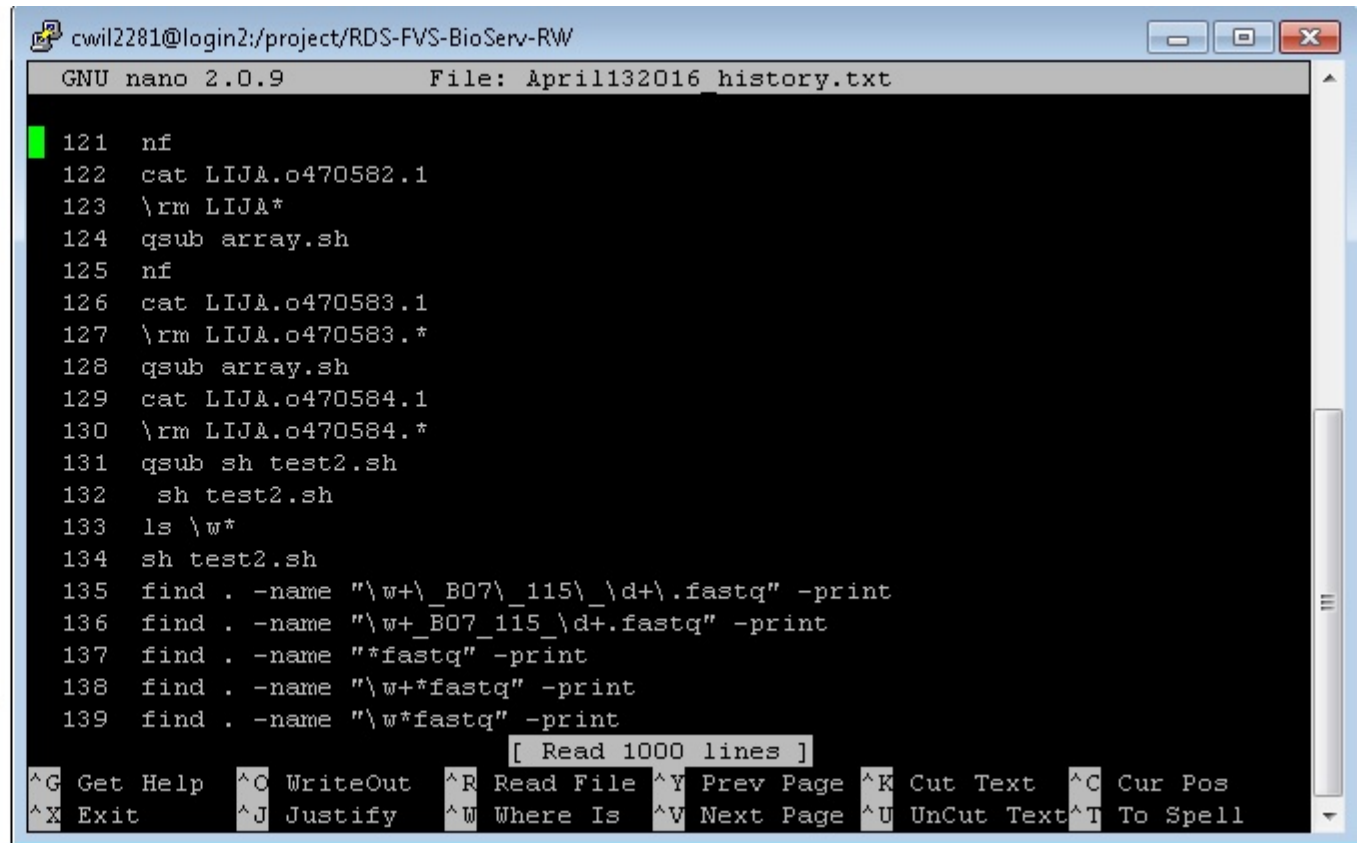
- If you have an X Server installed (eg Xwin, Xquartz):
 - GEdit and Emacs will run as a GUI (point-and-click)
 - Make sure to include `-X` or `-Y` flag in ssh connection command
 - Run GEdit or Emacs GUI with the ampersand ('&') character. This runs the program in the background, ie you can use your terminal and edit text at the same time
- If not:
 - Nano is the easiest to use if you are an inexperienced shell user

Nano in Xwin or Putty/Mac terminal



The screenshot shows the Nano editor interface within a terminal window. The title bar indicates the user is 'cwil2281' at 'login2:/project/RDS-FVS-BioServ-RW'. The editor's status bar shows 'GNU nano 2.0.9' and the file name 'File: April132016_history.txt'. The main text area contains a list of commands numbered 121 to 139, including 'nf', 'cat', 'rm', 'qsub', 'sh', and 'find'. A status bar at the bottom displays various keyboard shortcuts for actions like 'Get Help', 'WriteOut', 'Read File', 'Prev Page', 'Cut Text', 'Cur Pos', 'Exit', 'Justify', 'Where Is', 'Next Page', 'UnCut Text', and 'To Spell'. A message '[Read 1000 lines]' is visible above the shortcuts.

```
GNU nano 2.0.9 File: April132016_history.txt
121 nf
122 cat LIJA.o470582.1
123 \rm LIJA*
124 qsub array.sh
125 nf
126 cat LIJA.o470583.1
127 \rm LIJA.o470583.*
128 qsub array.sh
129 cat LIJA.o470584.1
130 \rm LIJA.o470584.*
131 qsub sh test2.sh
132 sh test2.sh
133 ls \w*
134 sh test2.sh
135 find . -name "\w+_B07_115_\d+.fastq" -print
136 find . -name "\w+_B07_115_\d+.fastq" -print
137 find . -name "*fastq" -print
138 find . -name "\w+*fastq" -print
139 find . -name "\w*fastq" -print
[ Read 1000 lines ]
^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```



This screenshot shows the same Nano editor interface as the first, but with a dark background theme. The terminal window title and file name are identical. The command list (lines 121-139) is also the same. The status bar at the bottom shows the same set of keyboard shortcuts. The '[Read 1000 lines]' message is present. The visual difference is the dark color scheme for the editor's background and the status bar.

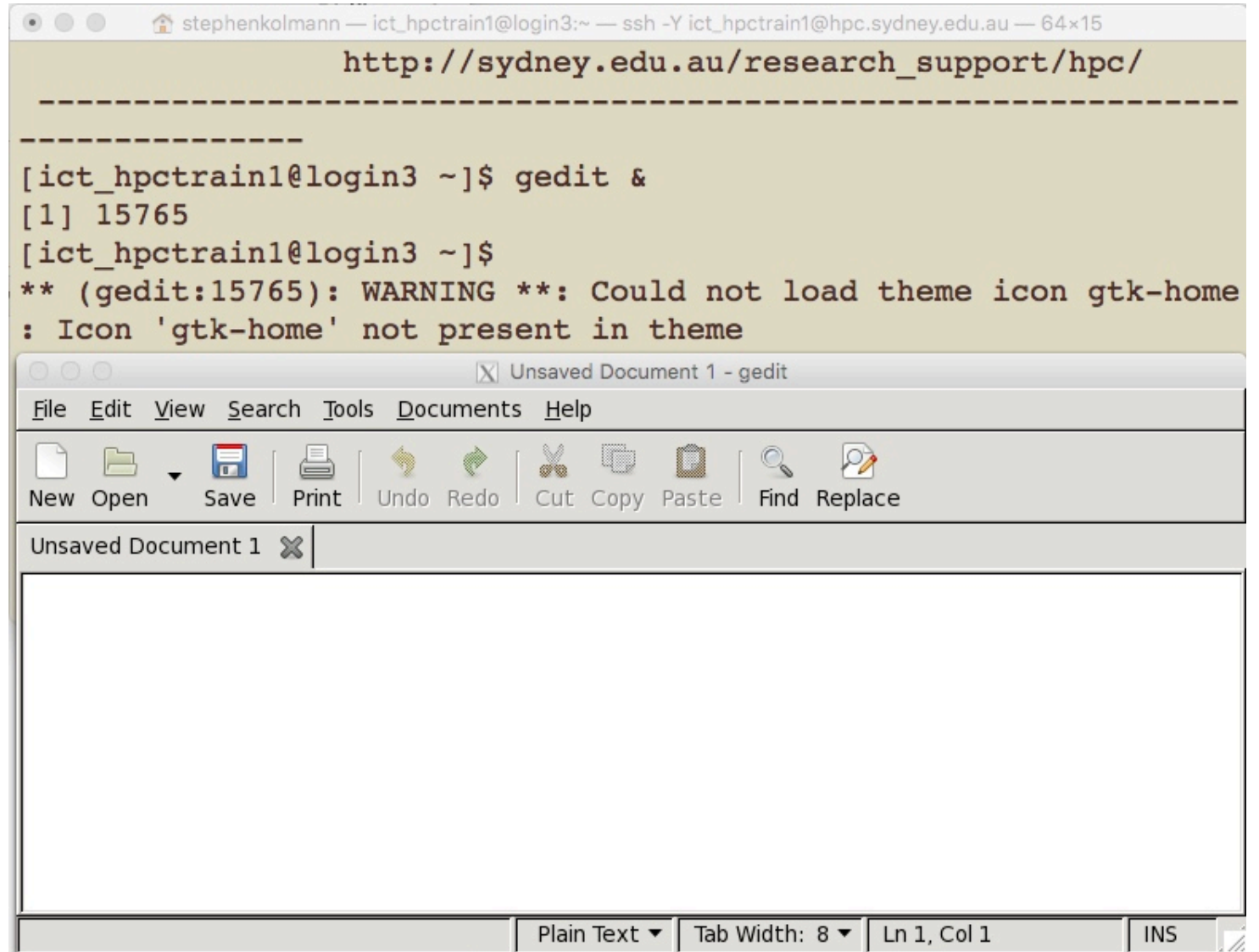
```
GNU nano 2.0.9 File: April132016_history.txt
121 nf
122 cat LIJA.o470582.1
123 \rm LIJA*
124 qsub array.sh
125 nf
126 cat LIJA.o470583.1
127 \rm LIJA.o470583.*
128 qsub array.sh
129 cat LIJA.o470584.1
130 \rm LIJA.o470584.*
131 qsub sh test2.sh
132 sh test2.sh
133 ls \w*
134 sh test2.sh
135 find . -name "\w+_B07_115_\d+.fastq" -print
136 find . -name "\w+_B07_115_\d+.fastq" -print
137 find . -name "*fastq" -print
138 find . -name "\w+*fastq" -print
139 find . -name "\w*fastq" -print
[ Read 1000 lines ]
^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

- Nano opens inside the terminal window
- Commands to save, close etc are typed
- Commands are shown in the editor at all times making it easy to remember them

Emacs (or GEdit) in Xwin

- Xwin can open GEdit in a GUI window – this means you can use your mouse to point-and-click commands from the menu if you prefer
- Including the ‘&’ (run command in the background) means your terminal remains active while GEdit is open, eg

`gedit &`



GEdit and nano: basic commands

Operation	Nano command	GEdit command
Open new empty file	<code>nano</code>	<code>gedit &</code>
Open new file + name it	<code>nano NewFile</code>	<code>gedit NewFile &</code>
Open existing file	<code>nano ExistingFile</code>	<code>gedit ExistingFile &</code>
Save	<code>control + o</code>	<code>control + s</code>
Close/exit	<code>control + x</code>	<code>control + q</code>

Note:

- Only use '&' if you have an X Server (eg Xwin, Xquartz) installed, or the editor will not open

Important note for Windows users

- Windows and Linux use different character encoding
- If you create a file on Windows and copy to Linux, this will very often end in a failure to read the file correctly (it depends on the software you use and the file you create)
- You might see an error like this when trying to run a script on HPC which is affected by this problem:

```
-bash: /var/spool/PBS/mom_priv/jobs/1397159.pbserver.SC:  
/bin/bash^M: bad interpreter: No such file or directory
```

- In this example, with an exit status of 126

Important note for Windows users

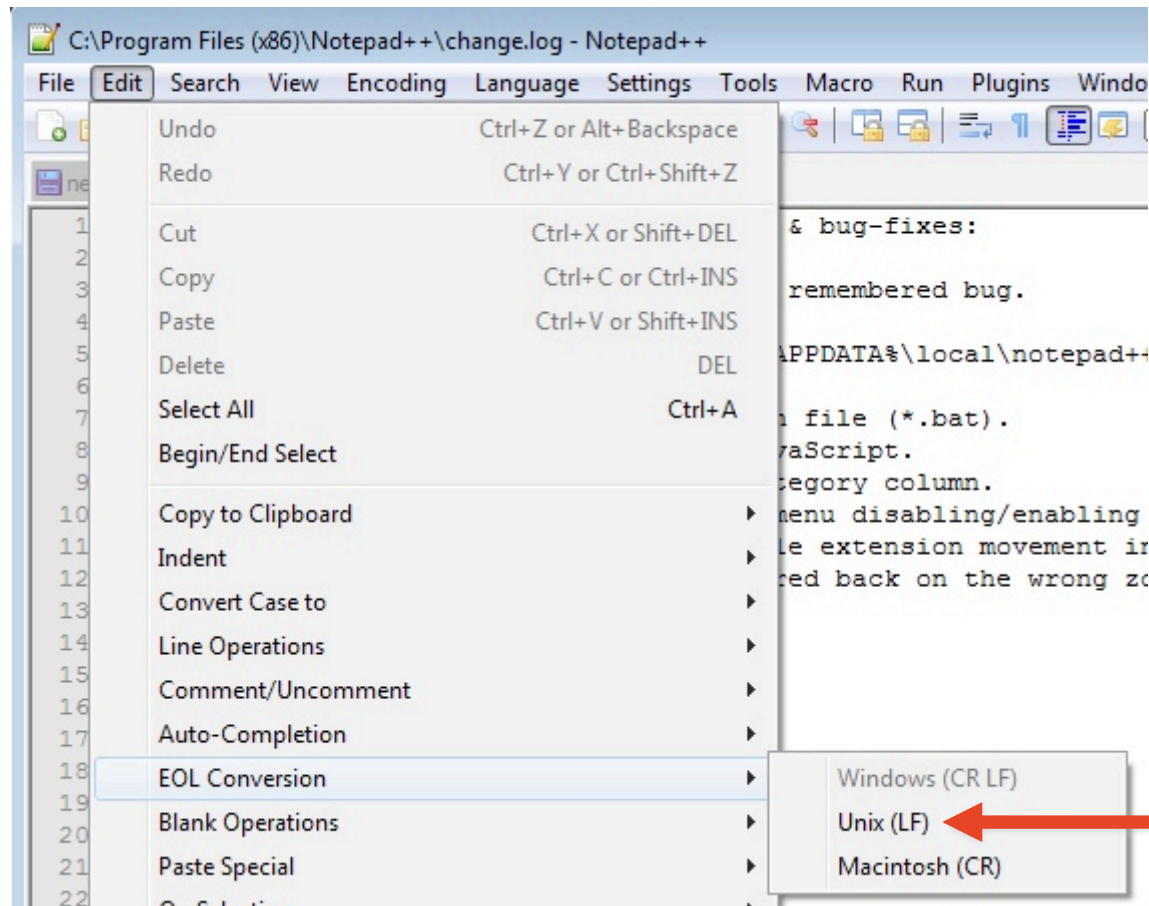
- In some cases (eg plain text files), this can be identified and corrected using 2 Linux commands:

file <filename>	- describes file type
dos2unix <filename>	- converts files to Unix/Linux format

```
[cwil2281@login2 ~]$ file from_excel.txt
from_excel.txt: ASCII text, with CRLF line terminators
[cwil2281@login2 ~]$ file from_HPC.txt
from_HPC.txt: ASCII text
[cwil2281@login2 ~]$ dos2unix from_excel.txt
dos2unix: converting file from_excel.txt to UNIX format ...
[cwil2281@login2 ~]$ file from_excel.txt
from_excel.txt: ASCII text
[cwil2281@login2 ~]$ █
```

Windows users

- Some text editors can save files with Artemis compatible line endings. eg Notepad++



Select Unix line endings
before saving your file

Important note for Windows users

In summary:

- Be aware of Windows vs Linux line endings
- If you encounter strange, seemingly inexplicable errors, investigate this as a source of the problem
- On HPC, use 'file' and 'dos2unix' to check and convert files – but remember that this does not solve the issue in all cases
- Try to perform creation and editing of scripts intended for HPC on the HPC directly

PBS scripts

➤ Open basic.pbs with your preferred text editor...

Nano:

```
nano basic.pbs
```

or

GEdit from Xwin terminal:

```
gedit basic.pbs &
```

❖ Three essential components of a PBS script:

1. **Directives:** instructions to PBS Pro, inc. resource requests and details about your job
2. **Module loads:** if your job needs some software, 'loading' the software adds it to your path for the current session
3. **Commands that make up the job:** can be anything you would enter into the terminal – Linux commands, calls to external software or scripts etc

```
index.pbs - emacs@login2
File Edit Options Buffers Tools Sh-Script Help

#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project
#PBS -N Index
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ

# Load modules
module load bwa
module load samtools

path=/project/<YourProject>/<YourName>

# BWA index:
bwa index -a bwtsw ${path}/canfam3_chr5.fasta

# SAMtools index:
samtools faidx ${path}/canfam3_chr5.fasta
```

1. Directives

2. Module loads

3. Commands to run the job

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project
#PBS -N Index
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultq
```

'#': a comment, **ignored** by the system.
Useful for descriptive notes to yourself

'#' then 'PBS' (no space): a **directive**, important instruction to the PBS scheduler

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project ← Short project name
#PBS -N Index
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ
```

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project ← Short project name
#PBS -N Index ← Job name (short)
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ
```

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project ← Short project name
#PBS -N Index ← Job name (short)
#PBS -l select=1:ncpus=1:mem=4GB ← Nodes, cores and RAM
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ
```

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project ← Short project name
#PBS -N Index ← Job name (short)
#PBS -l select=1:ncpus=1:mem=4GB ← Nodes, cores and RAM
#PBS -l walltime=00:10:00 ← Max run time hrs:mins:secs
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ
```

If you're unsure of run time: overestimate, because the system will kill your job if it goes over the requested time!

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project ← Short project name
#PBS -N Index ← Job name (short)
#PBS -l select=1:ncpus=1:mem=4GB ← Nodes, cores and RAM
#PBS -l walltime=00:10:00 ← Max run time hrs:mins:secs
#PBS -M your.name@sydney.edu.au
#PBS -m a ← Send yourself an email when the job
#PBS -q defaultQ ← aborts (a), begins (b) and ends (e). Optional.
```

If you're unsure of run time: overestimate, because the system will kill your job if it goes over the requested time!

1. Directives

```
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project ← Short project name
#PBS -N Index ← Job name (short)
#PBS -l select=1:ncpus=1:mem=4GB ← Nodes, cores and RAM
#PBS -l walltime=00:10:00 ← Max run time hrs:mins:secs
#PBS -M your.name@sydney.edu.au
#PBS -m a ← Send yourself an email when the job
#PBS -q defaultQ ← Queue to submit job to
               aborts (a), begins (b) and ends (e).
               Optional.
```

If you're unsure of run time: overestimate, because the system will kill your job if it goes over the requested time!

```
#PBS -l select=1:ncpus=1:mem=4GB
```

select = number of 'chunks' of compute resources your job requires

ncpus = number of CPUs/cores your job requires

mem = maximum amount of RAM your job requires

- The above example is a single-core job using a maximum of 4 GB RAM
- RAM is total RAM per chunk, so if you have a 4-core job that requires 4 GB per CPU, request 16 GB RAM
 - Max. RAM is enforced: your job will be killed by PBS Pro if it exceeds the requested amount
 - Min. RAM is 1 GB
- If your job can run on multiple CPUs, you can increase the value at 'ncpus'
- If your job can run across multiple nodes, you can increase the value at 'select'
- Only request 'select' and 'ncpus' > 1 if you are certain that your software or code can make use of the additional resources; over-requesting wastes resources and will likely cause your job to wait in queue longer than necessary. Be sure to include the right flags/syntax in your code to utilise extra cores/nodes
- For an MPI job, specifying a greater value at 'select' rather than at 'ncpus' is likely to result in shorter queue times, as the scheduler can more easily find many smaller chunks of available compute resources rather than fewer larger chunks, eg

select=1:ncpus=16:mem=64GB *versus* **select=16:ncpus=1:mem=4GB**

Queues on the Artemis HPC

- The nodes on Artemis are partitioned into different ‘queues’
 - Each queue is a collection of nodes dedicated to run different kinds of jobs and each has different maximum resources allowed on that particular queue
 - Some queues have a higher or lower ‘fair share’ penalty than others
 - If you leave off the ‘-q’ directive, your job will go to the default queue, which is called ‘defaultQ’
 - Depending on the resources you have requested, jobs submitted to ‘defaultQ’ might be placed on:
 - Small, Normal or Large
 - GPU
 - High memory
- #PBS -q defaultQ**
- High memory jobs are those that request > 20 GB RAM per core or > 123 GB RAM per chunk, up to a maximum of 6 TB RAM
 - GPU jobs are requested in a special way: **#PBS -l select=1:ncpus=1:ngpus=1**
 - There are 4 other queues you can request:
 - small-express, with **#PBS -q small-express**
 - scavenger, with **#PBS -q scavenger**
 - dtq, with **#PBS -q dtq**
 - interactive, this is via the command line for an interactive job (not submitted with a job script):
qsub -I -P <project> -l select=1:ncpus=1:mem=4GB,walltime=1:00:00

Artemis HPC queue resource limits

Queue	Max walltime	Max cores per job	Max cores per user	Memory per core	Memory per node	Fair share weighting	Request by
small	1 day	24	96	< 20 GB	<123 GB	1x	#PBS -q defaultQ
normal	7 days	96	96	< 20 GB	<123 GB	1x	#PBS -q defaultQ
large	21 days	288	288	< 20 GB	< 123 GB	1x	#PBS -q defaultQ
high mem	21 days	192	192	> 20 GB	123 GB – 6 TB	5x	#PBS -q defaultQ
gpu	7 days	120	120	N/A	< 123 GB	5x	#PBS -q defaultQ #PBS -l select=N:ncpus=N:ngpus=N
small- express	1 day	24	96	N/A	< 123 GB	5x	#PBS -q small-express
scavenger	2 days	288	288	N/A	< 123 GB	0x	#PBS -q scavenger
dtq	10 days	2	16	N/A	< 16 GB	0x	#PBS -q dtq
interactive	4 hours	4	4	N/A	< 123 GB	10x	Command-line interactive session, include '-I' flag

Artemis HPC Overall Resource Limits

- Maximum jobs queued per user: 200
- Maximum cores per user: 600
- If you need to submit more jobs at once, you can use *array jobs* (won't be covered today)
- Array jobs can have up to 1,000 sub-jobs.
- Therefore, if you take full advantage of array jobs, you could submit $200 \times 1,000 = 200,000$ jobs simultaneously!

What is the small-express queue?

- Urgent jobs for which you are prepared to pay a 5 times higher fair share penalty to have them leave the queue and start running more quickly
 - If you are a frequent user and always submit to small-express, your fair share penalty will accumulate rapidly and your priority in the queue will be decreased
 - This is to ensure that small-express is not abused
- ❖ We will submit our small sample jobs today to **small-express** to make sure they run quickly

What is the scavenger queue?

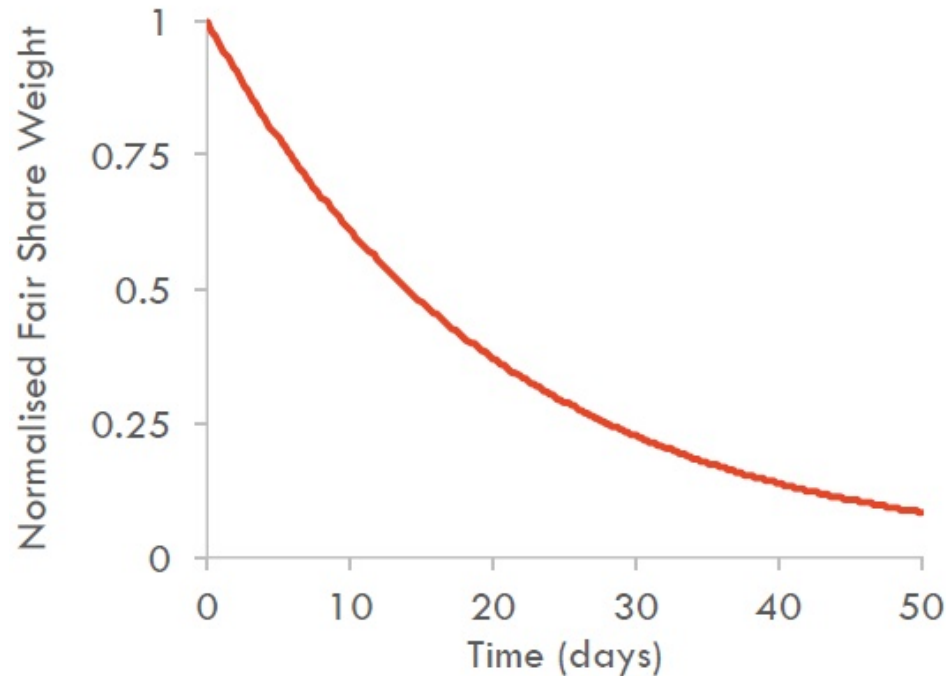
- Non-urgent jobs that you don't mind how quickly they complete
- These jobs have ZERO impact on your fair share!
- The down side is they are subject to termination if a non-scavenger job needs the resources your scavenger job is using
 - This means the job may be paused and resume later
 - If it is paused for too long it will be killed by PBS Pro and you will need to restart it

What is the data transfer queue (dtq)?

- A queue dedicated to running data transfer and file input/output jobs
- These jobs have ZERO impact on your fair share!
- The only queue on Artemis with direct access to RCOS
 - RCOS = Research Computing Optimised Storage, the NFS option of Research Data Store (RDS)
 - The workshop 'Data Transfer and RDS for HPC' will explain in more detail
- No compute jobs allowed
 - Data transfer or file management tasks only
 - Pure compute jobs will be terminated
 - Jobs can use CPU power to do file compression/encryption/data transfer

Fair share

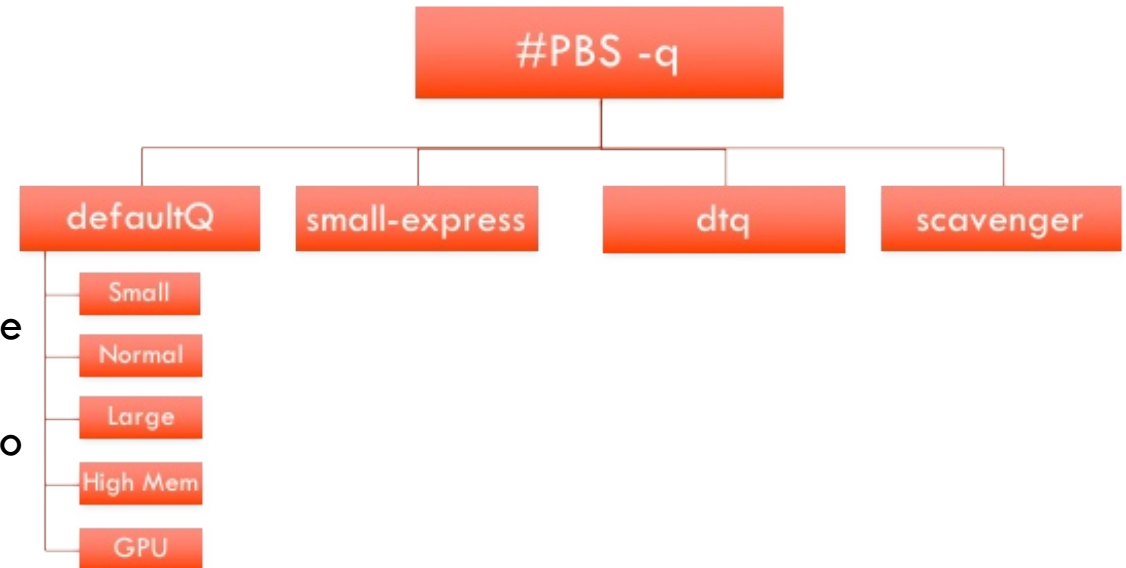
- Calculated at the level of 'project' (Faculty is no longer involved)
 - All users in the same project contribute to the same fair share weighting
- Your fair share weighting is increased as you use more computation time
 - Higher fair share weight means your priority in the Artemis queue decreases
- Half-life of your fair share weight is 2 weeks



Queues summary

- The complex queue structure is there for you to take advantage of IF YOU CHOOSE TO
- For a novice user or those who don't wish to understand the queues in depth,
just submit to defaultQ 😊

- For **urgent** jobs (that fit within the max parameters) you can submit to small-express and cop the **increased fair share** penalty
- For **non-urgent** jobs and users who care about fair share weight, you can submit to scavenger
- **High mem** jobs are automatically recognised by PBS Pro if they ask for > 123 GB RAM
- **GPU** jobs are automatically recognised by PBS Pro if they request 'ngpus'



```
index.pbs - emacs@login2
File Edit Options Buffers Tools Sh-Script Help

#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project
#PBS -N Index
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ

# Load modules
module load bwa
module load samtools

path=/project/<YourProject>/<YourName>

# BWA index:
bwa index -a bwtsw ${path}/canfam3_chr5.fasta

# SAMtools index:
samtools faidx ${path}/canfam3_chr5.fasta
```

1. Directives

2. Module loads

3. Commands to run the job

Within each session (connection to Artemis) you must **load** the **modules** you want to use. In other words – tell the system which programs you need to run your job. They will be added to your path for the current session.

2. Module loads

```
# Load modules  
module load bwa  
module load samtools
```

← Just a comment

← 2 modules (programs/software) loaded

- To see what modules are available on the HPC, run the command:

```
module avail
```

- To see what versions of a program are available (for example, Matlab):

```
module avail |& grep matlab
```

```
index.pbs - emacs@login2
File Edit Options Buffers Tools Sh-Script Help

#!/bin/bash
# Create indexes for reference sequence
#PBS -P Project
#PBS -N Index
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -M your.name@sydney.edu.au
#PBS -m abe
#PBS -q defaultQ

# Load modules
module load bwa
module load samtools

path=/project/<YourProject>/<YourName>

# BWA index:
bwa index -a bwtsw ${path}/canfam3_chr5.fasta

# SAMtools index:
samtools faidx ${path}/canfam3_chr5.fasta
```

1. Directives

2. Module loads

3. Commands to run the job

Any command can be entered in a PBS script. Which commands will depend on what kind of job you are running.

How do you know what commands to include?

1. Choose the program/s you want to run
2. Read the manual/user guide
3. Work out the commands and what parameters/arguments are required
4. Enter them into the PBS script

3. Commands to run the job

Any command can be entered in a PBS script. Which commands will depend on what kind of job you are running.

3. Commands to run the job

How do you know what commands to include?

1. Choose the program/s you want to run
2. Read the manual/user guide
3. Work out the commands and what parameters/arguments are required
4. Enter them into the PBS script

General minimum structure of a software call:
Program name, +/- functions, +/- arguments, input data

```
# BWA index:
```

```
bwa index -a bwtsw /PATH/canfam3_chr5.fasta
```

An argument (-a) and its value (bwtsw)

The program

A function of the program

The input file

Preparing a simple job

- Now that you understand what the aspects of a PBS script are and the Artemis queues that run your jobs, we will modify basic.pbs into a very simple job
 - Copying PBS directives from one job script to another is common practice, and saves you having to re-type everything
 - Always be sure to edit the directives to suit the job at hand
- In today's course you will submit 3 practice jobs
- For each job, you will need to:
 1. Open the .pbs script
 2. Make some changes to the script in a text editor
 3. Save the changes
 4. Submit the job
 5. Check the status of the job
 6. Check the output of the job

Preparing a simple job

- Now to edit your basic.pbs script:
 1. Adjust your project name at '-P' to reflect your Project
 - Training login users – remember your project is 'Training'
 2. Change the job name at '-N' to your first name
 3. The script currently requests one core – this is fine, our basic job won't multithread so leave ncpus=1
 4. The script currently requests 10 minutes wall time – this is too much, reduce it to 2 minutes
 5. What about RAM? We won't be doing heavy compute so reduce it to the minimum of 1 GB
 6. Insert your email address at -M
 7. Change the queue at '-q' to the small-express queue (if it isn't already small-express)
 8. Now delete everything below the directives, including module loads, comments and commands. We will replace this content with our own commands.

Preparing a simple job

9. Add this line to your script, replacing the project and <YourName> as appropriate:

```
cd /project/Training/<YourName>
```

PBS considers your 'working directory' to be your home directory, even if you are in project when you submit the job

Thus it is VIP to specify location, either by using 'cd' or full pathnames for input and output files

Preparing a simple job

9. Add this line to your script, replacing the project and <YourName> as appropriate:

```
cd /project/Training/<YourName>
```

PBS considers your 'working directory' to be your home directory, even if you are in project when you submit the job
Thus it is VIP to specify location, either by using 'cd' or full pathnames for input and output files

10. Now add some commands to the script: remember that any command you would normally put on the command line can be embedded within a script

```
mkdir New_job
```

```
cp basic.pbs New_job/copy.pbs
```

```
perl hello.pl <YourName>
```

Preparing a simple job

9. Add this line to your script, replacing the project and <YourName> as appropriate:

```
cd /project/Training/<YourName>
```

PBS considers your 'working directory' to be your home directory, even if you are in project when you submit the job
Thus it is VIP to specify location, either by using 'cd' or full pathnames for input and output files

10. Now add some commands to the script: remember that any command you would normally put on the command line can be embedded within a script

```
mkdir New_job  
cp basic.pbs New_job/copy.pbs
```

← Regular Linux commands can be included, just as they would on the command line

```
perl hello.pl <YourName>
```

← Calls to external scripts can be included

Preparing a simple job

11. Save your script , eg

Nano:

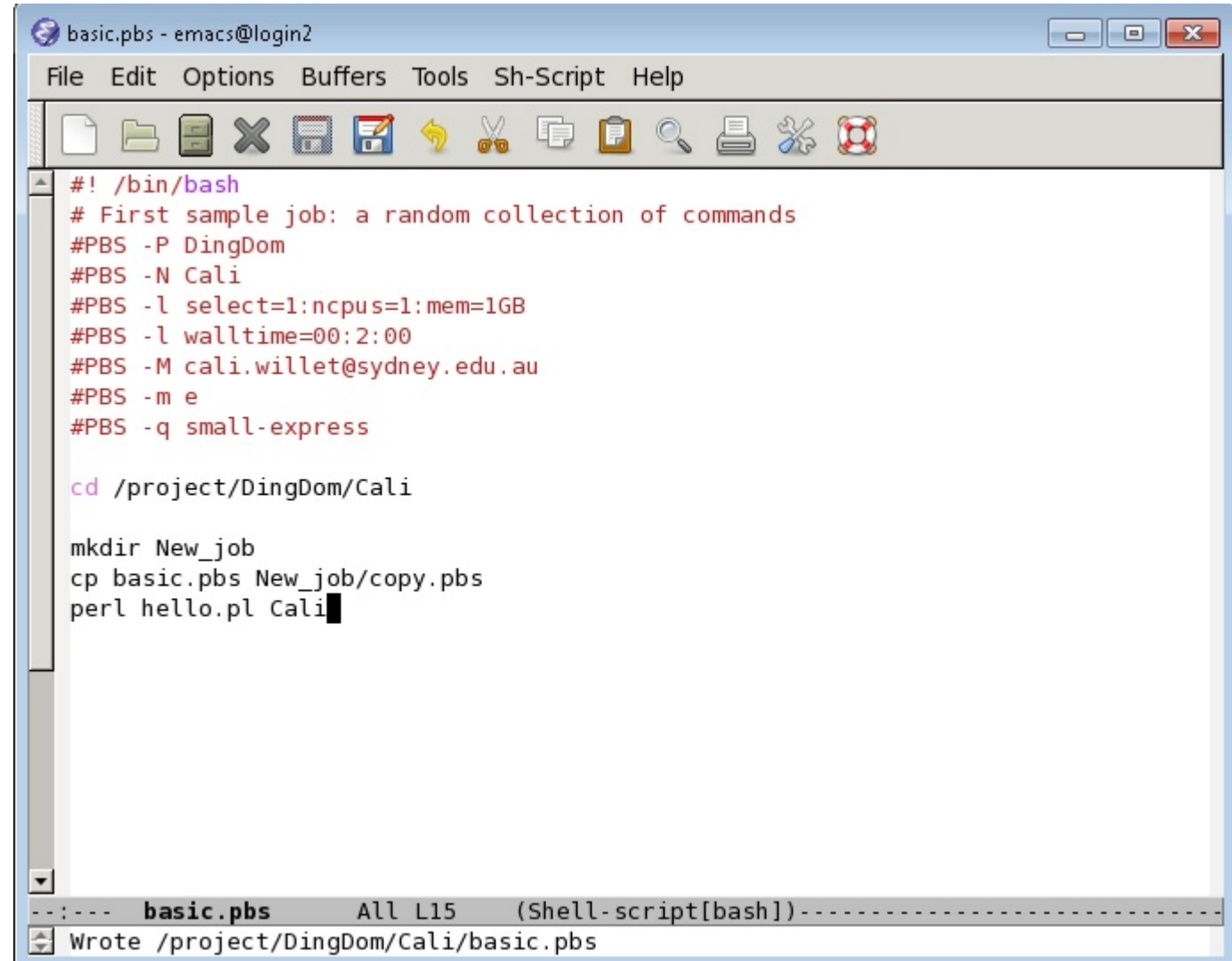
ctrl + o

Emacs (keystrokes):

ctrl + x, ctrl + s

Emacs (GUI menu):

File → Save



```
basic.pbs - emacs@login2
File Edit Options Buffers Tools Sh-Script Help

#!/bin/bash
# First sample job: a random collection of commands
#PBS -P DingDom
#PBS -N Cali
#PBS -l select=1:ncpus=1:mem=1GB
#PBS -l walltime=00:2:00
#PBS -M cali.willet@sydney.edu.au
#PBS -m e
#PBS -q small-express

cd /project/DingDom/Cali

mkdir New_job
cp basic.pbs New_job/copy.pbs
perl hello.pl Cali
```

--:-- basic.pbs All L15 (Shell-script[bash])-----
Wrote /project/DingDom/Cali/basic.pbs

Submitting & monitoring jobs

Checking job output & resource usage



Submitting a job

- Essential PBS commands:

`qsub job.pbs` Submit the job

`qstat -x jobID` Check the status of a submitted job ('x' flag allows historical query)

`qstat -xf jobID` As above, showing detailed job information

`qstat -u <unikey>` Query all queued and running jobs for user

`qdel jobID` Delete a submitted job

`qstat -T | grep <unikey>` Show estimated start time of queued jobs

Submitting a job

- Essential PBS commands:

`qsub job.pbs` Submit the job

- Submit your job with the 'qsub' command:

`qsub basic.pbs`

- Check it's status with the 'qstat' command:

`qstat -x <jobID>`

- ❖ Note the ID that was printed to screen:
*this is your **job ID** and can be used to
query your job's status*

- You can specify *jobID* with or without
the '.pbsserver' suffix

```
[cwil2281@login2 Cali]$ qsub basic.pbs
1107528.pbsserver
[cwil2281@login2 Cali]$ qstat -x 1107528
```

Job id	Name	User	Time Use	S	Queue
1107528.pbsserver	Cali	cwil2281	00:00:00	F	small-express

```
[cwil2281@login2 Cali]$
```

Checking on a job's output

- Great, it's finished!
 1. Did it do what you expected?
Need to know what you expected
 2. Did it successfully complete?
Can check your email report
 3. Did it run without errors?
Check the job logs and output files



www.dog-names-and-more.com

Checking on a job's output

1. Did it do what you expected?
 - Consider what each command in the job was supposed to do:

```
mkdir New_job
```

```
cp basic.pbs New_job/copy.pbs
```

We should see a new directory called 'New_job', containing a file called copy.pbs

➤ Check this with 'ls'

```
[cwil2281@login2 Cali]$ ls
align.pbs  Cali.e1107528  Cali.o1107528_usage  index.pbs
basic.pbs  Cali.o1107528  hello.pl             New_job
[cwil2281@login2 Cali]$ ls New_job/
copy.pbs
[cwil2281@login2 Cali]$
```

Checking on a job's output

1. Did it do what you expected?

```
perl hello.pl <YourName>
```

This command ran a perl script which prints your name to 'standard output':

- If we ran it on the command line, we would see the message in the terminal window
- Since it was run from within a PBS script, the message will be in the **standard output log file created by PBS**
- By default, PBS will create **three log files**:
 - one for standard output, named JobName.oJobID
 - one resource usage summary, named JobName.oJobID_usage
 - one for standard error, named JobName.eJobID

➤ Check if the perl script ran correctly with the '**cat**' command

```
-rw-r--r-- 1 cwil2281 RDS-FVS-DingDom-RW 478 Nov 30 15:04 Cali.o1107528_usage
-rw----- 1 cwil2281 RDS-FVS-DingDom-RW  31 Nov 30 15:04 Cali.o1107528
-rw----- 1 cwil2281 RDS-FVS-DingDom-RW  54 Nov 30 15:04 Cali.e1107528
[cwil2281@login2 Cali]$ cat Cali.o1107528
Hello, world! My name is Cali.
[cwil2281@login2 Cali]$
```

Checking on a job's output

2. Did it successfully complete?

- Since we included the `-M` and `-m` directives, we can check our email to view a report on the completed job
- Exit status of 0 is good
 - Anything else is bad
- You can also see exit status in the **usage log** or by running **`qstat -xf JobID`**



Checking on a job's output

3. Did it run without errors?

- Check your '.e' log file. For this job, it should be empty
- For some software, an error-free job will yield an empty error log
- For others, there may be content in the error log that is not necessarily indicative of an error
- So if you see an empty error log: great! If not, check what it contains
- Tips:
 - Run '`wc -l`' first to get a line count of the error log
 - If it's short, '`cat`' is an easy way to view it's contents
 - If it's long, '`more`' is preferable
 - Running '`tail`' on the log to view the last 10 lines can often quickly reveal a fatal error
 - Use '`grep -i error`', or '`grep -i warning`' etc to search the log for potential error messages

Checking on a job's output

3. Did it run without errors?

- Example: A job is killed by the system because it ran out of walltime



root <adm@mgmt1.hpc.syr>
PBS JOB 425901.mgmt1

To: Cali Willet

PBS Job Id: 425901.mgmt1
Job Name: Index_Cali
Execution terminated
Exit_status=271
resources_used.cput=0
resources_used.cput=00:00:10
resources_used.mem=77000kb
resources_used.ncpus=1
resources_used.vmem=311764kb
resources_used.walltime=00:00:42



```
[cwil2281@login1 Cali]$ tail Index_Cali.o425901
[bwa_index] Pack FASTA... 0.72 sec
[bwa_index] Construct BWT for the packed sequence...
[BWTIncCreate] textLength=80000000, availableWord=17629100
[BWTIncConstructFromPacked] 10 iterations done, 29079360 characters processed.
=>> PBS: job killed: walltime 10 exceeded limit 2
[cwil2281@login1 Cali]$
```



Submitting jobs without a PBS script

- Very simple jobs can be run without writing a PBS script
- For example, `hello.pl` could be submitted from the command line:

➤ `qsub -P Training -- /usr/local/perl/5.24.0/bin/perl \`
`/project/Training/<Your Name>/hello.pl <Your Name>`

1. Did it do what you expected?

The output will be written to `STDIN.o<jobid>`

2. Did it successfully complete?

`qstat -xf <jobid> | grep 'Exit_status'`

0 if successful, > 0 if there was an error

3. Did it run without errors?

`cat STDIN.e<jobid>`

Note: Only full path names work when using this method

Preparing a job using modules

- We will now look at a slightly more complex job that calls on other software to process input data and write output data
- Open index.pbs
 - This creates indexes of the canfam3_chr5.fasta DNA reference sequence
 - The indexes are required to perform the next job, which maps DNA sequence fragments back to the reference

Creating an analysis pipeline

- ❖ When building pipelines (series of tasks required for a complete analysis) for your own data analysis, you can choose to split out key stages into separate jobs, or include them all within the one job
- ❖ You should consider:
 - Which parts of a workflow need to be run once only, and which must be re-run for every new set of inputs?
 - Where would you like to perform 'checkpointing' before continuing the analysis?
 - This could mean checking results to decide if further steps are needed, deciding on parameters for the next stage of the pipeline, splitting your job into smaller stages to ensure they can run within the maximum walltime for the queue you want to run on, etc etc
 - Long running jobs that hold information in RAM for extended periods of time are recommended to periodically write the data to disk. This ensures that if the job fails (due to error, hardware malfunction etc) that all of the compute is not lost, and the job can be resumed by reading back in from disk instead of starting all over again
 - Are there stages of a workflow that use a lot more or less resources than other stages?
 - It is best to group portions of a pipeline into sections based on resource use. Eg if you had a 5-step analysis where only one step multi-threaded on 24 cores and the rest was single-core, for the duration of the 4 single-core steps you have 23 cores sitting idle on the machine (23 cores that you have queued longer for...)

Preparing a job using modules

➤ Now to edit your script:

1. Adjust your project name at '-P' to reflect your Project
- Training login users – remember your project is 'Training'
2. Change the job name at '-N' to 'Index_<yourName>'
3. The script currently requests one core – this is fine, our basic job won't multithread so leave `ncpus=1`
4. The script currently requests 10 minutes wall time – this is too much, reduce it to 5 minutes
5. The file to be indexed is small so reduce RAM to the minimum (1 GB)
6. Insert your email address at `-M`
7. Request your job be placed in the small-express queue (we want it to run right now)

Preparing a job using modules

7. Edit the value of the **variable** 'io' – you need to update 'Project' and 'YourName'

- A **variable** is a key concept in bash (Linux) and other programming languages. A variable simply stores some value
- We won't explore programming today, but variables are very handy for making your PBS scripts 'portable' (easy to reuse) from one experiment/run to the next

`io=/project/<Project>/<YourName>` → a variable containing the directory pathname for your input files

- By using `${io}` for input and output path names, we only have to type our directory path once!

➤ Edit the value of 'io' here

→ `io=/scratch/DingDom/Cali`

It's value will be read by the script here and here

```
# BWA index:
bwa index -a bwtsw ${io}/canfam3_chr5.fasta

# SAMtools index:
samtools faidx ${io}/canfam3_chr5.fasta
```

Submitting the job and checking status

- Save the script
- Submit the job:
`qsub index.pbs`
- Periodically check it's status using the 'qstat' command:
`qstat -x jobID`

Remember: jobID is a unique number assigned to your job by the system.

If you forget the jobID? Run this command:

```
qstat -u <unikey>
```

```
[cwil2281@login1 Cali]$ qstat -x 1107583
Job id          Name          User          Time Use S Queue
-----
1107583.pbserver Cali_Index    cwil2281      00:00:00 R small-express
[cwil2281@login1 Cali]$ qstat -x 1107583
Job id          Name          User          Time Use S Queue
-----
1107583.pbserver Cali_Index    cwil2281      00:00:00 R small-express
[cwil2281@login1 Cali]$ qstat -x 1107583
Job id          Name          User          Time Use S Queue
-----
1107583.pbserver Cali_Index    cwil2281      00:00:22 F small-express
[cwil2281@login1 Cali]$
```

R = running

F = finished

Check the completed job

- Before continuing with the next stage of the data analysis, you should always check the job ran successfully
 1. Did it do what you expected?
Need to know what you expected → we should see some new output files
 2. Did it successfully complete?
Check your email report or usage log → make sure your exit status is zero
 3. Did it run without errors?
Check the job log files → check the error log for errors and that the output is correct

Check the completed job

1. Did it do what you expected?

➤ Run the '**ls -lh**' command on your scratch directory

- Output should be 6 index files – these have the same prefix as the input sequence file, but with an additional suffix
- Don't view these, most are not human readable. We know they are OK if they exist, are not empty and the exit status and error logs reveal no errors

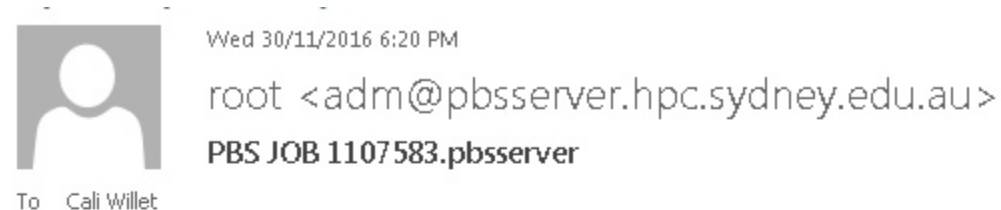
```
-rw-r--r-- 1 cwil2281 RDS-FVS-BioServ-RW 5.9K Sep 25 10:43 canfam3_chr5.fasta.amb  
-rw-r--r-- 1 cwil2281 RDS-FVS-BioServ-RW  40 Sep 25 10:43 canfam3_chr5.fasta.ann  
-rw-r--r-- 1 cwil2281 RDS-FVS-BioServ-RW 39M Sep 25 10:43 canfam3_chr5.fasta.bwt  
-rw-r--r-- 1 cwil2281 RDS-FVS-BioServ-RW  19 Sep 25 10:43 canfam3_chr5.fasta.fai  
-rw-r--r-- 1 cwil2281 RDS-FVS-BioServ-RW 9.6M Sep 25 10:43 canfam3_chr5.fasta.pac  
-rw-r--r-- 1 cwil2281 RDS-FVS-BioServ-RW 20M Sep 25 10:43 canfam3_chr5.fasta.sa
```

6 index files

Check the completed job

2. Did it successfully complete?

- Check your email summary or usage log for an exit status of zero



PBS Job Id: 1107583.pbsserver
Job Name: Cali_Index
Execution terminated
Exit_status=0
resources_used.cput=00:00:22
resources_used.mem=175760kb
resources_used.ncpus=1
resources_used.vmem=175760kb
resources_used.walltime=00:00:26



The information in the job summary about what 'mem' and 'walltime' resources were used can help you decide on appropriate resource requests for future similar jobs.

You can also get this information from the usage log.

The completion email is handy to give you a heads up when your job has completed, especially for queued or long running jobs.

Check the completed job

3. Did it run without errors?

➤ Check your standard error log file

- Is it empty? If not, investigate... (it shouldn't be empty for this job as BWA program writes things to standard error)
- If it's not too long, view contents with 'cat'
- You could search for error messages using 'grep', a pattern matching search command

➤ Enter this command (replace your name and job ID)

grep -i error Index_<YourName>.eJobID

This performs a case-insensitive (the **i** flag) for the word 'error' in the log file

```
[cwil2281@login1 Cali]$ grep -i error Index_Cali.e425902  
[cwil2281@login1 Cali]$
```

No errors found 😊

Check the completed job: usage logs

4. Check actual usage versus requested resources
 - Artemis produces a usage log file for all completed jobs
 - Similar to the email report, but with additional information making it easier to determine whether your resource requests were appropriate
 - Why is this important?
 - For you: over-requesting can result in longer queue time – use information in your usage log to adjust directives for your next similar job
 - For others: over-requesting leads to inaccurate estimations of queued job start times
 - Usage log naming convention:
JobName.oJobID_usage

Check the completed job: usage logs

4. Check actual usage versus requested resources

```
[cwil2281@login1 Cali]$ cat Cali_Index.o1107583_usage
Job Id: 1107583.pbserver for user cwil2281 in queue small-express
Job Name: Cali_Index
Project: RDS-FVS-DingDom-RW
Exit Status: 0
Walltime requested: 00:05:00 :      Walltime used: 00:00:26
  Cpus requested: 1 :
    Cpu Time: 00:00:16 :      Cpu percent: 101
    Mem requested: 1gb :      Mem used: 156696kb
    VMem requested: None :      VMem used: 156696kb
    PMem requested: None :      PMem used: None
[cwil2281@login1 Cali]$
```

Way too much wall time! If you were to rerun this job, you would use this information to make more appropriate requests next time

Check the completed job: usage logs

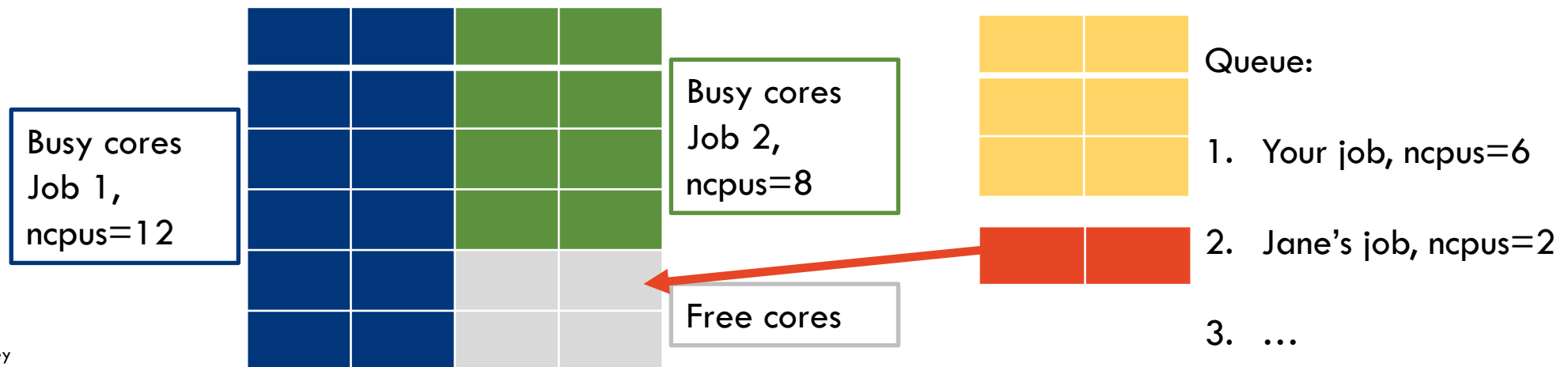
4. Check actual usage versus requested resources

```
[cwil2281@login1 Cali]$ cat Cali_Index.o1107583_usage
Job Id: 1107583.pbserver for user cwil2281 in queue small-express
Job Name: Cali_Index
Project: RDS-FVS-DingDom-RW
Exit Status: 0
Walltime requested: 00:05:00 :      Walltime used: 00:00:26
  Cpus requested: 1 :
    Cpu Time: 00:00:16 :      Cpu percent: 101
  Mem requested: 1gb :      Mem used: 156696kb
  VMem requested: None :      VMem used: 156696kb
  PMem requested: None :      PMem used: None
[cwil2281@login1 Cali]$
```

153 MB used – we can't reduce this job to less than 1 GB, but for real jobs you should check 'mem used' to make sure your request is appropriate

Queued jobs

- Hopefully our jobs won't queue today since they are tiny and running on small-express
- Why do jobs with fewer cores/RAM generally leave the queue sooner than larger jobs?
 - This is **not** a sign of the system favouring small jobs over large – we have dedicated nodes for running large jobs and there is no difference in priority of a large versus small jobs (except for small-express which is weighted high)
 - It is a product of the fact that on a large cluster running many different types of jobs, it is easier for the scheduler to find small free spaces than it is to find large free spaces
 - An over-simplified example: a 24-core node is running 1 x 12-core job and 1 x 8-core job. This leaves 4 free cores. You have submitted a 6-core job and are waiting in queue. Later, Jane submits a 2-core job which runs before yours, because it can fit on the 4 free cores while yours cannot
 - This **does not mean** it is always better to submit single-core jobs compared to multi-core jobs



Queued jobs

❖ Why are jobs queued?

- The system does not have the amount of resources you requested spare right now
- Order in the queue is affected by your fair share weighting

❖ When will your job come off the queue and start running?

- As soon as the resources you need are available, considering your place in the queue

❖ How to avoid/minimise queue time?

- Don't over-request resources
 - Processors: only request what you need
 - For multi-threading or MPI jobs: use trial and error to find a good balance between speed (number of CPUs requested) and queue time
 - Yes, this is indeed complicated by cluster load at the time you submit
 - Walltime and RAM: try and make reasonably educated requests (plus a little bit spare just in case, esp. time) – your usage logs are useful here

Queued jobs

- ❖ How to tell if a job has been queued?

`qstat -x JobID`

```
[cwil2281@login2 Training_content]$ qstat -x 240757.mgmt1
Job id          Name          User          Time Use S Queue
-----
240757.mgmt1    Index          cwil2281          0 Q compute
[cwil2281@login2 Training_content]$
```

Q = queued

Queued jobs

❖ Can I get an estimate of start time?

- The system can **estimate** a start time **assuming** all currently running and queued jobs run to completion, using **all** the resources that they have requested
- This estimate is a **worst-case scenario**: your job will usually always start well ahead of that time because:
 - Some jobs will be terminated or removed from the queue by the user
 - Some jobs will fail early due to an error
 - Most jobs will run for less walltime than requested

`qstat -T | grep yourUnikey`

```
[cwil2281@login2 Training_content]$ qstat -T | grep cwil2281
240776.mgmt1    cwil2281 compute Index          --    1 12    -- 02:00 Q Sa 18
[cwil2281@login2 Training_content]$
```

Estimated start time
Sat at 6pm

Delete a job

- You may want to delete a job if:
 - It has been queued and you would like to reconfigure your requests to try and leave the queue earlier
 - You realise you have made a mistake in the script
 - You simply no longer want to run it
- You can stop a job that is running or in the queue with this command:

```
qdel jobID
```

Preparing another job using modules

- We will now edit and submit align.pbs
 - This job uses the index files created by index.pbs to align the 'fastq' DNA sequence files against the 'fasta' reference sequence file
 - It is essential that the index job ran correctly for this job to run (so check now if you haven't already)
- Open align.pbs

Preparing another job using modules

➤ Edit the script:

1. Adjust your project name at '-P' to reflect your Project
- Training login users – remember your project is 'Training'
2. Change the job name at '-N' to 'Align_yourName'
3. The script currently requests 2 processors (ncpus=2) – the software 'BWA' can multithread, so we need to **make sure that we tell BWA to use the 2 processors we've requested**
4. The script currently requests 20 minutes wall time – probably too much, reduce to 10 mins
5. For a ~3 GB mammalian genome, BWA uses just over 3 GB RAM per core, but this is only 1 chromosome so reduce it to 1 GB per core
6. Insert your email address at -M
7. Edit the value of the variable 'io' – you need to update Project and YourName
8. Note that the final output files are copied to your project directory into a new directory called 'Output' inside the directory you're working in
9. Save the script

```
io=/project/Training/test
mkdir ${io}/Output
```

Handy use of a variable

```
#Load modules:
module load bwa
module load samtools
```

```
#Align reads to reference with BWA:
```

```
bwa mem -M -t 2 -R '@RG\tID:134\tPL:illumina\tPU:CONNEACXX\tSM:MS_134' \
    ${io}/canfam3_chr5.fasta ${io}/134_R1.fastq.gz \
    ${io}/134_R2.fastq.gz | samtools view -bSho ${io}/Output/134.bam -
```

```
#Sort by genomic coordinate with SAMtools:
```

```
samtools sort ${io}/Output/134.bam ${io}/Output/134_sorted
```

```
#Index alignment with SAMtools:
```

```
samtools index ${io}/Output/134_sorted.bam
```

```
io=/project/Training/test
mkdir ${io}/Output
```

Handy use of a variable

```
#Load modules:
module load bwa
module load samtools
```

The backslash means that the command is continued on a new line - tells the shell to ignore the newline character

```
#Align reads to reference with BWA:
```

```
bwa mem -M -t 2 -R '@RG\tID:134\tPL:illumina\tPU:CONNEACXX\tSM:MS_134' \
    ${io}/canfam3_chr5.fasta ${io}/134_R1.fastq.gz \
    ${io}/134_R2.fastq.gz | samtools view -bSho ${io}/Output/134.bam -
```

```
#Sort by genomic coordinate with SAMtools:
```

```
samtools sort ${io}/Output/134.bam ${io}/Output/134_sorted
```

```
#Index alignment with SAMtools:
```

```
samtools index ${io}/Output/134_sorted.bam
```

```
io=/project/Training/test
mkdir ${io}/Output
```

Handy use of a variable

```
#Load modules:
module load bwa
module load samtools
```

```
#Align reads to reference with BWA:
```

```
bwa mem -M -t 2 -R '@RG\tID:134\tPL:illumina\tPU:CONNEACXX\tSM:MS_134' \
    ${io}/canfam3_chr5.fasta ${io}/134_R1.fastq.gz \
    ${io}/134_R2.fastq.gz | samtools view -bSho ${io}/Output/134.bam -
```

```
#Sort by genomic coordinate with SAMtools:
```

```
samtools sort ${io}/Output/134.bam ${io}/Output/134_sorted
```

```
#Index alignment with SAMtools:
```

```
samtools index ${io}/Output/134_sorted.bam
```

The backslash means that the command is continued on a new line - tells the shell to ignore the newline character

Be aware of how much disk space you're using. /project/Training has a 1 TB limit. Check how much disk space your files are using with the `ls -lh` command.

Preparing another job using modules

- Have a look at the three commands that comprise this job
 - Looking at 'Align reads to reference' command, can you see where we have directed BWA to use 2 processors?
 - The 't' flag specifies 'threads' (processors/cores) to BWA. How multithreading is specified is specific to each software, so make sure you check the software documentation
 - **Always make sure the number of cores your directives request are utilised by the software** – BWA does not know to run on 2 cores just because the PBS directive has requested them

```
# Align reads to reference with BWA:
```

```
bwa mem -M -t 2 -R '@RG\tID:134\tPL:illumina\tPU:CONNEACXX\tSM:MS_134' \  
  ${io}/canfam3_chr5.fasta ${io}/134_R1.fastq.gz \  
  ${io}/134_R2.fastq.gz | samtools view -bSho ${io}/134.bam -
```

Preparing another job using modules

- Have a look at the three commands that comprise this job
 - Also note the directory paths pointing to the file inputs and outputs
 - By using full pathnames (instead of using `cd` as in the basic.pbs script) we can explicitly use different locations on the system
 - **Triple check that you have no typos in your directory paths** – if you try and write to a location that doesn't exist, your job will terminate!
- Example error when file/pathnames are incorrect:

```
[cwil2281@login1 Cali]$ tail Align.e425940
[M::mem_pestat] low and high boundaries for pro
[M::mem_pestat] skip orientation FF
[M::mem_pestat] skip orientation RR
[M::mem_process_seqs] Processed 198020 reads in 19.303 CPU sec, 9.231 real sec
[samopen] SAM header is present: 1 sequences.
[main_samview] fail to open "}/Output/134.bam" for writing.
open: No such file or directory
[bam_sort_core] fail to open file /project/RDS-FVS-BioServ-RW/Cali/Output/134.bam
open: No such file or directory
[bam_index_build2] fail to open the BAM file.
[cwil2281@login1 Cali]$
```

Align reads to reference with BWA:

```
bwa mem -M -t 2 -R '@RG\tID:134\tPL:illumina\tPU:CONNEACXX\tSM:'
stq.gz | samtools view -bSho $pPath}/Output/134.bam -
```

Some quick revision!

- Take 15 minutes to work through the revision activity
 - Write down your answers as you work through the tasks
 - We will then go through the answers together



Revision activity

1. Submit align.pbs (also, submit the job now)

2. Check the status of the job

3. How might you know when the job is finished?

4. How much space are you allocated in /project, and how do you check how much you have used?

5. What log files are produced after a job completes, and what do they contain?

6. What do the following directives indicate to the scheduler:

a) **-P** _____

b) **-N** _____

c) **-q** _____

7. Write the directive requesting 10.5 hours wall time

8. Write the directive to specify a 6-core job using 2 GB RAM per core

9. Your job uses the software 'beast'. What needs to come before your command to run beast?

10. Where should all important input and output data be stored long-term?

Align.pbs - check for errors

- First, use 'qstat' to check that your job has finished
 - `qstat -x <jobID>` (the "F" in the "S" column means it finished)

1. Did it do what you expected?

Within your Output directory, you should see an alignment file (bam), a sorted bam and it's index

2. Did it successfully complete?

Check for exit status of zero

3. Did it run without errors?

Check the error log

Note that the error log is not empty – BWA sends it's output to the error log

```
[cwil2281@login1 Cali]$ qstat -x 1107606
Job id          Name          User          Time Use S Queue
-----
1107606.pbserver Cali_Align      cwil2281      00:05:30 F small-express
[cwil2281@login1 Cali]$ ls -lh Output/
total 268M
-rw-r--r-- 1 cwil2281 RDS-FVS-DingDom-RW 268M Nov 30 19:29 134_sorted.bam
-rw-r--r-- 1 cwil2281 RDS-FVS-DingDom-RW 117K Nov 30 19:29 134_sorted.bam.bai
```

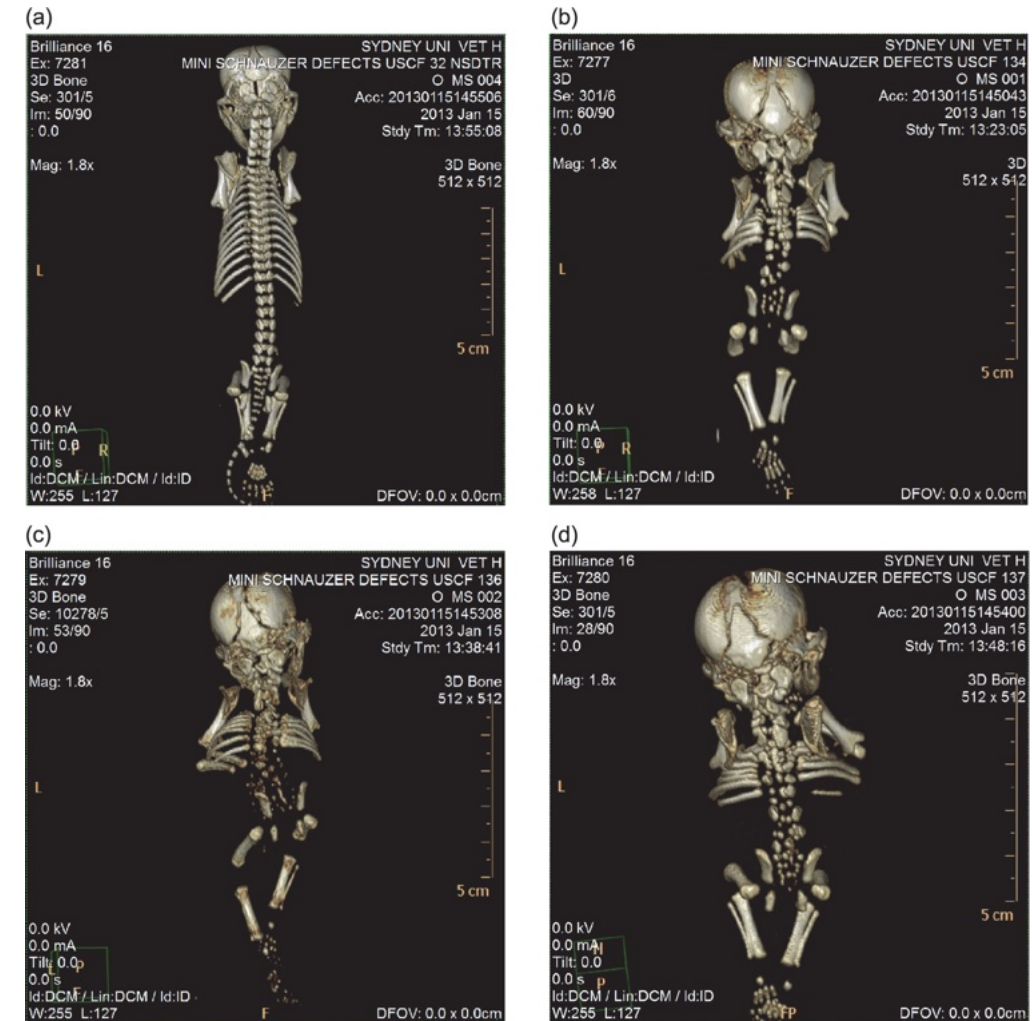
Align.pbs - check usage

- Check your usage log

Were your resource requests appropriate?

```
[cwil2281@login1 Cali]$ cat Cali_Align.o1107633_usage
Job Id: 1107633.pbsserver for user cwil2281 in queue small-express
Job Name: Cali_Align
Project: RDS-FVS-DingDom-RW
Exit Status: 0
Walltime requested: 00:05:00 :      Walltime used: 00:03:16
Cpus requested: 2 :
Cpu Time: 00:02:43 :      Cpu percent: 352
Mem requested: 2gb :      Mem used: 474364kb
VMem requested: None :      VMem used: 474364kb
PMem requested: None :      PMem used: None
[cwil2281@login1 Cali]$
```

Much better than last time! Walltime can fluctuate depending on system usage – we have over-requested but only by 2 minutes



Willet et al 2015, PLoS ONE 10(2): e0117055a

The sequence alignment you just created reveals the causal mutation for canine spondylocostal dysostosis (genomic position cfa5:32,945,846)

Some final tips about log files

- You can have both standard output and standard error send to the same log file (.o), instead of 2 separate logs, by adding this directive:

```
#PBS -j oe
```

- You can change the default name of your log files like this:

```
#PBS -o output_MyJob.log
```

```
#PBS -e error_MyJob.log
```

- Error and output logs are only readable by you (see permissions). To enable others in your project to read your logs (handy) add this directive:

```
#PBS -W umask=022
```

```
| -rw----- 1 cwil2281 RDS-FVS-DingDom-RW      0 Nov 30 20:31 Cali_Align.o1107633  
| -rw----- 1 cwil2281 RDS-FVS-DingDom-RW 23219 Nov 30 20:33 Cali_Align.e1107633  
| -rw-r--r-- 1 cwil2281 RDS-FVS-DingDom-RW   484 Nov 30 20:34 Cali_Align.o1107633_usage
```

Some final tips about variables

- PBS has a range of really useful ‘environment variables’
 - These are variables that are filled automatically when you submit a job
 - They are not covered today in the interest of time, however I encourage you to check them out to see how they could help simplify your scripting
- For example:

PBS_O_WORKDIR	- filled with the full pathname of the directory you are in when you submit your job
NCPUS	- the number of cpus requested

Backing up research data

- **Artemis is not backed up**

- Store raw data, final output, job scripts etc on the research data store!

To learn how to back up your data and move data to and from Artemis HPC, come to the Data Transfer session this afternoon!



Data transfer and RDS for HPC

Stephen Kolmann
Cali Willet

Hayim Dar
Nathaniel Butterworth

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Sydney Informatics Hub
A Core Research Facility



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SYDNEY



Image: thecultureconcept.com

Overview of today's course

- Research Data Store Introduction
- “Classic RDS” vs “RCOS”
- Data transfer between your computer, the research data store and Artemis
- Artemis data transfer queue (dtq)
- Getting accounts on HPC and RDS via RDMP



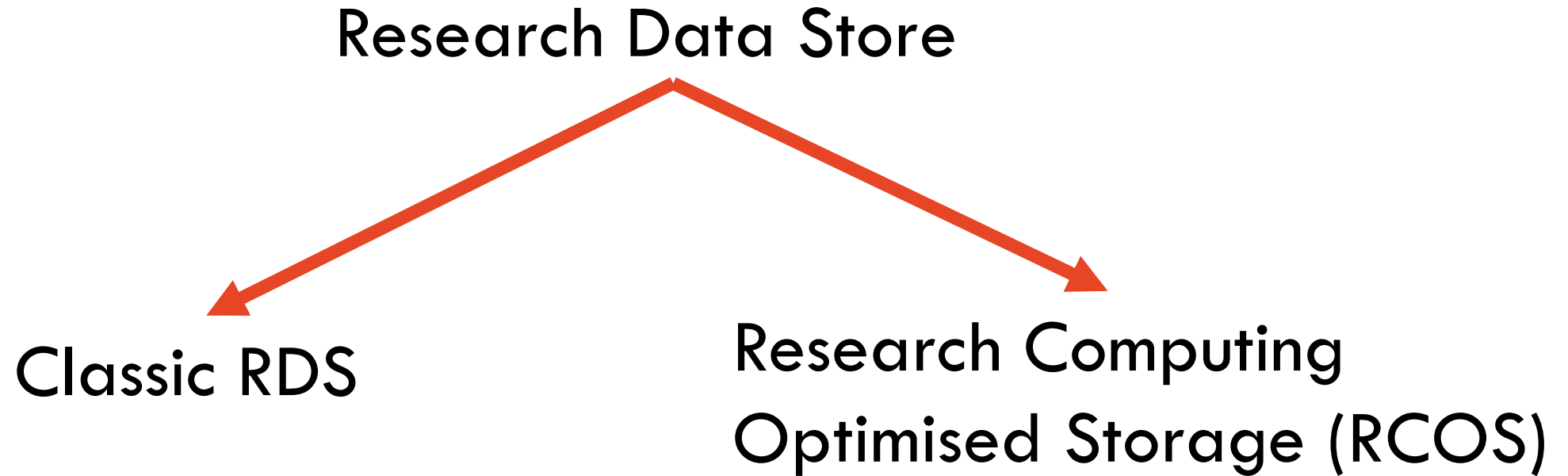
Research Data Store (RDS)



- The research data store is a central place to store digital data generated as part of your research
- **No cost to USyd researchers!**
 - Access requires a UniKey and valid RDMP
- Request as much storage as you need, it is free (if it ever says there is a charge, it is wrong)
- Backed up
 - No need for an external hard drive backup!
- Secure
 - Only people you specify will have access. Not sending data to overseas/cloud storage providers with possibly non-regulatory compliant storage

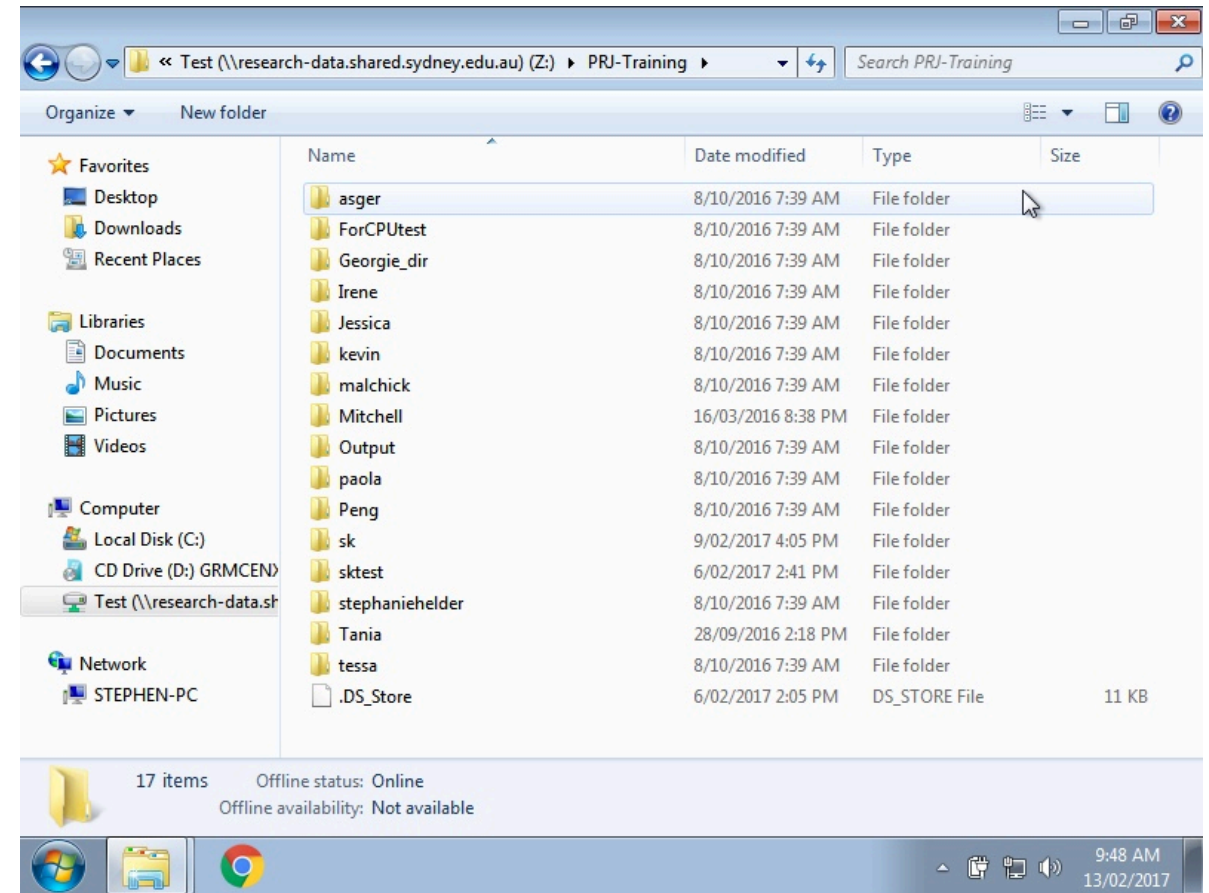
Research Data Store (RDS)

- Two types of Research Data Store:
- Each project must choose one:



Option 1: Classic RDS

- The best option if you don't use HPC
- Map a network drive and browse/transfer files in a graphical file explorer
- Open files in programs (eg Matlab) without transferring data to your local hard disk
- CIFS/Samba file access
 - Network accessible storage



Option 2: RCOS

- Use SFTP (secure File Transfer Protocol) to transfer data to transfer data
- Command-line driven: Good for Linux and Artemis users
- RCOS is mounted on Artemis (NFS v4)
 - Transfer data to/from Artemis with `cp/mv` commands
- To open files on your computer, you must transfer them to your local machine first (not network accessible)

```
vlan-2689-10-17-109-95:~ stephenkolmann$ sftp ict_hpctrain1@hpc.sydney.edu.au
ict_hpctrain1@hpc.sydney.edu.au's password:
Connected to hpc.sydney.edu.au.
sftp> ls
montyhall.m                                montyhall.out                                runmonty.pbs
runmonty.pbs.e1054450                      runmonty.pbs.e1268904                      runmonty.pbs.o1054450
runmonty.pbs.o1054450_usage                runmonty.pbs.o1268904                      runmonty.pbs.o1268904_usage
sftp> get montyhall.m
Fetching /home/ict_hpctrain1/montyhall.m to montyhall.m
/home/ict_hpctrain1/montyhall.m              100% 527    0.5KB/s   00:00
sftp> ll
Applications      R      computations      jnk      scaling
Desktop           RDSMount          dedalus_testing   jobs_running  sjk.kdbx
Documents         VirtualBox VMs    dell             junkdir      sourcecode
Downloads         acetalddehyde.xyz  django           mech_eng_abaqus  usage.png
Library           artemis.pub       get-pip.py       montyhall.m    veracrypt
Movies            bin               hello_world.c.txt pgadmin.log
Music             bt.tar.gz         hw48.pbs.txt     python_google
Pictures          cleantraining.sh  initdb           rocks_servers
Public            code              iso              runmonty.pbs
sftp> put bt.tar.gz
Uploading bt.tar.gz to /home/ict_hpctrain1/bt.tar.gz
bt.tar.gz              100% 45KB 44.6KB/s   00:00
sftp>
vlan-2689-10-17-109-95:~ stephenkolmann$
```

Getting an account on HPC and RDS

1. Fill out Research Data Management Plan (RDMP) at <https://rdmp.sydney.edu.au>
For more information on data management: http://sydney.edu.au/research_support/data/
 2. Sent to Lead Chief Investigator (LCI) for approval
 3. Sent to Research Data Support team for final approval
 4. Sent to ICT and HPC administrators who create your accounts
 5. You will receive an email when your accounts are created
-
- ❖ The whole process takes a few business days
 - ❖ If you experience a delay, in most cases it is because the LCI has not approved it
 - You can check up on the status of your RDMP by emailing researchdatasupport@sydney.edu.au

Getting an account on HPC and RDS

- Key points re RDMP:
 - Abbreviated project name is used to name HPC and RDS directories, so make sure its meaningful/memorable to you (these are unique University-wide)
 - At least one LCI per project, must have Academic Profile Online
 - All collaborators + LCI can read and write to HPC and RDS Project directories
 - Must select 'Yes' at HPC question to have HPC access created for that project
 - You can have multiple projects
 - **Storage on RDS:** choose 'classic RDS' (CIFS) or RCOS (NFS)

Data Transfer

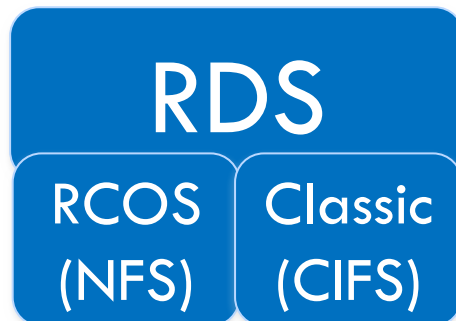
Overview of routes and tools for data transfer between different sources and destinations



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Data transfer methods

- Depending on where your data is located and where you want to copy it to will dictate which transfer method to use
- **A few things to keep in mind:**
 - Artemis HPC and RDS are physically separate machines
 - Artemis HPC is NOT backed up, RDS is
 - All raw data and important output should be stored on RDS and only copied to HPC when it is required for computational analysis



<http://www.clker.com/clipart-14276.html>, <https://www.hsn.com/products/dell-2tb-nal-hard-drive/7889994>, http://www.123rf.com/stock-photo/internet_icon.html, notechlead.com/networking/dell-brings-new-hpc-for-research-manufacturing-and-genomics-

Data analysis workflow

- 1) Raw data obtained
- 2) Copy it from source to RDS
- 3) Copy it from RDS to HPC scratch directory for analysis
- 4) Run compute job, sending output to scratch directory
- 5) Check the job has completed successfully and the output is correct
- 6) Copy final output and scripts to RDS for long-term, safe storage
- 7) If space permits and you require certain output files for downstream computational analysis, copy them to your project directory
- 8) Confirm that you have all raw data and output files backed up on RDS
- 9) Delete raw data, scripts and output files from scratch to RDS



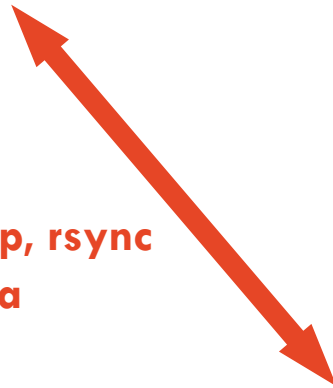
Data transfer between your computer and HPC



Local computer

Image: freelargeimages.com

scp, sftp, rsync
FileZilla



Artemis HPC

Image: itnews.com

Data transfer between your computer and 'classic' RDS (CIFS)



Artemis HPC

Image: itnews.com

Data transfer between HPC and 'classic' RDS (CIFS)



Local computer

Image: freelargeimages.com

Research data store

'classic' RDS

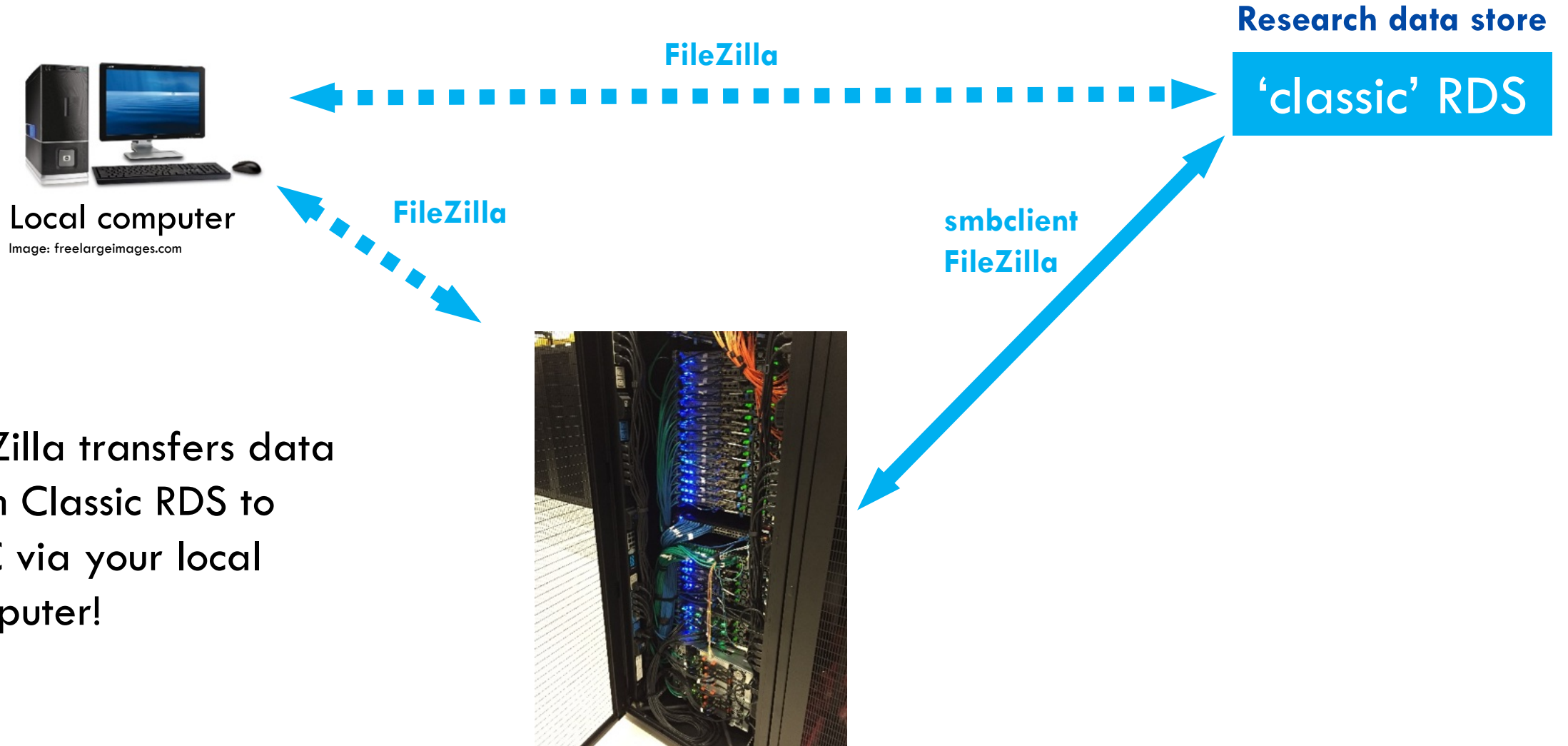
smbclient
FileZilla



Artemis HPC

Image: itnews.com

Data transfer between HPC and 'classic' RDS (CIFS)



FileZilla transfers data from Classic RDS to HPC via your local computer!

Data transfer between your computer and RCOS (NFS)

Research data store



Local computer

Image: freelargeimages.com

scp, sftp, rsync
FileZilla



Artemis HPC

Image: itnews.com

RCOS

Data transfer between HPC and RCOS (NFS)

Research data store



Local computer

Image: freelargeimages.com



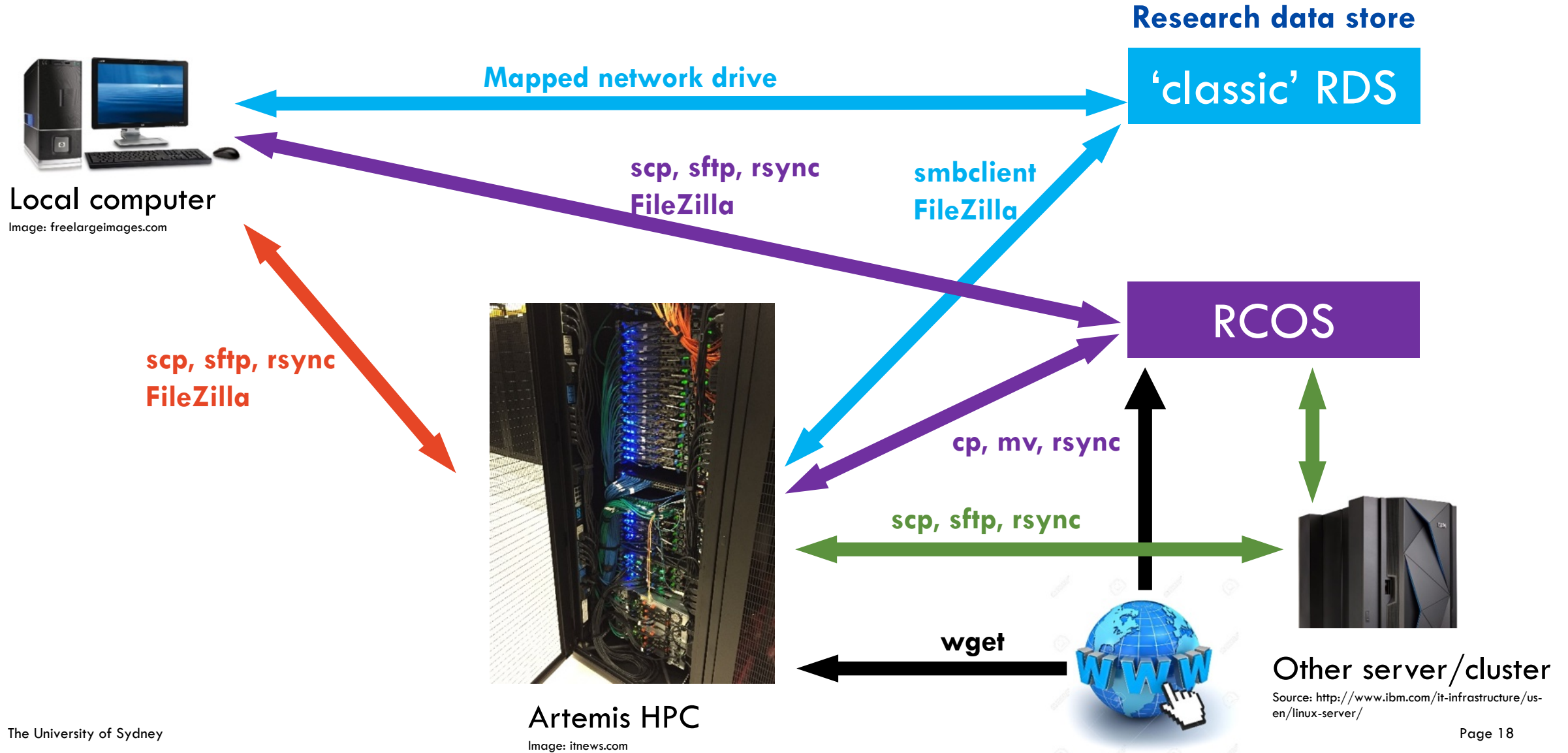
Artemis HPC

Image: itnews.com

RCOS

Artemis mount
enables cp, mv,
rsync

Data transfer methods summary



Data transfer: some practice

1. From local computer to HPC with scp, sftp, rsync or FileZilla
 - Any data
2. Web to RCOS with wget
 - Input data (Sample_data.tar.gz)
3. From RCOS to HPC with cp
 - Input Data (Sample_data.tar.gz)
4. From HPC to RCOS with rsync
 - Output files (Extracted data from Sample_data.tar.gz)
5. From local computer to RCOS with scp, sftp, rsync or FileZilla
 - Any data
6. From HPC to classic RDS with smbclient
 - See slides at the end of this document

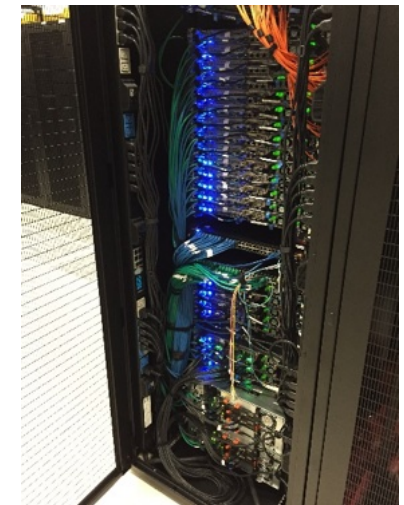
1. Local to HPC using FileZilla



- Upload data to HPC with FileZilla
 - Mac or Cygwin/Ubuntu Windows users, if you are comfortable with scp or sftp you can use either of those tools instead
 - Use any data you like for this exercise. For example, you could download this file:
ftp://ftp.ncbi.nlm.nih.gov/SampleData/Genomes/Analyzed_RNAseq/SRR1259333.bw

- FileZilla is a free file transfer client
 - Available from <https://filezilla-project.org/>
 - Windows, Linux or Mac

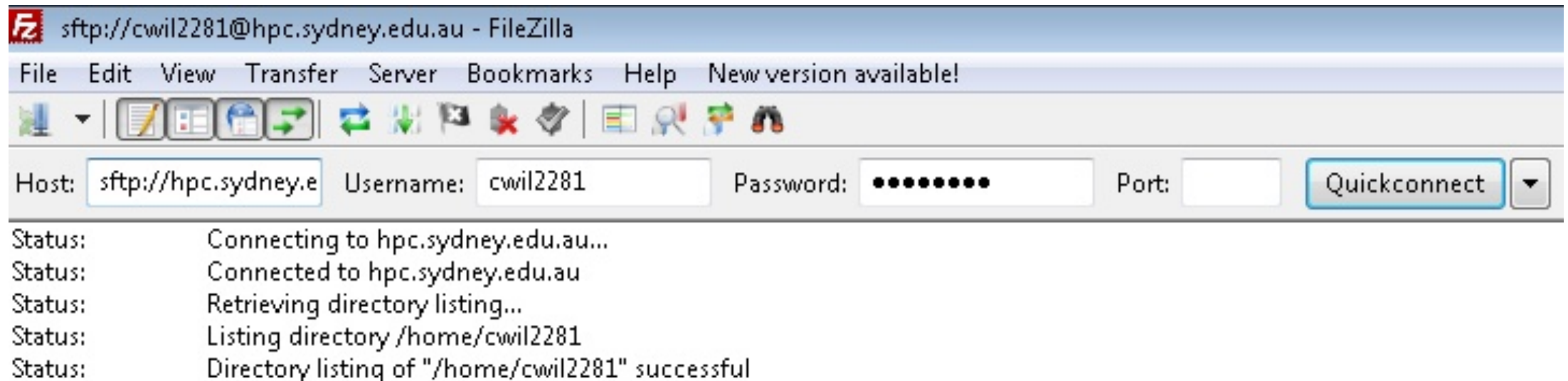
- Open FileZilla by clicking the windows icon, then type FileZilla, then click the FileZilla icon



Artemis HPC
Image: itnews.com

1. Local to HPC using FileZilla

- In the 'host' field, enter: **sftp://hpc.sydney.edu.au**
- In the 'Username' field, enter your unikey (or ict_hpctrain<N>)
- In the 'Password' field, enter your unikey password (or ict training password)
- You can leave port blank (default = 22)
- Click 'Quick connect'



1. Local to HPC using FileZilla

- The middle area of the FileZilla window is divided into four sections:

The screenshot displays the FileZilla client interface. The top bar shows the 'Local site' as the local machine and the 'Remote site' as '/home/cwil2281'. The interface is divided into four main sections:

- Top Left:** A tree view of the local PC's file system. It includes Desktop, My Documents, Computer, and various drives (C:, E: (Data2), F:, G: (\\research-data.shared.sydney.edu.au\FVS\GRP-Genopheno), H: (\\fs1.shared.sydney.edu.au\cwil2281\$), I: (\\research-data.shared.sydney.edu.au\FVS), and Q: (Lenovo_Recovery)). A red label 'Folders on your local PC' is overlaid on this section.
- Top Right:** A tree view of the remote HPC file system. It shows a 'home' directory containing a 'cwil2281' folder. A red label 'Folders on Artemis HPC' is overlaid on this section.
- Bottom Left:** A table listing the contents of the selected local folder (C:). The table has columns for Filename, Filesize, Filetype, and Last modified. A red label 'Files in the selected PC folder' is overlaid on this section.
- Bottom Right:** A table listing the contents of the selected remote folder (cwil2281). The table has columns for Filename, Filesize, Filetype, Last modified, Permissions, and Owner. A red label 'Files in the selected Artemis folder' is overlaid on this section.

At the bottom of the window, there is a status bar showing '14 files and 11 directories. Total size: 26,859 bytes' and a table with columns for Server/Local file, Direction, Remote file, Size, and Priority.

Filename	Filesize	Filetype	Last modified
C:		Local Disk	
E: (Data2)		Local Disk	
F:		CD Drive	
G: (\\research-data.shared.sydney.edu.au\FVS\GRP-Geno...		Network Drive	
H: (\\fs1.shared.sydney.edu.au\cwil2281\$)		Network Drive	
I: (\\research-data.shared.sydney.edu.au\FVS)		Network Drive	

Filename	Filesize	Filetype	Last modified	Permissions	Owr
..					
Script.pl	0	PL File	24/08/2015 12:42:57 PM	-rw-r--r--	cwil.
Samples.txt	0	Text Docu...	24/08/2015 12:42:57 PM	-rw-r--r--	cwil.
Sample_sheet.csv	0	Microsoft ...	24/08/2015 12:42:57 PM	-rw-r--r--	cwil.
Empty.txt	0	Text Docu...	24/08/2015 12:21:33 PM	-rw-r--r--	cwil.

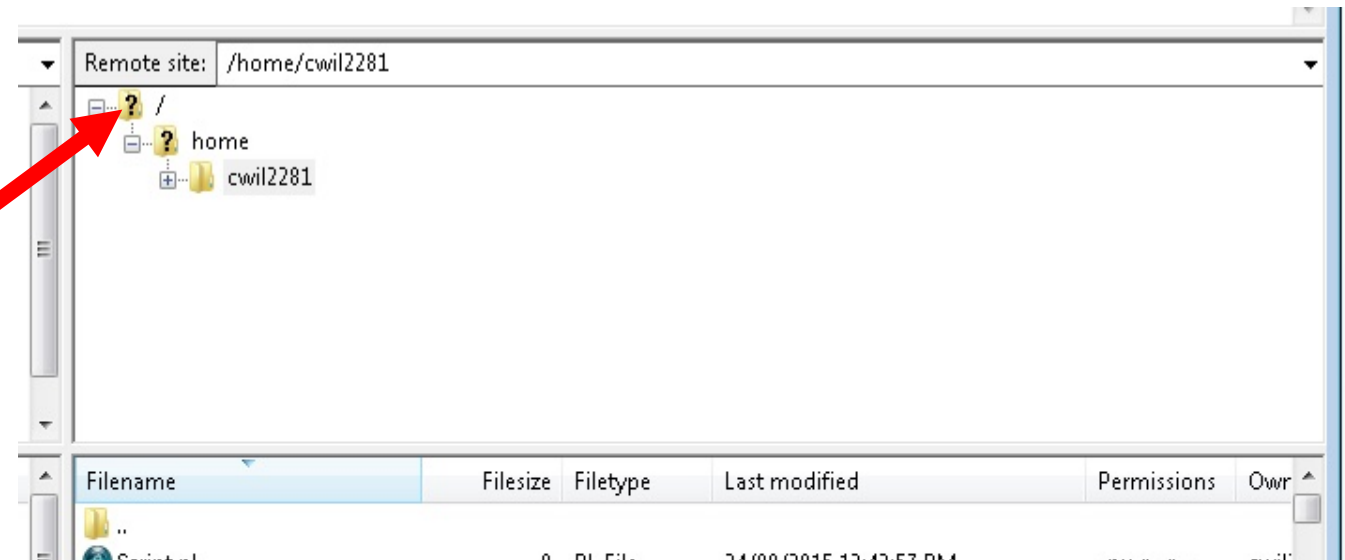
1. Local to HPC using FileZilla

- To copy from local to HPC, first open the HPC directory that you want to copy the file to
 - Navigate to `/project/Training/`
 - Make a directory for yourself (right click in the lower-right file list box and select “Create Directory”)

You can navigate to `/project/Training` in one of two ways:

- Expand out the list of Artemis directories from the root by clicking
- Type the full directory path in the ‘Remote site’ text box

Single-click
the question
mark to
expand list of
directories



1. Local to HPC using FileZilla

- Next, locate the data to be uploaded using the left-hand panes of FileZilla
- Locate your data on your computer in the left hand side of FileZilla
- Upload the file to Artemis using your preferred method:
 - Double click
 - Right click → upload
 - Drag and drop
- You will see the copied file appear in your HPC directory (lower right pane)
- Optional: start an Artemis terminal session and confirm the file was transferred by running 'ls' on the directory containing the transferred file

1. Local to HPC on the command line

- From the Mac terminal, or from a Windows Cygwin or Ubuntu (W10) terminal, command line transfer is an alternative to FileZilla

Example scp command:

```
scp /path/to/file/SRR1259333.bw <unikey>@hpc.sydney.edu.au:/project/Training/<YourName>
```

Example rsync command:

```
rsync /path/to/file/SRR1259333.bw <unikey>@hpc.sydney.edu.au:/project/Training/<YourName>
```

Example sftp commands:

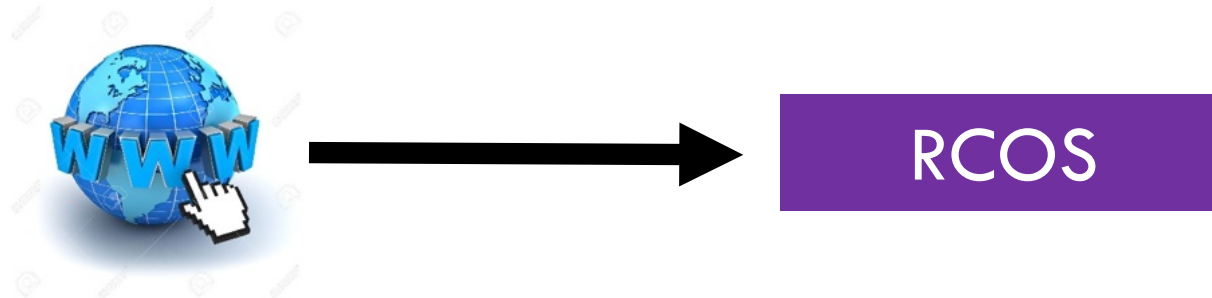
```
sftp <unikey>@hpc.sydney.edu.au  
cd /project/Training/<YourName>  
put /path/to/file/SRR1259333.bw  
ctrl + d
```

2. From the web to RCOS

- Next we will use 'wget' to download data from the internet to RCOS

Important points regarding RCOS and the Artemis HPC:

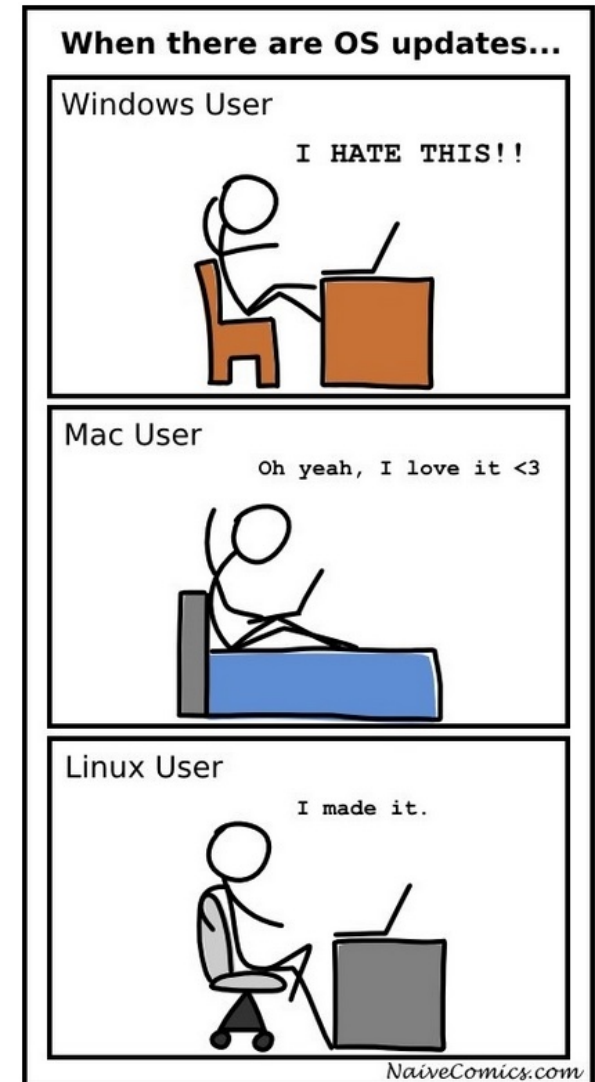
- RCOS has been **mounted** on the Artemis HPC: this means that it is accessible from an Artemis session under the drive /rds
- The **same project name** as used in /project and /scratch space is used, with the addition of a prefix 'PRJ-'
- Eg, if your short project name is Training, your RCOS directory on HPC is
/rds/PRJ-Training
- You can change into it using 'cd', move data to and from it with 'cp', 'mv', 'rsync' etc
- One thing you **cannot** do: read files on RCOS from a job running on the compute nodes
- This means that **input data for a compute job must first be copied to HPC** (/scratch or /project area)



Accessing the Artemis HPC from your computer

- **Mac/Linux:** use the existing Terminal application
- **Windows:** install a terminal client – many available with various features, eg:

Client	License	Benefit
Xwin	Provided by USyd	Edit text in a graphical window
Cygwin	Free	Approximates a native Linux environment on Windows PC
Putty	Free	Easily change size and colour of text



Connecting to Artemis from: Mac

- Optional: Install XQuartz: <https://www.xquartz.org/>
- Go to Finder → Applications → Utilities
- Double-click Terminal
- Type the following into Terminal then hit enter:

`ssh -Y ict_hpctrain<N>@hpc.sydney.edu.au`

NOTE:

If you see text enclosed in arrows, this means replace the text with content relevant to you

NOTE:

The -Y option only works if you installed XQuartz

Connecting to Artemis from: Mac

- Optional: Install XQuartz: <https://www.xquartz.org/>
- Go to Finder → Applications → Utilities
- Double-click Terminal
- Type the following into Terminal then hit enter:

`ssh -Y ict_hpctrain<N>@hpc.sydney.edu.au`

NOTE:

The `-Y` option only works if you installed XQuartz

- Type the password then press 'enter'

NOTE:

If you see text enclosed in arrows, this means replace the text with content relevant to you

```
Stephens-MacBook-Pro:~ stephenlevo$ ssh -Y ict_hpctrain10@hpc.sydney.edu.au
The authenticity of host 'hpc.sydney.edu.au (10.250.96.204)' can't be established.
RSA key fingerprint is da:c3:1e:26:24:fc:f9:1c:74:99:99:fb:85:26:d7:50.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'hpc.sydney.edu.au,10.250.96.204' (RSA) to the list of known hosts.
ict_hpctrain10@hpc.sydney.edu.au's password:
Last login: Fri Apr 24 10:39:55 2015 from vlan-2657-10-16-233-8.5 student-wireless-sydney.edu.au
-----
Welcome to the University of Sydney HPC Linux Cluster - Artemis
-----

** NOTE: All jobs must request a project the user is a member of.
Please specify -P <project> on the command line,
or #PBS -P <project> in your PBS scripts.

** Please don't run jobs on the login node!!! Use the queuing system

** Repeat! If you run jobs on the login nodes they will be subject to
random termination if they cause load problems. Please don't do it.

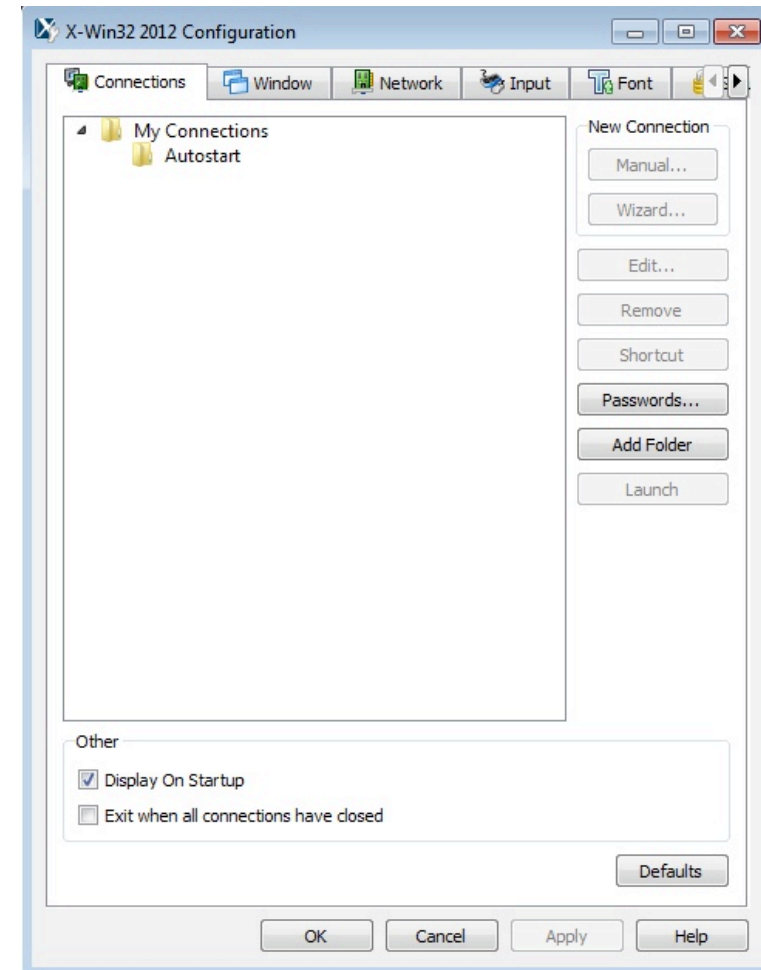
[ict_hpctrain10@login2 ~]$
```

If connecting for the first time you will be asked if you trust the authenticity of host – type 'yes' then hit enter key

Logon success

Connecting to Artemis from: Windows with Xwin

- Download and install X-Win: <http://softserv.usyd.edu.au/data/X-Win32/X-Win32-14.0/>
- Click the windows icon in the lower-right hand corner of your screen
- Type X-Win into the search bar
- Click the X-Win32 or X-Config options
- A configuration window will appear

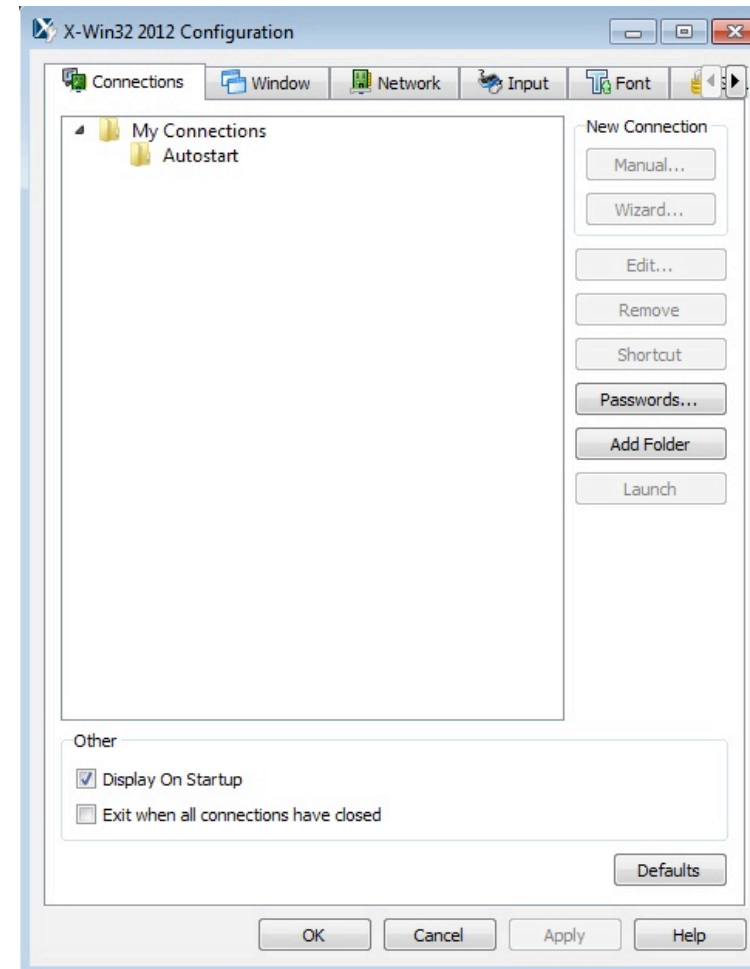


Where to download software for your own Windows machine?

- Xwin: <http://softserv.usyd.edu.au/data/X-Win32/X-Win32-14.0/>
 - Requires unikey credentials
 - Follow the instructions to install license server details
- Putty: <http://www.putty.org/>
 - Download and run 'putty.exe'
- Cygwin: <https://www.cygwin.com/>
 - Download and run 'setup-x86_64.exe' (64 bit OS) or 'setup-x86.exe' (32 bit OS)
 - During setup, select additional packages to install. Recommended: openssh, x-org server, xinit

Connecting to Artemis from: Windows with Xwin

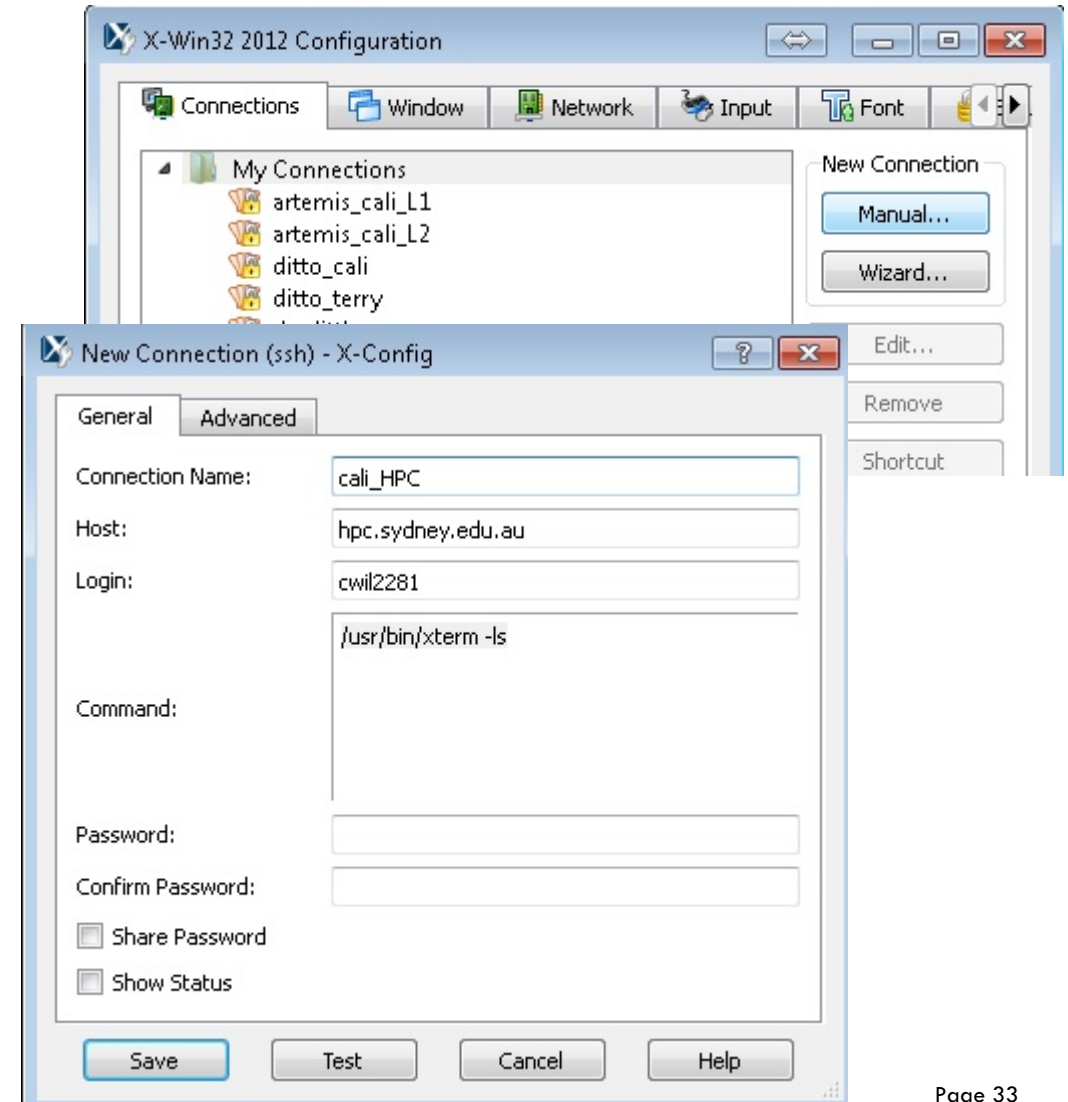
- Click the windows icon in the lower-right hand corner of your screen
- Type X-Win into the search bar
- Click the X-Win32 or X-Config options
- A configuration window will appear



Connecting to Artemis from: Windows with Xwin

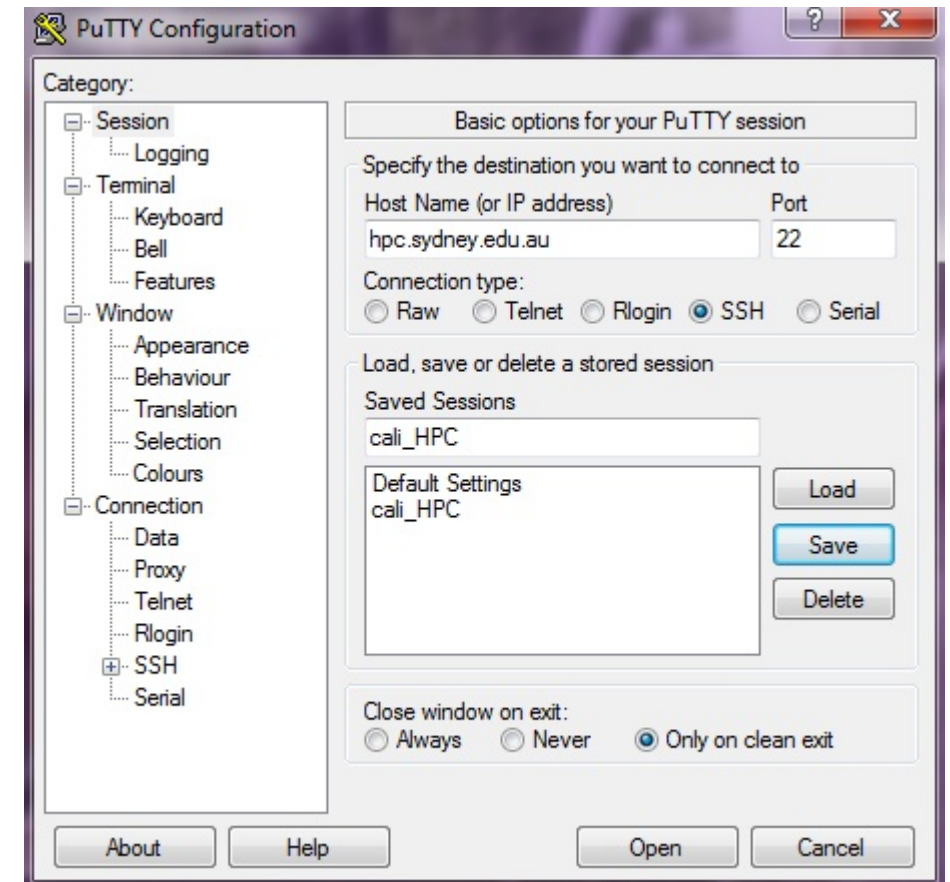
- Along the right hand side under 'New Connection' select 'Manual'
- Select 'ssh' for connection method then click 'Next'
- In the 'Connection Name' field, enter any descriptive name eg `<name_HPC>`
- In the 'Host' field, enter:
`hpc.sydney.edu.au`
- In the 'Login' field, enter your unikey
 - Training unikey: `ict_hpctrain<N>`
- In the 'Command' field, enter:
`/usr/bin/xterm -ls`
- Click 'Save' then 'Launch'

Letter 'l'
not numeral 1



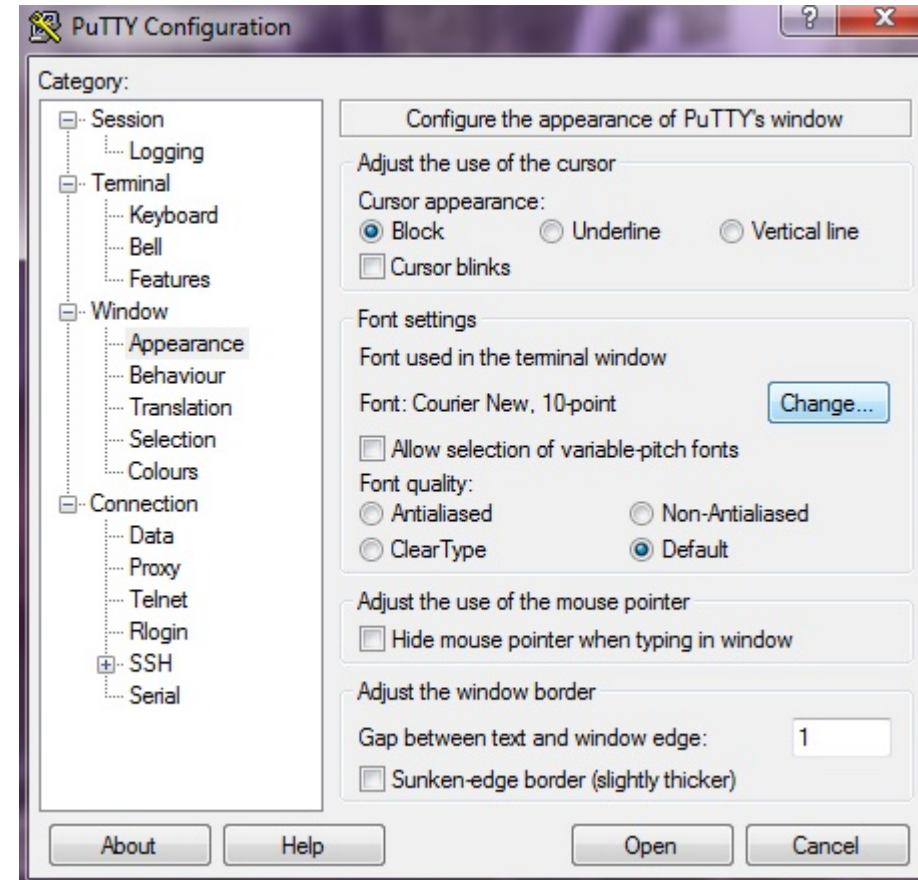
Connecting to Artemis from: Windows with Putty

- Double click the Putty desktop icon
A configuration window will appear
- In the 'Host Name' field, enter **hpc.sydney.edu.au**
- Leave port as 22 (default)
- Type a name for your connection into 'Saved Sessions' and click 'Save'
- Click 'Open'
- At 'login as' enter your unikey (or ict_hpctrain<N>)
- Enter your unikey (or training) password



Adjusting the terminal font size/colour on Putty

1. Open your Putty configuration window again
2. Click on your saved session name
3. Click 'Load'
4. Under 'Window', select 'Appearance'
5. Under 'Font Settings', click 'Change'
6. Change the font as desired
7. Click 'OK'
8. Click 'Session'
9. Click 'Save'
10. Click 'Open' to launch your new-look terminal



2. From the web to RCOS

- In your HPC terminal, change into your RCOS directory*

**Not all HPC accounts have RCOS. If you have a 'classic RDS' account, please copy the data to your project area instead. Instructions for using 'smbclient' (similar to ftp) to transfer data between 'classic RDS' and HPC are at the end of this presentation.*

```
cd /rds/PRJ-Training
```

- Make a new directory and call it your name

```
mkdir <YourName>_data
```

- Change into your new directory

```
cd <YourName>_data
```

- In a web browser (any EXCEPT Internet Explorer), open <http://hp580.angis.org.au/~cali/>

- Right click the file 'Sample_data.tar.gz' and select 'copy link address' or 'copy link location'

- In your terminal, type 'wget' then a space, paste the copied link** then press 'enter'

***Control + V does not mean copy to Linux! To copy into a Linux terminal on Windows: laptop pad is usually right click, mouse is usually centre click (wheel). You might need to press Shift+Insert to paste. To copy into Mac terminal, use Command + V*

```
wget http://hp580.angis.org.au/~cali/Sample_data.tar.gz
```

- ❖ Successful transfer should show a progress bar followed by a message: **"Sample_data.tar.gz" saved**

3. RCOS to HPC using cp

- We will transfer Sample_data.tar.gz from RCOS to Artemis and perform our "data processing" (which is simply extracting the archive today!)

- On Artemis, change directory to your Artemis /project directory:

```
mkdir /project/Training/<YourName>_data  
cd /project/Training/<YourName>_data
```

- Copy Sample_data.tar.gz from /rds to your current working directory:

```
cp /rds/PRJ-Training/<YourName>_data/Sample_data.tar.gz .
```

- Extract the files from Sample_data.tar.gz and move them to your directory:

```
tar -zxf Sample_data.tar.gz  
mv Sample_data/* .  
rmdir Sample_data
```

Don't forget to
include the full-stop!

RCOS

cp, mv, rsync

Artemis HPC

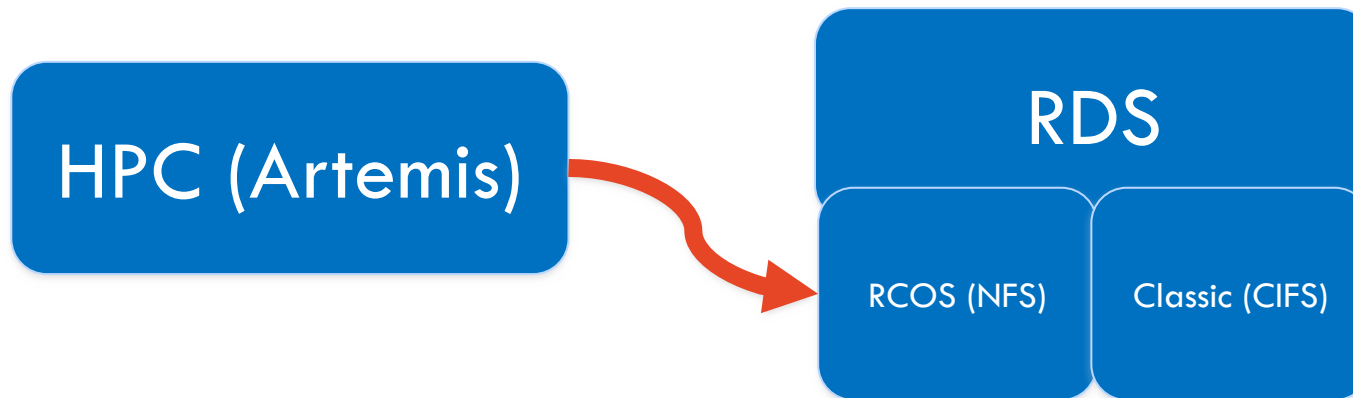
Image: itnews.com



Moving data from HPC to RDS

Now that Artemis has processed our data, we must transfer our output elsewhere.

- Remember: Artemis is for processing data, not storing data
- Artemis storage is not backed up
- Move your important data to the Research Data Store
- Clean up your Artemis directories regularly



Data transfer: some practice

1. From local computer to HPC with scp, sftp, rsync or FileZilla
 - Any data 
2. Web to RCOS with wget
 - Input data (Sample_data.tar.gz) 
3. From RCOS to HPC with cp
 - Input Data (Sample_data.tar.gz) 
4. From HPC to RCOS with rsync
 - Output files (Extracted data from Sample_data.tar.gz)
5. From local computer to RCOS with scp, sftp, rsync or FileZilla
 - Any data
6. From HPC to classic RDS with smbclient
 - See slides at the end of this document

4. HPC to RCOS: rsync

- This is the reverse of step 3
- The pathname is: `/rds/PRJ-Training`
- You can back up data there with `cp`, `mv` or `rsync` commands
- `rsync` can preserve permissions and file creation/modification times and only copies files that are new or have changed



Artemis HPC

Image: itnews.com

RCOS

Artemis mount
enables `cp`, `mv`,
`rsync`

➤ Backup your final output files from `/project` to RCOS using `rsync`:

```
rsync -rt /project/Training/<YourName>_data/* /rds/PRJ-Training/<YourName>_data
```

- `r` = recursive (copies directories too), `t` = preserves timestamps
- Run `ls` on RCOS to confirm transfer

5a. local to RCOS: FileZilla

- Upload any local data to RCOS

FileZilla:

1. Open FileZilla and enter into hostname field:
`sftp://rcos-int.sydney.edu.au`
`sftp://rcos.sydney.edu.au`
2. Fill in your UniKey and password and click 'quick connect'
3. On the right hand panes (RCOS) find your project
 - Either type the path into 'Remote site' field, or use the mouse to traverse the folder icons
4. On the left hand side (local) find the unzipped files you copied from the HPC
5. Click one file, hold down control and then select the other 2
 - Copy to RCOS by either drag-and-drop or right click → upload (double-click won't work for multiple files)
6. You will see a status of the transfer down the bottom, and the files will appear in the folder on the right when the transfer is complete



Local computer

Image: freelargeimages.com

FileZilla

RCOS

5b. local to RCOS: sftp

- Upload any local data to RCOS from **Mac, Windows (with Cygwin) or Windows 10 (with the beta Ubuntu terminal)**

sftp:

1. Open a terminal on your **local** computer
2. Change into the directory containing the 3 unzipped files you downloaded from HPC
3. Connect to RCOS over sftp and upload the 3 files:

```
sftp <unikey>@rcos-int.sydney.edu.au  
cd /rds/PRJ-Training  
mput *
```



sftp

RCOS

Internal hostname:
`rcos-int.sydney.edu.au`
External hostname:
`rcos.sydney.edu.au`

5c. local to RCOS: rsync

- Upload any local data to RCOS from **Mac, Windows Cygwin or Windows 10 Ubuntu**

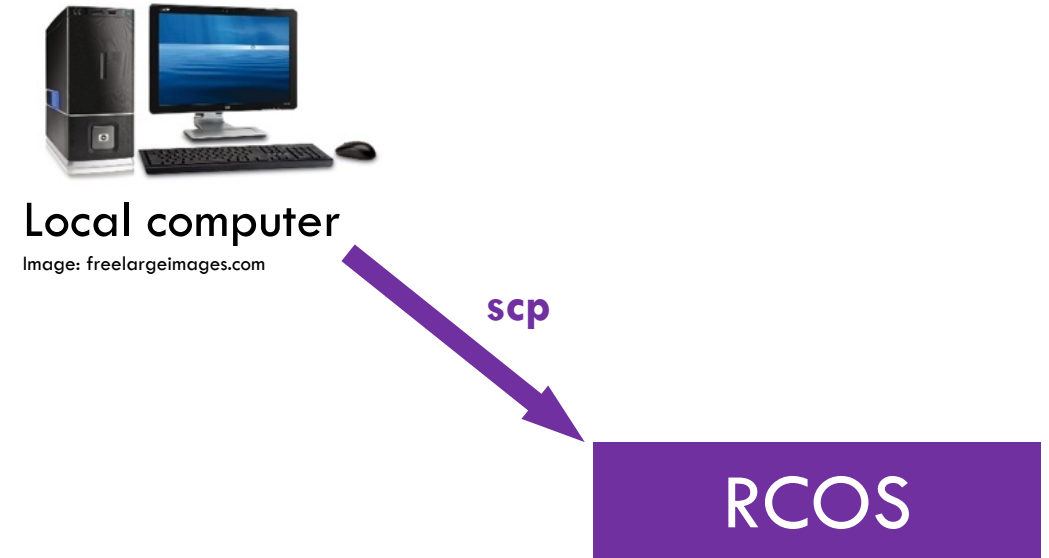
rsync:

1. Open a terminal on your **local** computer
2. Change into the directory containing the files you downloaded from HPC
3. Use rsync (or scp) to upload the 3 scripts to RCOS:

```
rsync * <unikey>@rcos-int.sydney.edu.au:/rds/PRJ-Training
```

Tip: to recursively copy a directory, add the 'r' flag to rsync, eg

```
rsync -r Directory <unikey>@rcos-int.sydney.edu.au:/rds/PRJ-Training
```



Data transfer: some practice

1. From local computer to HPC with scp, sftp or FileZilla 
 - Any data
2. Web to RCOS with wget 
 - Input data (Sample_data.tar.gz)
3. From RCOS to HPC with cp 
 - Input Data (Sample_data.tar.gz)
4. From HPC to RCOS with rsync 
 - Output files (Extracted data from Sample_data.tar.gz)
5. From local computer to RCOS with scp, rsync, sftp or FileZilla 
 - Output files (Extracted data from Sample_data.tar.gz)
6. From HPC to classic RDS with smbclient
 - See slides at end of this document

6. HPC to classic RDS: smbclient

- 'smbclient' is a command-line tool similar to sftp to transfer between a Linux system (Artemis) and a Windows system (RDS)
- FileZilla could also be used, but this is much slower
 - FileZilla moves data between HPC and RDS via your PC/laptop creating a network bottleneck
- See slides at the end for instructions

Research data store

'classic' RDS

smbclient
FileZilla



Artemis HPC

Image: itnews.com

Data transfer: some practice

1. From local computer to HPC with scp, sftp or FileZilla 
 - Any data
2. Web to RCOS with wget 
 - Input data (Sample_data.tar.gz)
3. From RCOS to HPC with cp 
 - Input Data (Sample_data.tar.gz)
4. From HPC to RCOS with rsync 
 - Output files (Extracted data from Sample_data.tar.gz)
5. From local computer to RCOS with scp, sftp rsync, or FileZilla 
 - Output files (Extracted data from Sample_data.tar.gz)
6. From HPC to classic RDS with smbclient 
 - See slides at the end of this document

Artemis Data Transfer Queue

Schedule data transfers using a data transfer PBS script

Create data transfer jobs using dt-script



THE UNIVERSITY OF
SYDNEY

Data transfer queue on Artemis

- There is an Artemis queue dedicated to running data transfer jobs
 - dtq
- This queue is for I/O intensive jobs (such as transferring data to and from Artemis)
- Commands such as **cp**, **tar**, **scp**, **rsync** can all be used in this queue
- This is the only queue that can read to or write from RCOS (/rds)
- This queue also has internet access, so you can run (for example) wget commands in a dtq script

Data transfer queue on Artemis

- An example data transfer script is below. Don't worry about making this now. It's just an example 😊

```
#!/bin/bash
#PBS -P Training
#PBS -l select=1:ncpus=1:mem=2gb
#PBS -l walltime=00:05:00
#PBS -q dtq

cd /project/Training/<YourName>_data
rsync /rds/PRJ-Training/<YourName>_data/* .
```

Using the data transfer queue on Artemis

- The end result: Data was copied from RCOS to Artemis. No big deal right?
 - We already know how to copy data from RCOS to Artemis
- BUT
 - You can schedule a job to start straight after copying data to Artemis
 - You don't have to worry about your computer going to “sleep”
 - This is especially useful if you need to transfer large amounts of data to Artemis before running a compute job
 - Your job will automatically begin at 5am when the data transfer completes 😊

Automation using the data transfer queue on Artemis

- We can do a data transfer and compute job using two, linked PBS scripts:
 1. A script that downloads data from the internet, another cluster or server to Artemis using download.pbs (which is a dtq queue script)
 2. A script that runs a job using the downloaded data as input: index.pbs (which is a small-express queue script)

➤ Copy download.pbs and index.pbs to your own dtq_testing folder:

```
cd /project/Training/<YourName>_data
mkdir dtq_testing
cd dtq_testing
rsync /project/Training/data_transfer_scripts/* .
```

Automation using the data transfer queue on Artemis

- Script 1: download.pbs

```
download.pbs X
#!/bin/bash
#PBS -P Training
#PBS -l select=1:ncpus=1:mem=2gb
#PBS -l walltime=00:05:00
#PBS -q dtq

cd $PBS_O_WORKDIR
wget http://hp580.angis.org.au/~cali/Sample_data.tar.gz
tar -zxf Sample_data.tar.gz
cp Sample_data/* .
rm Sample_data/*
rmdir Sample_data
```

Changes directory to the directory where you submit this script

This script grabs input data from the internet, extracts it, and reorganises it so it is ready for the analysis in the next script!

Automation using the data transfer queue on Artemis

- Script 2: index.pbs

```
index.pbs X
#!/bin/bash
# Create indexes for reference sequence
#PBS -P Training
#PBS -N Index
#PBS -l select=1:ncpus=1:mem=4GB
#PBS -l walltime=00:10:00
#PBS -q small-express

# Load modules
module load bwa
module load samtools

io=$PBS_O_WORKDIR

# BWA index:
bwa index -a bwtsv ${io}/canfam3_chr5.fasta

# SAMtools index:
samtools faidx ${io}/canfam3_chr5.fasta
```

This script performs analysis on the data we set up in Script 1

Automation using the data transfer queue on Artemis

- Submit the two scripts to the PBS Scheduler

```
qsub download.pbs
```

```
1265834.pbsserver
```

```
qsub -W depend=afterok:1265834 index.pbs
```



Replace this number with
the the job id assigned to
your job

- -W depend=afterok is a qsub option that says “only run index.pbs if download.pbs terminated successfully”
- index.pbs won't run unless the files were successfully downloaded and extracted
- Check job status

```
qstat -xu <unikey>
```

Automation using the data transfer queue on Artemis

```
[abcd1234@login3 data_transfer_scripts]$ qstat -xu abcd1234
```

pbsserver:

Job ID	Username	Queue	Jobname	SessID	NDS	TSK	Req'd Memory	Req'd Time	Elap S	Time
1426229.pbsserv	abcd1234	dtq	download.p	110451	1	1	2gb	00:05	R	00:00
1426230.pbsserv	abcd1234	small-ex	Index	--	1	1	4gb	00:10	H	--

- “H” state means the job is “held”. It won’t start until it is manually released, or until it’s job dependencies are met.
- We used `-W depend=afterok:1426229`, so index.pbs won’t start until download.pbs successfully completed.

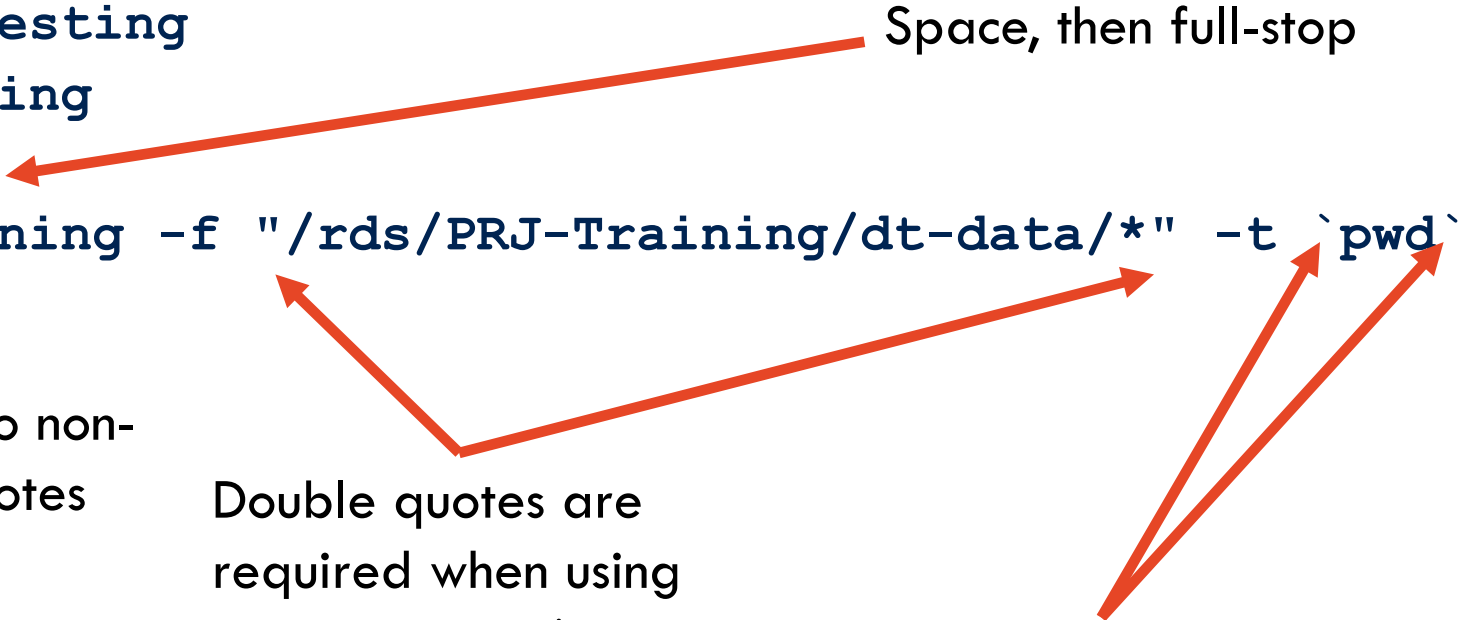
Data transfer queue helper script – dt-script

- There is a script available to help make using dtq simpler
 - dt-script
- We could have done the previous jobs with a single dt-script command!
- dt-script creates the dtq job for you. Specify where to copy data from and to, your project name, and (optionally) a PBS job to run after the data transfer completes.
- Syntax:
`dt-script -P <Project> -f <from> -t <to> -j job.pbs`
- `-P`, `-f` and `-t` are compulsory; `-j` is optional.
- There are other options available to dt-script. To see these, run the command:
`dt-script --help`
- Advanced users can make a copy of dt-script and tailor it to their needs

Data transfer queue helper script – dt-script

- Submit a copy/compute job using dt-script:

```
mkdir dt-script-testing  
cd dt-script-testing  
cp ../index.pbs .  
dt-script -P Training -f "/rds/PRJ-Training/dt-data/*" -t `pwd`  
-j index.pbs
```



Space, then full-stop

The above command uses two non-interchangeable types of quotes

Double quotes are required when using wildcards (eg *)

These quotes execute the command inside the quotes. In this case, pwd prints your current working directory

Data transfer queue helper script – dt-script

- Submit the dt-script job:

```
dt-script -P Training -f "/rds/PRJ-Training/dt-data/*" -t `pwd`  
-j index.pbs
```

- This dt-script command submits two jobs: a data transfer job and a compute job
- The data transfer script was created automatically

```
qstat -u <unikey>
```

Job ID	Username	Queue	Jobname	SessID	NDS	TSK	Req'd Memory	Req'd Time	S	Elap Time
1277947.pbsserv	skol2049	dtq	dt-script	--	1	1	4gb	24:00	Q	--
1277948.pbsserv	skol2049	small-ex	Index	--	1	1	4gb	00:10	H	--

The "H" means the job is in the "held" state. It will start once the dtq job successfully finishes.



Copy important data back to RCOS

- **Verify your output data is OK** before transferring back to RCOS or another computer
 - A machine cannot do this for you (in most cases). The onus is on **you** the researcher to ensure correct output before backing up your data and cleaning it off /project or /scratch

➤ To copy data back to RCOS, you can use dt-script:

```
mkdir /rds/PRJ-Training/<YourName>_data/dtq_output  
dt-script -P Training \  
-f `pwd` \  
-t "/rds/PRJ-Training/<YourName>_data/dt-output/*"
```

No backslashes
required if you type
this all on one line

Some quick revision!

- Take 15 minutes to work through the revision activity
 - Write down your answers as you work through the tasks
 - We will then go through the answers together



Revision activity

1. What types of Research Data Storage are available?

2. What commands/programs can you use to transfer data from Artemis to RCOS?

3. What is the name of the Artemis data transfer queue?

4. What is the fair share weighting of dtq?

5. What three flags must you always include in a data transfer script command?

6. What PBS directive tells the scheduler only to run a job after another job has successfully terminated?

7. What data transfer script flag lets you specify a PBS script to run after the data transfer job finishes?

8. What is the maximum number of CPUs you can request in a dtq job?

9. What is the maximum walltime for dtq?

10. Where should all important input and output data be stored long-term?

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1. What types of Research Data Storage are available?

Classic RDS and RCOS

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-P -f -t

**e.g. dt-script -P Training \
-f "/scratch/Training/mydata" \
-t "/rds/PRJ-Training"**

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-j

**e.g. dt-script -P Training \
-f "/scratch/Training/mydata" \
-t "/rds/PRJ-Training" \
-j job.pbs**

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-t "/rds/PRJ-Training" \
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8. What is the maximum number of CPUs you can request in a dtq job?

Two

9. What is the maximum walltime for dtq?

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10 days

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-f "/scratch/Training/mydata" \
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**e.g. dt-script -P Training \
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Two

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10 days

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Research Data Store

HPC commands summary

PBS directives

-P = project

-N = job name

-l select=1:ncpus=1:mem=1GB = request nodes, processors and total RAM

-l walltime=00:10:00 = walltime in hours:mins:secs

-M = email, use with **-m**

-m = mail when job: a = aborts, b = begins, e = ends

-j oe = create one job log containing both STDOUT and STDERR

-q = specify queue (can omit to submit to defaultQ)

Disk quota

pquota = show space allowed and used in all your accessible directories

PBS commands

qsub job.pbs = submit a job

qstat -x jobID = check status of specific job

qstat -u unikey = check status of all your submitted jobs

qstat -T | grep unikey = check estimated start time of all your submitted jobs

qdel jobID = delete job

qsub -W depend=afterok:1234567 job.pbs = submit a job and start it only if job 1234567 terminated successfully

dt-script --help = shows you how to use dt-script

smbclient and sftp commands

put File = transfer 1 file from HPC to RDS

get File = transfer 1 file from RDS to HPC

mput * = transfer multiple files from HPC to RDS using star wild card to match filenames

mget * = transfer multiple files from RDS to HPC using star wild card to match filenames

lcd = local cd

!pwd = local pwd

!ls = local ls

smbclient only

prompt = toggle on/off prompt

recurse = toggle on/off recurse

mput Directory* = run after the above 2 commands, will recursively copy Directory and all its files and subfolders (retaining directory structure) from HPC to RDS

mget Directory* = as above, from RDS to HPC

scp

Copy data to host: **scp data user@host:destinationDir**

Get data from host: **scp user@host:data destinationDir**

Thank you!

- That completes your introduction to using Artemis HPC
- Thanks to all for your attendance and participation
- Thanks also to Cali Willet for designing the courses!

Where to get help:

- Submit a HPC service request:
 - Go to the ICT Self-Service Portal at https://sydney.service-now.com/selfservice/ict_services.do
 - Select ICT Services → Research → High Performance Computing Request

- Please feel free to contact us with any questions:

stephen.kolmann@sydney.edu.au, hayim.dar@sydney.edu.au

We would also welcome any feedback about the training

- Research Data Support: RDMP form, data management

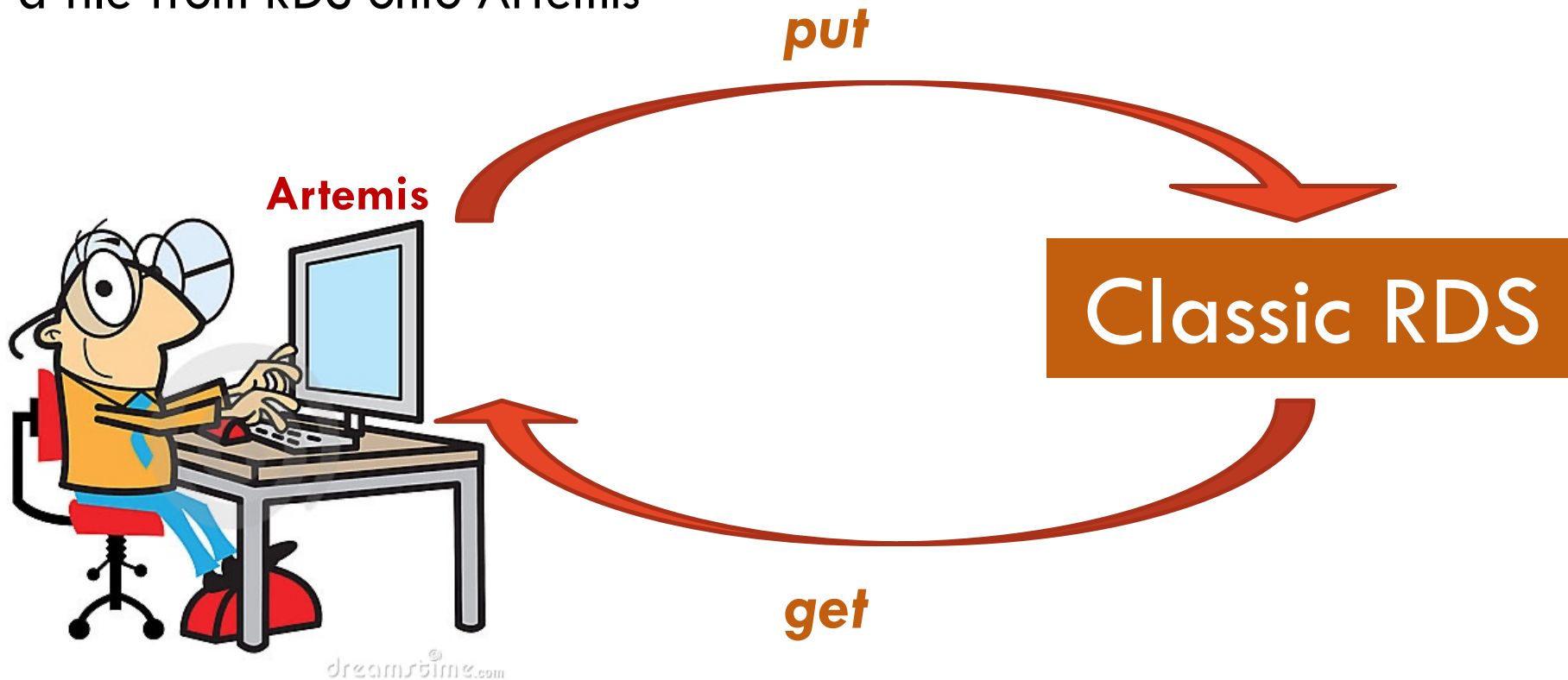
researchdatasupport@sydney.edu.au



Image: iconshut.com

7. Data transfer: HPC to Classic RDS - smbclient

- **'put'** a file from Artemis onto RDS
- **'get'** a file from RDS onto Artemis



Data transfer with smbclient: location matters!

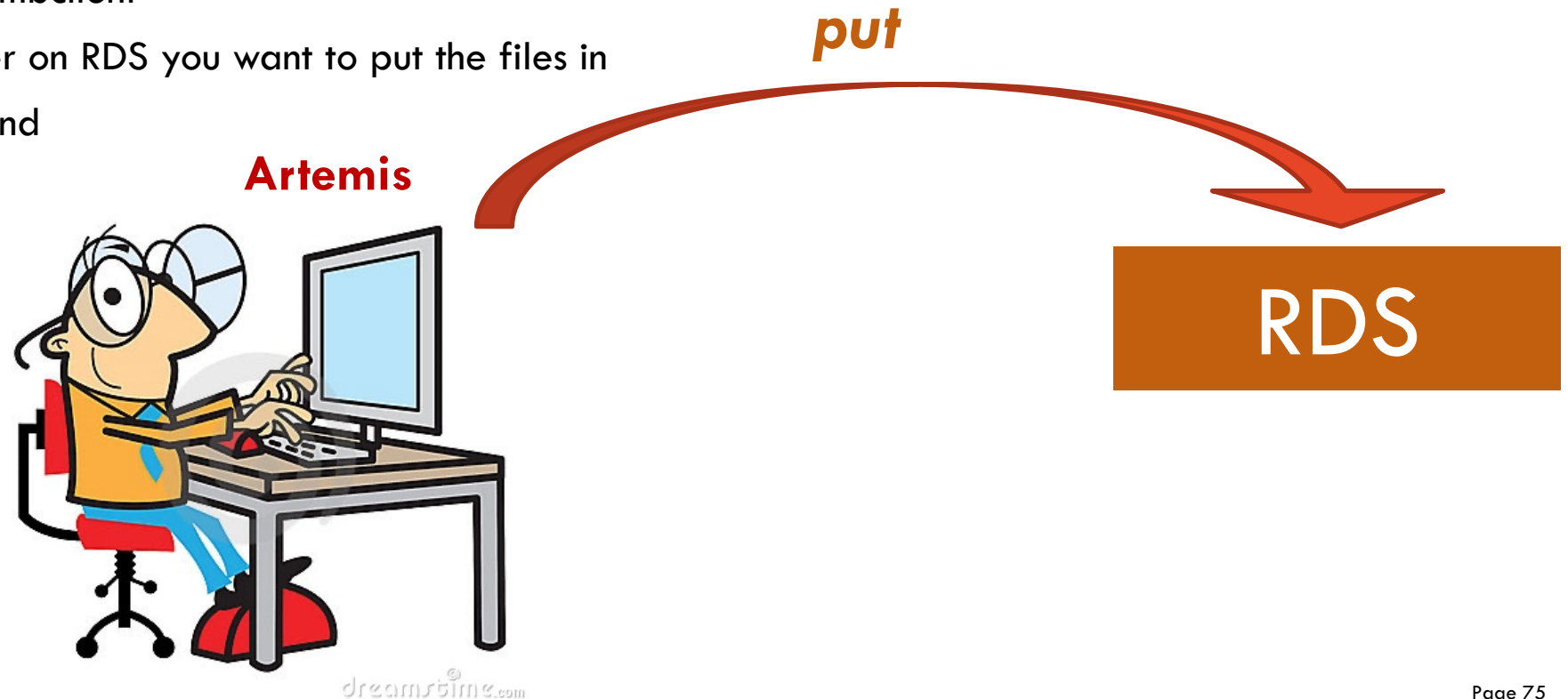
- **'put'** a file from Artemis onto RDS

Step 1: change into the directory on Artemis that contains the file/files you want to copy to RDS

Step 2: connect to RDS with smbclient

Step 3: change into the folder on RDS you want to put the files in

Step 4: issue the 'put' command



Data transfer with smbclient: location matters!

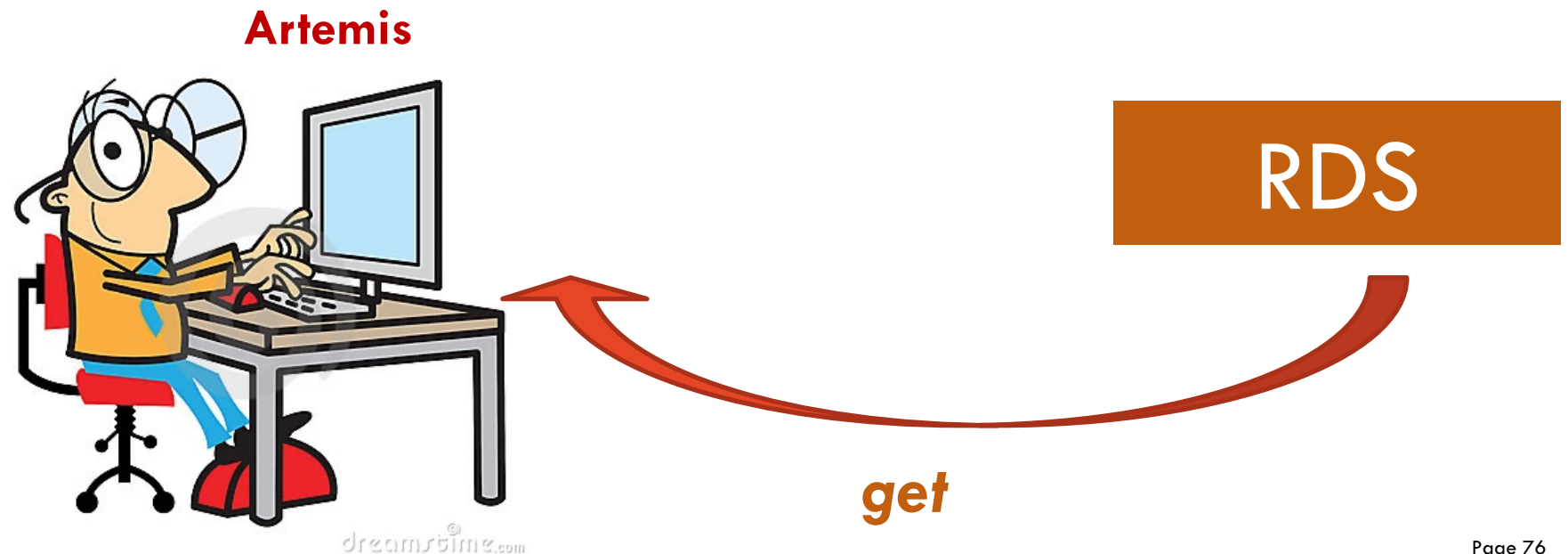
- **'get'** a file from RDS onto Artemis

Step 1: change into the directory on Artemis where you want the files to end up

Step 2: connect to RDS with smbclient

Step 3: change into the folder on RDS that contains the file/files you want to get

Step 4: issue the 'get' command



Data transfer with smbclient: location matters!

- If you need to change or query a directory on Artemis (local) **while logged in to RDS** (remote) with smbclient, you need to use modified commands:

Operation	Classic RDS	Artemis
Change directory	<code>cd Directory</code>	<code>lcd Directory</code>
Present working directory	<code>pwd</code>	<code>!pwd</code>
List directory contents	<code>ls</code>	<code>!ls</code>

Data transfer: HPC to Classic RDS - smbclient

➤ 'put' canfam3_chr5.fasta from Artemis onto RDS

1. In your HPC terminal, change into your directory that contains the file

➤ `cd /project/Training/<YourName>`

2. Enter the following command to connect to RDS via smbclient:

```
smbclient //research-data.shared.sydney.edu.au/<FAC>  
-U <UniKey> -W SHARED
```

*Make sure to replace '<FAC>' with your 3 letter Faculty abbreviation, eg SMS, FAE, FVS

***Training login users:** use 'Test' in place of 'FAC'

*Make sure to replace '<UniKey>' with your UniKey (or your HPC training login eg ict_hpctrain10)

3. Enter your UniKey (or training login) password when prompted

Data transfer: HPC to Classic RDS - smbclient

➤ 'put' canfam3_chr5.fasta from Artemis onto RDS

4. Run 'ls' to list the RDS project folders (prefix 'PRJ') you have permission to access

```

cwil2281@login2:~
[cwil2281@login2 ~]$ smbclient //research-data.shared.sydney.edu.au/FVS -U cwil2281 -W SHARED
Enter cwil2281's password:
Domain=[SHARED] OS=[Unix] Server=[DellFluidFS]
smb: \> ls
.                D          0 Thu Jul  2 18:35:58 2015
..               D          0 Thu Jul  2 18:35:58 2015
GRP-Genopheno    D          0 Thu Jun 25 09:36:59 2015
README.txt       A       3514 Fri Jun 20 11:37:11 2014
.DS_Store        AH     10244 Wed Jul  8 17:03:29 2015
➔ PRJ-Cribbing    D          0 Fri Jul  3 12:21:32 2015
.clusterConfig   DHS          0 Fri Jun 20 13:00:19 2014
➔ PRJ-DOG_DEAF    D          0 Fri Aug 21 12:12:40 2015
➔ PRJ-BioServ     D          0 Fri Aug 21 12:05:35 2015
Error in dskattr: NT_STATUS_NOT_SUPPORTED
smb: \>

```

Data transfer: HPC to Classic RDS - smbclient

- 'put' canfam3_chr5.fasta from Artemis onto RDS
- 5. Change into the RDS project folder you want to copy the files to

cd PRJ-Project

*Make sure to replace 'project' with your abbreviated project name

***Training login users**, your project is 'Training'

```
[ict_hpctrain10@login2 RDS-ICT-HPCTRaining-RW]$ smbclient //research-data.shared.sydney.edu.au/Test -W SHARED -U ict_hpctrain10
Enter ict_hpctrain10's password:
Domain=[SHARED] OS=[Unix] Server=[DellFluidFS]
smb: \> ls
.                D          0 Thu Jan 28 09:20:32 2016
..               D          0 Thu Jan 28 09:20:32 2016
PRJ-Training     D          0 Wed Jan 20 18:15:30 2016
README.txt       A        3514 Fri Jun 20 11:37:11 2014
.DS_Store        AH       10244 Mon Nov 30 11:31:43 2015
.clusterConfig   DHS          0 Mon Nov 23 09:19:34 2015
transactionManager.log.filepart A 25720290570 Tue May 26 14:42:29 2015
Error in dskattr: NT_STATUS_NOT_SUPPORTED
smb: \> cd PRJ-Training\
smb: \PRJ-Training\>
```

Data transfer: HPC to Classic RDS - smbclient

- 'put' canfam3_chr5.fasta from Artemis onto RDS

6. Extra step for training login users:

- Make a directory inside PRJ-Training, call it your name
- Change into it
- Use it for the rest of the data transfer activities

7. Use the 'put' command to copy canfam3_chr5.fasta to RDS

put canfam3_chr5.fasta

```
smb: \PRJ-Training\stephen\> put canfam3_chr5.fasta
putting file canfam3_chr5.fasta as \PRJ-Training\stephen\canfam3_chr5.fasta (51231.6 kb/s)
(average 51231.6 kb/s)
```

Multiple transfer: HPC to Classic RDS - smbclient

➤ 'mput' the two .pbs files from Artemis onto Classic RDS:

1. Verify your **local** directory (on Artemis) contains the .pbs data files

!pwd

!ls

```
[smb: \PRJ-Training\stephen\> !pwd
/project/Training/stephen
[smb: \PRJ-Training\stephen\> !ls
134_R1.fastq.gz align.pbs hello.pl transfer
134_R2.fastq.gz canfam3_chr5.fasta index.pbs
```

Multiple transfer: HPC to Classic RDS - smbclient

- 'mput' the two .pbs files from Artemis onto Classic RDS:
- 2. Put both .pbs files onto Classic RDS using the star wild card:
`mput *.pbs`
- 3. You will be prompted 'put file align.pbs?' → enter 'y' for yes, 'n' for no

```
[smb: \PRJ-Training\stephen\> mput *.pbs  
[Put file index.pbs? y  
putting file index.pbs as \PRJ-Training\stephen\index.pbs (18.7 kb/s) (average 18.7 kb/s)  
[Put file align.pbs? y  
putting file align.pbs as \PRJ-Training\stephen\align.pbs (29.3 kb/s) (average 24.3 kb/s)
```

Multiple transfer: HPC to Classic RDS - smbclient

➤ 'mput' the two .pbs files from Artemis onto Classic RDS:

4. Run the '**ls**' command to see that your files were successfully copied to RDS:

```
[smb: \PRJ-Training\stephen\> ls
.                D           0  Tue Feb 14 13:02:03 2017
..               D           0  Tue Feb 14 13:01:48 2017
align.pbs        A          720  Tue Feb 14 12:59:57 2017
index.pbs        A          402  Tue Feb 14 12:59:56 2017
canfam3_chr5.fasta A 40500003  Tue Feb 14 12:57:23 2017
Error in dskattr: NT_STATUS_NOT_SUPPORTED
```

5. Type 'q' then hit enter (or use ctrl + d) to **quit smbclient**

Data transfer: Classic RDS to HPC - smbclient

➤ 'get' bt.tar.gz from RDS onto Artemis

1. Enter the following command at the Artemis shell prompt (or use the tip):

```
smbclient //research-data.shared.sydney.edu.au/FAC  
-U UniKey -W SHARED
```

2. Enter your UniKey password when prompted

3. Use 'cd' to change into the RDS project folder with the file:

```
cd PRJ-Training/ict
```

Tip: Use the 'up' arrow to retrieve the smbclient command you entered earlier, or run `history | grep smbc` then run `!n` 😊

Data transfer: Classic RDS to HPC - smbclient

➤ 'get' bt.tar.gz from Classic RDS onto Artemis

5. Use 'get' to copy bt.tar.gz to your 'Transfer' directory

get bt.tar.gz

You should see 'getting file...'

```
[smb: \> cd PRJ-Training  
[smb: \PRJ-Training\> get bt.tar.gz  
getting file \PRJ-Training\bt.tar.gz of size 45646 as bt.tar.gz  
(810.5 KiloBytes/sec) (average 810.5 KiloBytes/sec)
```

Multiple transfer: Classic RDS to HPC - smbclient

➤ 'mget' both .pbs files from Classic RDS to HPC

1. **Turn off prompt by typing 'prompt'** then hit enter
2. Use 'mget' to copy both .pbs files to your Artemis directory

mget *.pbs

- Notice how both files are transferred without you needing to type 'y' for each

```
smb: \PRJ-Training\ict\> mget *.pbs
getting file \PRJ-Training\ict\align.pbs of size 720 as align.pbs (78.1 KiloBytes/sec)
(average 707.5 KiloBytes/sec)
getting file \PRJ-Training\ict\index.pbs of size 402 as index.pbs (43.6 KiloBytes/sec)
(average 625.6 KiloBytes/sec)
```

3. Confirm your 'get' and 'mget' commands have worked by running local ls:

!ls

'prompt' asks you before transferring every file in a multiple transfer- this is by default on.
To toggle it off (recommended for numerous and/or large files), simply enter the command 'prompt'

Multiple transfer: recursive copy with smbclient

- If you want to copy a directory and its contents, you need to perform a **recursive copy**
 - To do this you need to have 'recurse' on (off by default)
 - Ideal to also have 'prompt' off (on by default)
- This will also retain the directory structure of the directory you are copying
 - Very handy for copying whole projects or essential directory structures
- Make sure you are one directory level above the directory you want to copy

Multiple transfer: recursive copy with smbclient

➤ Recursively copy the 'bt' directory from Artemis HPC to Classic RDS

1. Exit smbclient by typing 'q'

q

2. Extract the bt.tar.gz file:

tar -zxf bt.tar.gz

You should now have a directory in your current directory on Artemis called "bt":

```
[ict_hpctrain1@login3 stephen]$ tar -zxf bt.tar.gz
[ict_hpctrain1@login3 stephen]$ ls
134_R1.fastq.gz  align.pbs  bt.tar.gz  hello.pl
134_R2.fastq.gz  bt        canfam3_chr5.fasta  index.pbs
```

Multiple transfer: recursive copy with smbclient

- Recursively copy the 'Output' directory from Classic RDS to Artemis HPC

Opposite to 'prompt', '**recurse**' is by default off. Toggle it on by entering 'recurse'.

3. Log into smbclient again:

```
smbclient //research-data.shared.sydney.edu.au/Test -U  
ict_hpctrain1 -W SHARED
```

3. Change directories to your directory on Classic RDS

```
cd PRJ-Training/<YourName>
```

3. Toggle prompt off by typing '**prompt**'

4. Toggle on recursive copy by typing '**recurse**'

Multiple transfer: recursive copy with smbclient

➤ Recursively copy the 'Output' directory from Classic RDS to Artemis HPC

5. Recursively copy the 'bt' directory and all it's contents to RDS: `mput bt`

```
smb: \> cd PRJ-Training
smb: \PRJ-Training\> cd stephen
smb: \PRJ-Training\stephen\> prompt
smb: \PRJ-Training\stephen\> recurse
smb: \PRJ-Training\stephen\> put bt
bt does not exist
smb: \PRJ-Training\stephen\> mput bt
putting file bt/btC16.pbs as \PRJ-Training\stephen\bt\btC16.pbs (29.2 kb/s) (average 29.2 kb/s)
putting file bt/C16.o1016707 as \PRJ-Training\stephen\bt\C16.o1016707 (264.9 kb/s) (average 147.0 kb/s)
putting file bt/bt.C.16 as \PRJ-Training\stephen\bt\bt.C.16 (6897.6 kb/s) (average 3522.3 kb/s)
```

Classic RDS to Local Computer: Map a network Drive

- To transfer data between your local computer and Classic RDS, you can map a network drive and drag-and-drop files
- Detailed instructions are available on the University Website:
http://staff.ask.sydney.edu.au/app/answers/detail/a_id/174/~how-do-i-map-my-network-drive-so-i-can-access-my-network-storage%3F

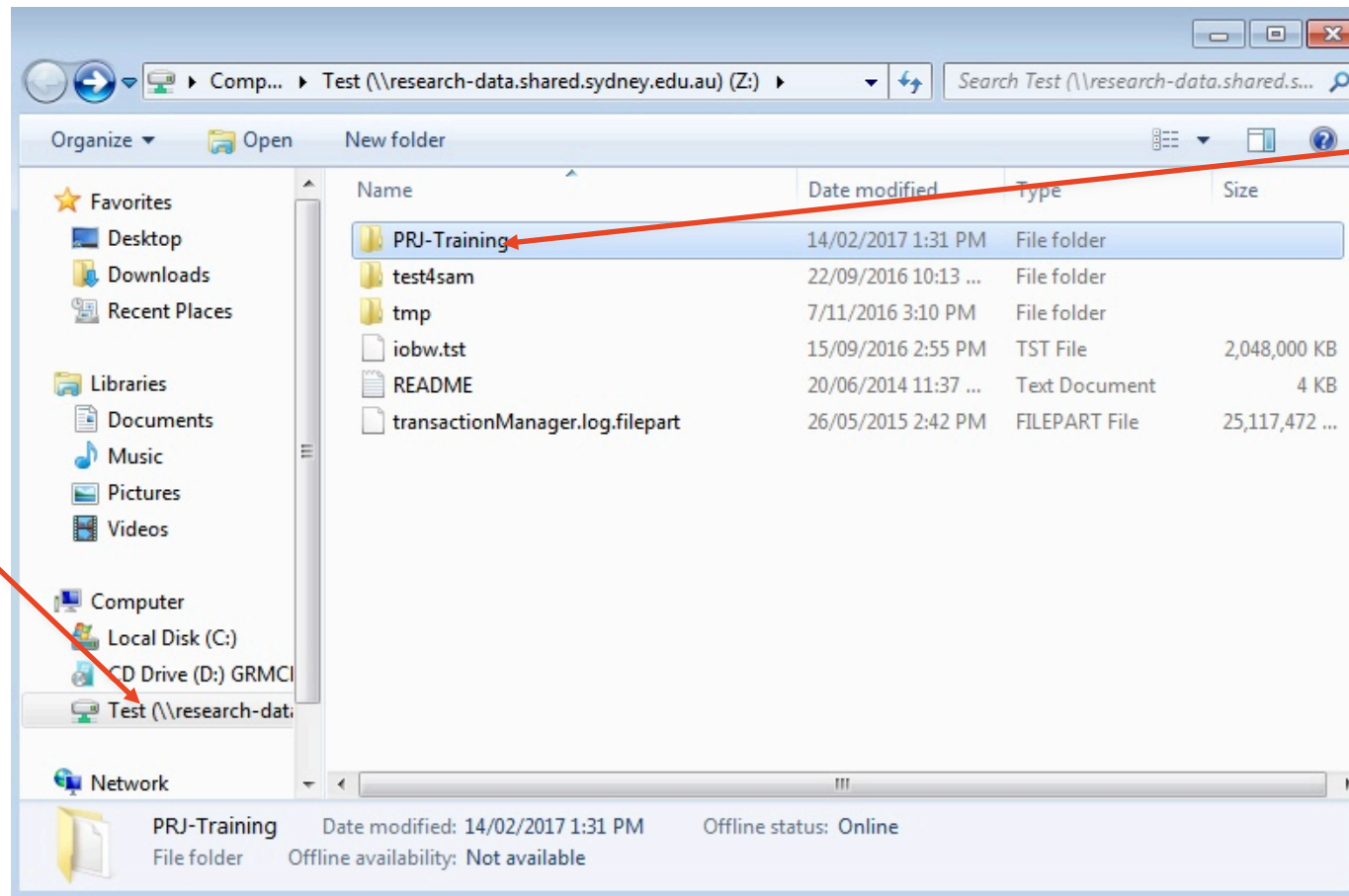
Classic RDS to Local Computer: Map a network Drive

- For Windows 7 and Classic RDS using the HPC training accounts:
 - From the start menu click **Computer**
 - When the Computer Window displays, click the **Map network drive button**.
 - Select a letter from the dropdown list in the Drive field
 - Enter the path of the shared drive you wish to connect to in the **Folder field**:
\\research-data.shared.sydney.edu.au\Test
 - Tick the **Connect using different credentials** box
 - Enter **mcs\UniKey** and your **password**. Use your ict_hpctrain unikey and password.
 - Click **OK**

Classic RDS to Local Computer: Map a network Drive

- If successful, you should see PRJ-Training in a file explorer window
- You can transfer data here just like you would on a normal windows computer

Classic RDS
is here



Your Classic
RDS directory is
here